

JensenNarleskiSaraf_proj

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2025-03-12

Is there a significant difference between exposures for pollutants based on activity? recreational, commercial, industrial, agricultural comparing specific pollutants (ANOVA) comparing the NC air quality levels for exposure to the real levels that we are seeing.

next steps: - picking pollutants (2) (formaldehyde, diesel PM, maybe another) - sorting columns into categories - find NC air quality standards for selected pollutants

Unemployment Data: https://hdpulse.nlmhd.nih.gov/data-portal/social/table?age=916&age_options=age25_1&demo=00012&demo_options=education_3&race=00&race_options=race_7&sex=0&sex_options=sexboth_1&socialtopic=090&socialtopic_options=social_6&statefips=37&statefips_options=area_states

Population Density: <https://beautifydata.com/united-states-population/census/2020/population-density-by-county/north-carolina>

Income: https://hdpulse.nlmhd.nih.gov/data-portal/social/table?age=001&age_options=ageall_1&demo=00010&demo_options=income_3&race=00&race_options=race_7&sex=0&sex_options=sexboth_1&socialtopic=030&socialtopic_options=social_6&statefips=37&statefips_options=area_states

Setup

```
# Load packages
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
## here() starts at /Users/jaclynrose/Desktop/ENV710/JensenNarleskiSaraf_proj
```

```

library(knitr)
library(readxl)
library(patchwork)

# Load data
exposure <- read.csv("ExposureConcentrationsRaw.csv")
# Loading in dataset with urban/rural classification for each county in NC
urban_rural <- read_excel("urban_rural.xlsx")
# dataset with unemployment and median family income
census <- read_excel("census_combined.xlsx")

```

Figure 1: National Air Toxics Assessment (NATA) Emission Sources

On Road	Non-Road	Non-Point Stationary
Light Duty Gas On-Network	Recreational & Pleasure Craft	Industrial
Light Duty Diesel On-Network	Construction	Oil & Gas
Light Duty Gas Off-Network	Commercial Lawn & Garden	Commercial Cooking
Light Duty Diesel Off-Network	Residential Lawn & Garden	Solvents and Coatings
Heavy Duty Gas On-Network	Agricultural Equipment	Bulk Terminal Storage & Transfer (stage 1)
Heavy Duty Diesel On-Network	Commercial Equipment	Fuel Combustion (except residential wood)
Heavy Duty Gas Off-Network	Commercial Marine Vessel C1/C2	Waste Disposal
Heavy Duty Diesel Off-Network	Commercial Marine Vessel C3	Agricultural Livestock
Heavy Duty Diesel Hoteling	Commercial Marine Vessels Underway	Miscellaneous Nonindustrial
Refueling	Locomotives	
	Airports (point source)	
	Railroads (point source)	
	All Other Non-Road Equipment	

Data Tidying & Wrangling Code

```

# Select columns and rename for clarity
exposure_df <- exposure %>%
  select(County, Population, Pollutant.Name, Total.Exposure.Conc,
         OR.LightDuty.OffNetwork.Gas.Exposure.Conc,
         OR.LightDuty.OffNetwork.Diesel.Exposure.Conc,
         OR.HeavyDuty.OffNetwork.Gas.Exposure.Conc,
         OR.HeavyDuty.OffNetwork.Diesel.Exposure.Conc,
         OR.HeavyDuty.OnNetwork.Diesel.Exposure.Conc,
         OR.LightDuty.OnNetwork.Gas.Exposure.Conc,

```

```

OR.LightDuty.OnNetwork.Diesel.Exposure.Conc,
OR.HeavyDuty.OnNetwork.Gas.Exposure.Conc,
OR.HeavyDuty.Hoteling.Exposure.Conc,
OR.Refueling.Exposure.Conc,
NR.Recreational.inc.PleasureCraft.Exposure.Conc,
NR.Construction.Exposure.Conc,
NR.Agriculture.Exposure.Conc,
NR.CommercialEquipment.Exposure.Conc,
NR.Locomotives.Exposure.Conc,
NR.Point.Airports.Exposure.Conc,
NR..Point.Railyards.Exposure.Conc) %>%
rename("OR.Light.Off.Gas" = OR.LightDuty.OffNetwork.Gas.Exposure.Conc,
"OR.Light.Off.Diesel" = OR.LightDuty.OffNetwork.Diesel.Exposure.Conc,
"OR.Heavy.Off.Gas" = OR.HeavyDuty.OffNetwork.Gas.Exposure.Conc,
"OR.Heavy.Off.Diesel" = OR.HeavyDuty.OffNetwork.Diesel.Exposure.Conc,
"OR.Light.On.Gas" = OR.LightDuty.OnNetwork.Gas.Exposure.Conc,
"OR.Light.On.Diesel" = OR.LightDuty.OnNetwork.Diesel.Exposure.Conc,
"OR.Heavy.On.Gas" = OR.HeavyDuty.OnNetwork.Gas.Exposure.Conc,
"OR.Heavy.On.Diesel" = OR.HeavyDuty.OnNetwork.Diesel.Exposure.Conc,
"OR.Heavy.Idling" = OR.HeavyDuty.Hoteling.Exposure.Conc,
"OR.Refueling" = OR.Refueling.Exposure.Conc,
"NR.Recreational.Vehicles" =
  NR.Recreational.inc.PleasureCraft.Exposure.Conc,
"NR.Construction" = NR.Construction.Exposure.Conc,
"NR.Agriculture" = NR.Agriculture.Exposure.Conc,
"NR.Commerical.Equip" = NR.CommercialEquipment.Exposure.Conc,
"NR.Locomotives" = NR.Locomotives.Exposure.Conc,
"NR.Airpoints" = NR.Point.Airports.Exposure.Conc,
"NR.Railyards" = NR..Point.Railyards.Exposure.Conc) %>%
mutate(County = as.factor(County))

```

Exposure Data

```

# Making rural/urban data into a factor
urban_rural$URBAN_RURAL <- as.factor(urban_rural$URBAN_RURAL)
# Filtering census data
census_select <- census %>%
  select(County,
    `Median Family Income (Dollars)` :
    `Population Density (Persons Per SqMile)`)
# Making income a numeric class
census_select$`Median Family Income (Dollars)` <- as.numeric(gsub("[^0-9]", "",
  census_select$`Median Family Income (Dollars)`)
census_select$`Population Density (Persons Per SqMile)` <- as.numeric(gsub("[^0-9.]", "", census_select$`Population Density (Persons Per SqMile)`)

```

Urban/Rural & Census Data

```
formaldehyde <- exposure_df %>%
  filter(Pollutant.Name == "FORMALDEHYDE") %>%
  group_by(County) %>%
  summarise(across(where(is.numeric), sum, na.rm = TRUE)) %>%
  ungroup()
```

Formaldehyde Data

```
## Warning: There was 1 warning in 'summarise()'.
## i In argument: 'across(where(is.numeric), sum, na.rm = TRUE)'.
## i In group 1: 'County = Alamance'.
## Caused by warning:
## ! The '...' argument of 'across()' is deprecated as of dplyr 1.1.0.
## Supply arguments directly to '.fns' through an anonymous function instead.
##
## # Previously
##   across(a:b, mean, na.rm = TRUE)
##
## # Now
##   across(a:b, \(x) mean(x, na.rm = TRUE))
```

```
# subtracting entire state
formaldehyde <- formaldehyde[-34,]
# adding columns that sum all of the nonroad and onroad exposures for each county
# formaldehyde dataframe still has all of the exposure activities so it can be
# used for other analysis if we want
formaldehyde <- formaldehyde %>%
  mutate(Onroad_total = select(., OR.Light.Off.Gas:OR.Refueling) %>%
    rowSums(na.rm = TRUE)) %>%
  mutate(Nonroad_total = select(., NR.Recreational.Vehicles:NR.Railyards) %>%
    rowSums(na.rm = TRUE))

# joining urban/rural column
formaldehyde_select <- left_join(formaldehyde, urban_rural, by = "County")
# selecting necessary columns
formaldehyde_select <- formaldehyde_select %>%
  select(County, Population, Total.Exposure.Conc, Onroad_total:URBAN_RURAL)
# joining with census data
formaldehyde_complete <- left_join(formaldehyde_select,
  census_select, by = "County")

# making county into factor
formaldehyde_complete$County <- as.factor(formaldehyde_complete$County)
```

```
benzene <- exposure_df %>%
  filter(Pollutant.Name == "BENZENE") %>%
  group_by(County) %>%
  summarise(across(where(is.numeric), sum, na.rm = TRUE)) %>%
  ungroup()
```

```

# subtracting entire state
benzene <- benzene[-34,]
# adding columns that sum all of the nonroad and onroad exposures for each county
benzene <- benzene %>%
  mutate(Onroad_total = select(., OR.Light.Off.Gas:OR.Refueling) %>%
    rowSums(na.rm = TRUE)) %>%
  mutate(Nonroad_total = select(., NR.Recreational.Vehicles:NR.Railyards) %>%
    rowSums(na.rm = TRUE))

# joining urban/rural column
benzene_select <- left_join(benzene, urban_rural, by = "County")
# selecting necessary columns
benzene_select <- benzene_select %>%
  select(County, Population, Total.Exposure.Conc, Onroad_total:URBAN_RURAL)
# joining with census data
benzene_complete <- left_join(benzene_select, census_select, by = "County")
# making county into factor
benzene_complete$County <- as.factor(benzene_complete$County)

```

Benzene Data

```

## Diesel
diesel <- exposure_df %>%
  filter(Pollutant.Name == "DIESEL PM") %>%
  group_by(County) %>%
  summarise(across(where(is.numeric), sum, na.rm = TRUE)) %>%
  ungroup()
# subtracting entire state
diesel <- diesel[-34,]
# adding columns that sum all of the nonroad and onroad exposures for each county
diesel <- diesel %>%
  mutate(
    Onroad_total =
      select(., OR.Light.Off.Gas:OR.Refueling) %>%
      rowSums(na.rm = TRUE)) %>%
  mutate(Nonroad_total = select(., NR.Recreational.Vehicles:NR.Railyards) %>%
    rowSums(na.rm = TRUE))

# joining urban/rural column
diesel_select <- left_join(diesel, urban_rural, by = "County")
# selecting necessary columns
diesel_select <- diesel_select %>%
  select(County, Population, Total.Exposure.Conc, Onroad_total:URBAN_RURAL)

# joining with census data
diesel_complete <- left_join(diesel_select, census_select, by = "County")
diesel_complete$County <- as.factor(diesel_complete$County)

## Can still join the cancer data in if we want to

```

Diesel Data

Analysis

```
# descriptive statistics for results section:

mean_form <- mean(formaldehyde_complete$Total.Exposure.Conc)
sd_form <- sd(formaldehyde_complete$Total.Exposure.Conc)
mean_b <- mean(benzene_complete$Total.Exposure.Conc)
sd_b <- sd(benzene_complete$Total.Exposure.Conc)
mean_D <- mean(diesel_complete$Total.Exposure.Conc)
sd_d <- sd(diesel_complete$Total.Exposure.Conc)

mean_population <- mean(formaldehyde_complete$Population)
sd_pop <- sd(formaldehyde_complete$Population)
mean_population_density <-
  mean(formaldehyde_complete$`Population Density (Persons Per SqMile)`)
sd_density <-
  sd(formaldehyde_complete$`Population Density (Persons Per SqMile)`)
mean_income <- mean(formaldehyde_complete$`Median Family Income (Dollars)`)
sd_income <- sd(formaldehyde_complete$`Median Family Income (Dollars)`)
mean_unemploy <- mean(formaldehyde_complete$`Unemployment(Percent)`)
sd_unemploy <- sd(formaldehyde_complete$`Unemployment(Percent)`)
```

Summary Statistics

```
formaldehyde_onroad <- formaldehyde_complete %>%
  ggplot(aes(x = Onroad_total)) +
  geom_histogram(bins = 60, fill = "skyblue2", color = "slateblue4") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold",
    color = "slateblue4"),
    axis.title.x = element_text(face = "bold", color = "slateblue4"),
    axis.title.y = element_text(face = "bold", color = "slateblue4"),
    panel.background = element_rect(fill = "aliceblue"),
    panel.grid.major = element_line(color = "lavenderblush3"),
    panel.grid.minor = element_line(color = "lavenderblush2")) +
  labs(title = "Onroad Exposure",
    x = "Exposure Concentration (ug/m3)",
    y = "Count")

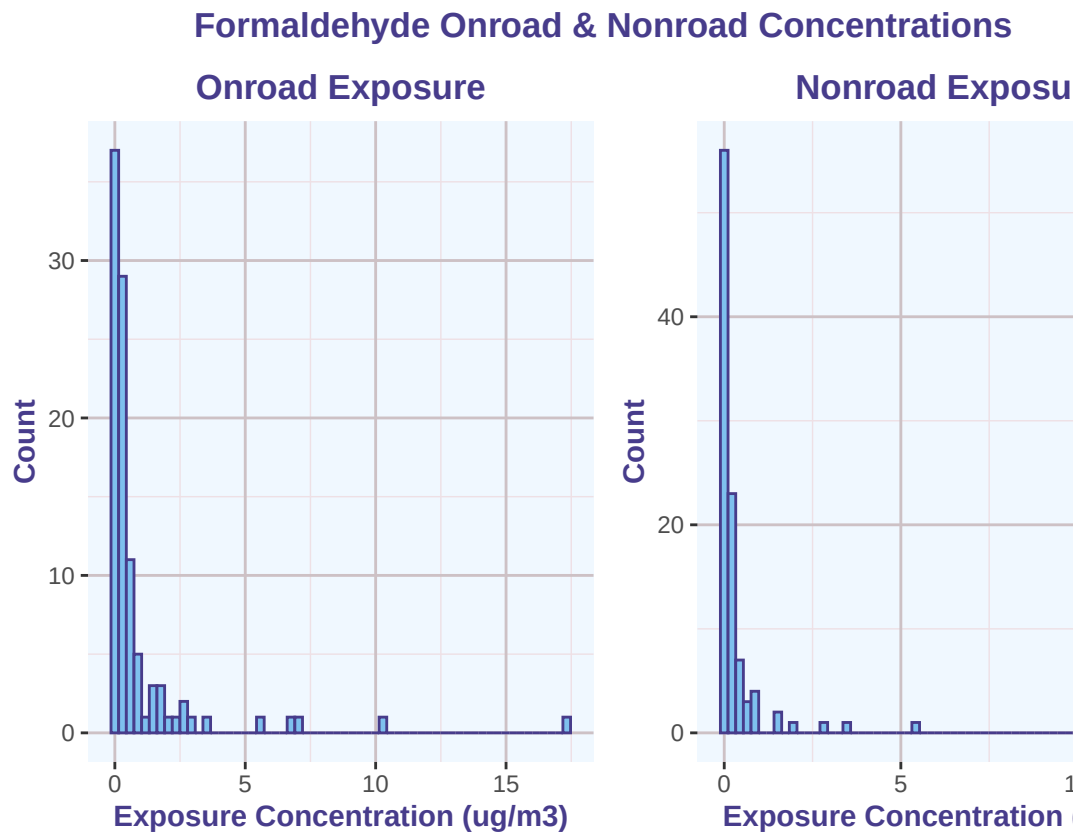
formaldehyde_nonroad <- formaldehyde_complete %>%
  ggplot(aes(x = Nonroad_total)) +
  geom_histogram(bins = 60, fill = "skyblue2", color = "slateblue4") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold",
    color = "slateblue4"),
    axis.title.x = element_text(face = "bold", color = "slateblue4"),
    axis.title.y = element_text(face = "bold", color = "slateblue4"),
    panel.background = element_rect(fill = "aliceblue"),
    panel.grid.major = element_line(color = "lavenderblush3"),
    panel.grid.minor = element_line(color = "lavenderblush2")) +
```

```

labs(title = "Nonroad Exposure",
      x = "Exposure Concentration (ug/m3)",
      y = "Count")

form_plot <- (formaldehyde_onroad + formaldehyde_nonroad) +
  plot_annotation(title = "Formaldehyde Onroad & Nonroad Concentrations ",
                  theme = theme(
                    plot.title = element_text(hjust = 0.5, face = "bold",
                                              color = "slateblue4")))
plot(form_plot)

```



Onroad & Nonroad Data

```

benzene_onroad <- benzene_complete %>%
  ggplot(aes(x = Onroad_total)) +
  geom_histogram(bins = 60, fill = "palegreen2", color = "darkgreen") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold",
                                  color = "darkgreen"),
        axis.title.x = element_text(face = "bold", color = "darkgreen"),
        axis.title.y = element_text(face = "bold", color = "darkgreen"),
        panel.background = element_rect(fill = "azure"),
        panel.grid.major = element_line(color = "azure3"),
        panel.grid.minor = element_line(color = "azure2")) +
  labs(title = "Onroad Exposure",
       x = "Exposure Concentration (ug/m3)",
       y = "Count")

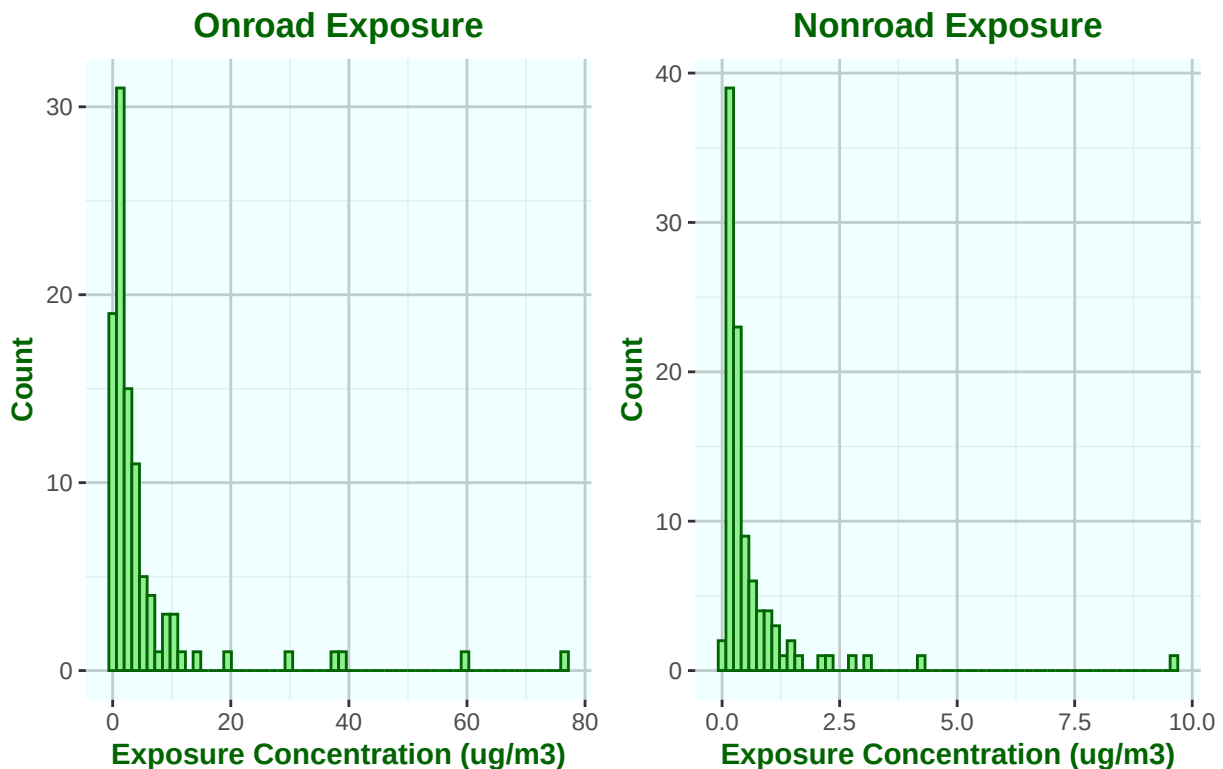
```

```
benzene_nonroad <- benzene_complete %>%
  ggplot(aes(x = Nonroad_total)) +
  geom_histogram(bins = 60, fill = "palegreen2", color = "darkgreen") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold",
    color = "darkgreen"),
    axis.title.x = element_text(face = "bold", color = "darkgreen"),
    axis.title.y = element_text(face = "bold", color = "darkgreen"),
    panel.background = element_rect(fill = "azure"),
    panel.grid.major = element_line(color = "azure3"),
    panel.grid.minor = element_line(color = "azure2")) +
  labs(title = "Nonroad Exposure",
    x = "Exposure Concentration (ug/m3)",
    y = "Count")

ben_plot <- (benzene_onroad + benzene_nonroad) +
  plot_annotation(title = "Benzene Onroad & Nonroad Concentrations ",
    theme = theme(
      plot.title = element_text(hjust = 0.5, face = "bold",
        color = "darkgreen")))

plot(ben_plot)
```

Benzene Onroad & Nonroad Concentrations



```
diesel_onroad <- diesel_complete %>%
  ggplot(aes(x = Onroad_total)) +
  geom_histogram(bins = 60, fill = "palevioletred", color = "violetred4") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold",
```



```

        color = "violetred4"),
axis.title.x = element_text(face = "bold", color = "violetred4"),
axis.title.y = element_text(face = "bold", color = "violetred4"),
panel.background = element_rect(fill = "seashell"),
panel.grid.major = element_line(color = "mistyrose3"),
panel.grid.minor = element_line(color = "mistyrose2")) +
labs(title = "Onroad Exposure",
x = "Exposure Concentration (ug/m3)",
y = "Count")

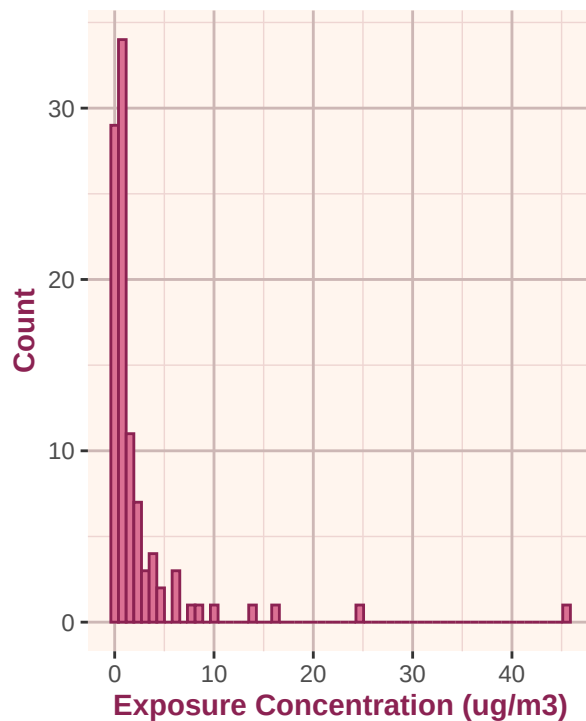
diesel_nonroad <- diesel_complete %>%
ggplot(aes(x = Nonroad_total)) +
geom_histogram(bins = 60, fill = "palevioletred", color = "violetred4") +
theme(plot.title = element_text(hjust = 0.5, face = "bold",
                                color = "violetred4"),
axis.title.x = element_text(face = "bold", color = "violetred4"),
axis.title.y = element_text(face = "bold", color = "violetred4"),
panel.background = element_rect(fill = "seashell"),
panel.grid.major = element_line(color = "mistyrose3"),
panel.grid.minor = element_line(color = "mistyrose2")) +
labs(title = "Nonroad Exposure",
x = "Exposure Concentration (ug/m3)",
y = "Count")

diesel_plot <- (diesel_onroad + diesel_nonroad) +
plot_annotation(title = "Diesel PM Onroad & Nonroad Concentrations ",
theme = theme(
plot.title = element_text(hjust = 0.5, face = "bold",
color = "violetred4")))
plot(diesel_plot)

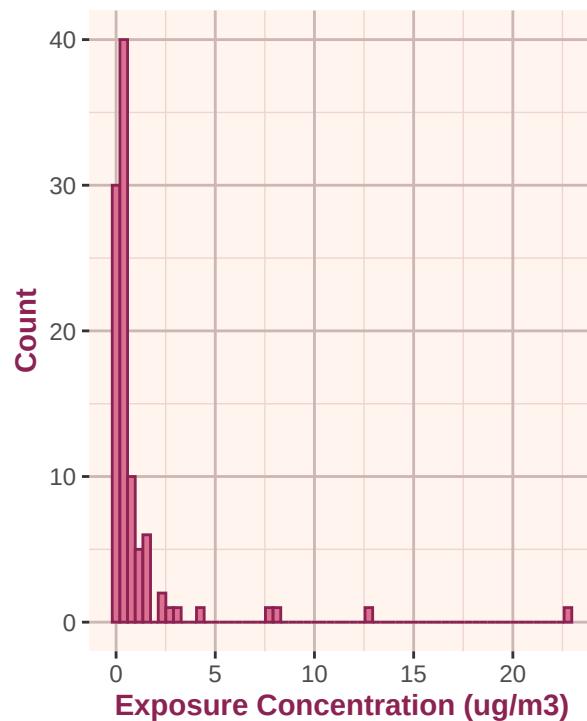
```

Diesel PM Onroad & Nonroad Concentrations

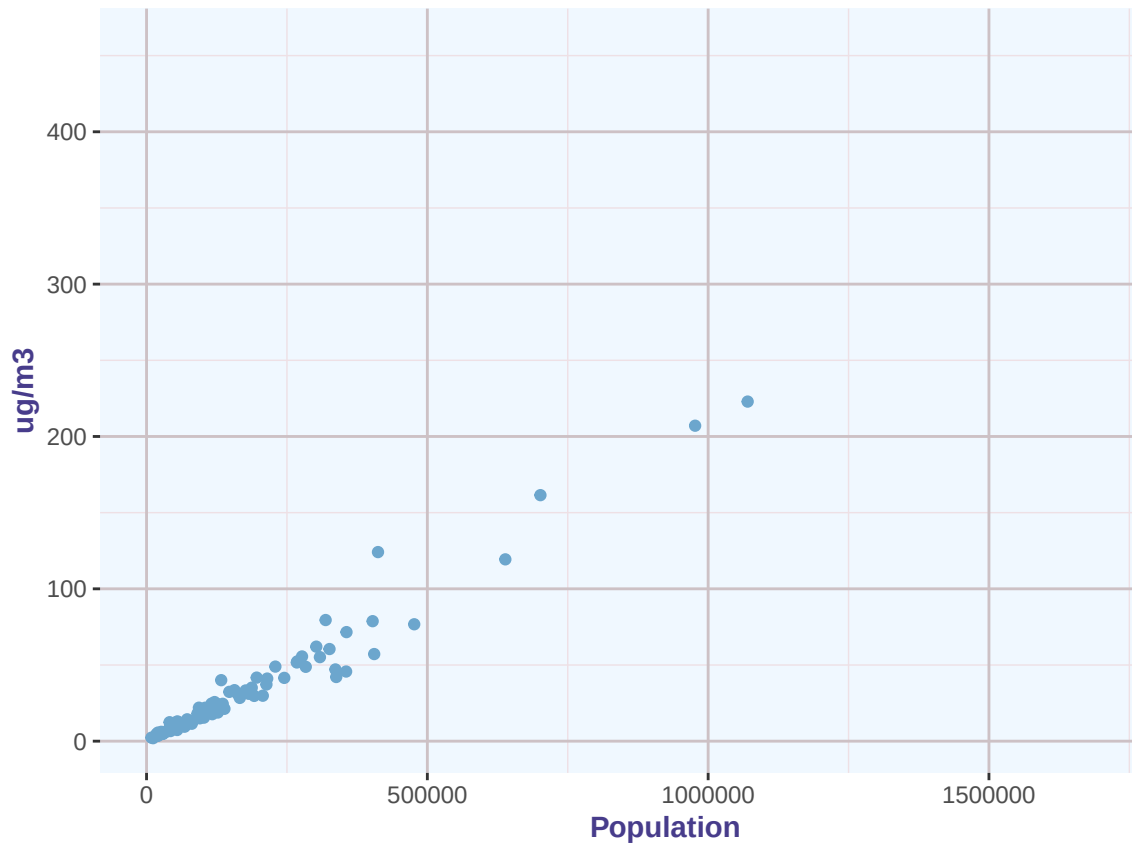
Onroad Exposure



Nonroad Exposure

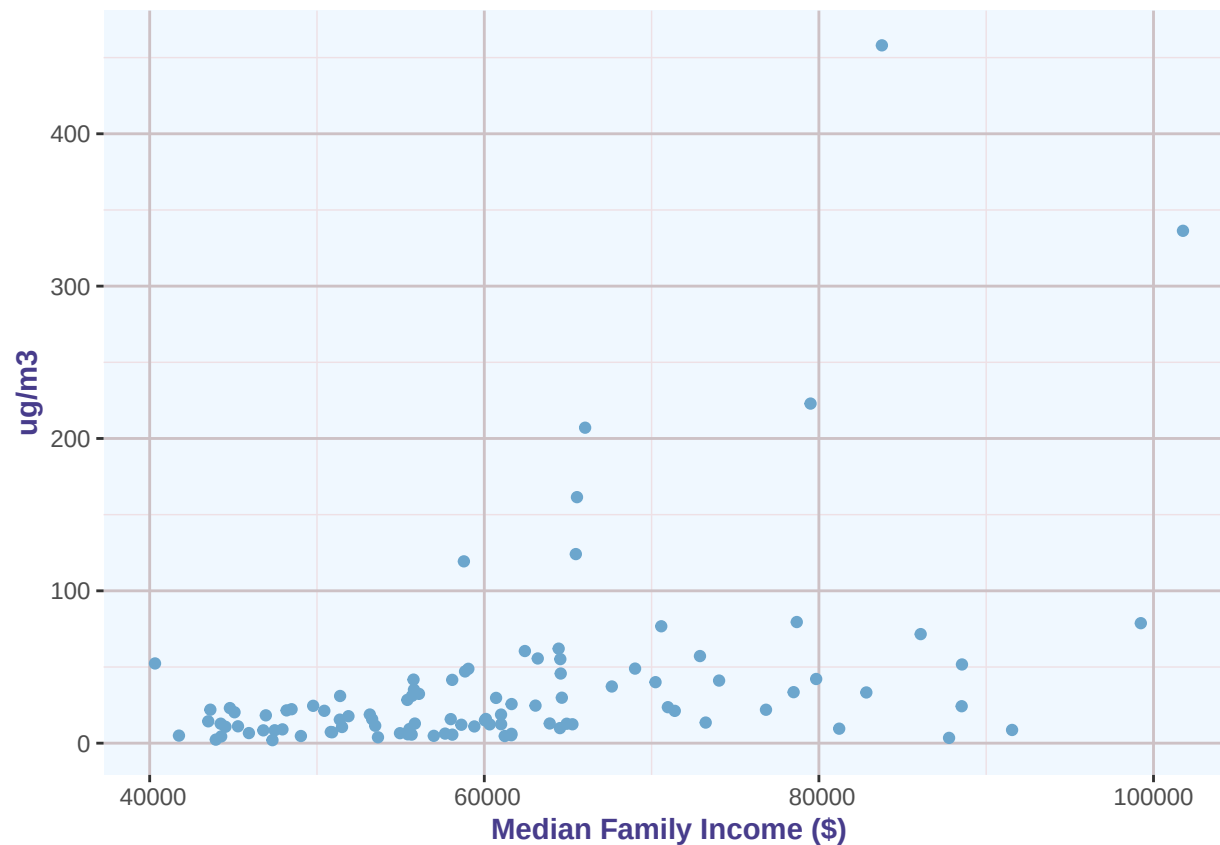


```
# Visualizing the relationship between response and descriptor variables
(scatter_population <- ggplot(formaldehyde_complete,
                             aes(x = Population, y = Total.Exposure.Conc)) +
  geom_point(color = "skyblue3") +
  theme(axis.title.x = element_text(face = "bold", color = "slateblue4"),
        axis.title.y = element_text(face = "bold", color = "slateblue4"),
        panel.background = element_rect(fill = "aliceblue"),
        panel.grid.major = element_line(color = "lavenderblush3"),
        panel.grid.minor = element_line(color = "lavenderblush2")) +
  labs(x = "Population",
       y = "ug/m3"))
```

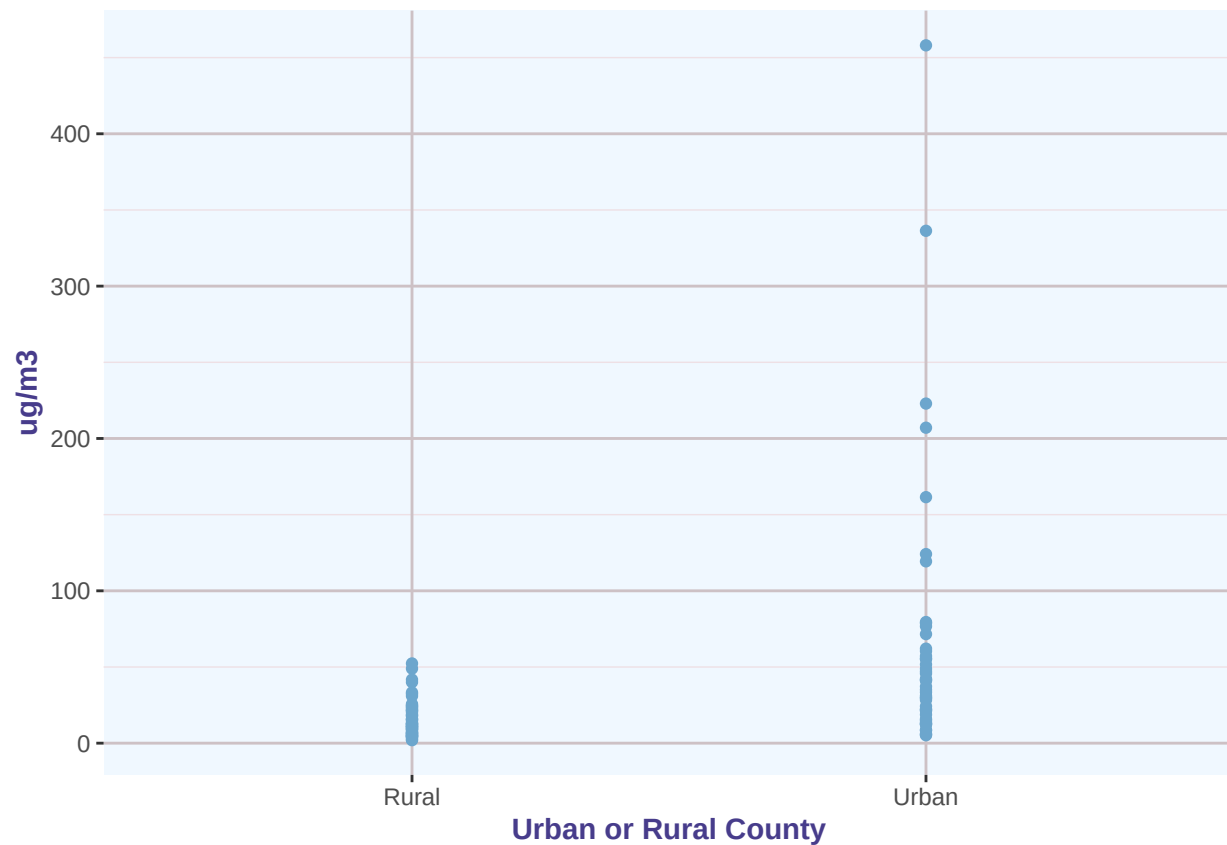


Formaldehyde Model

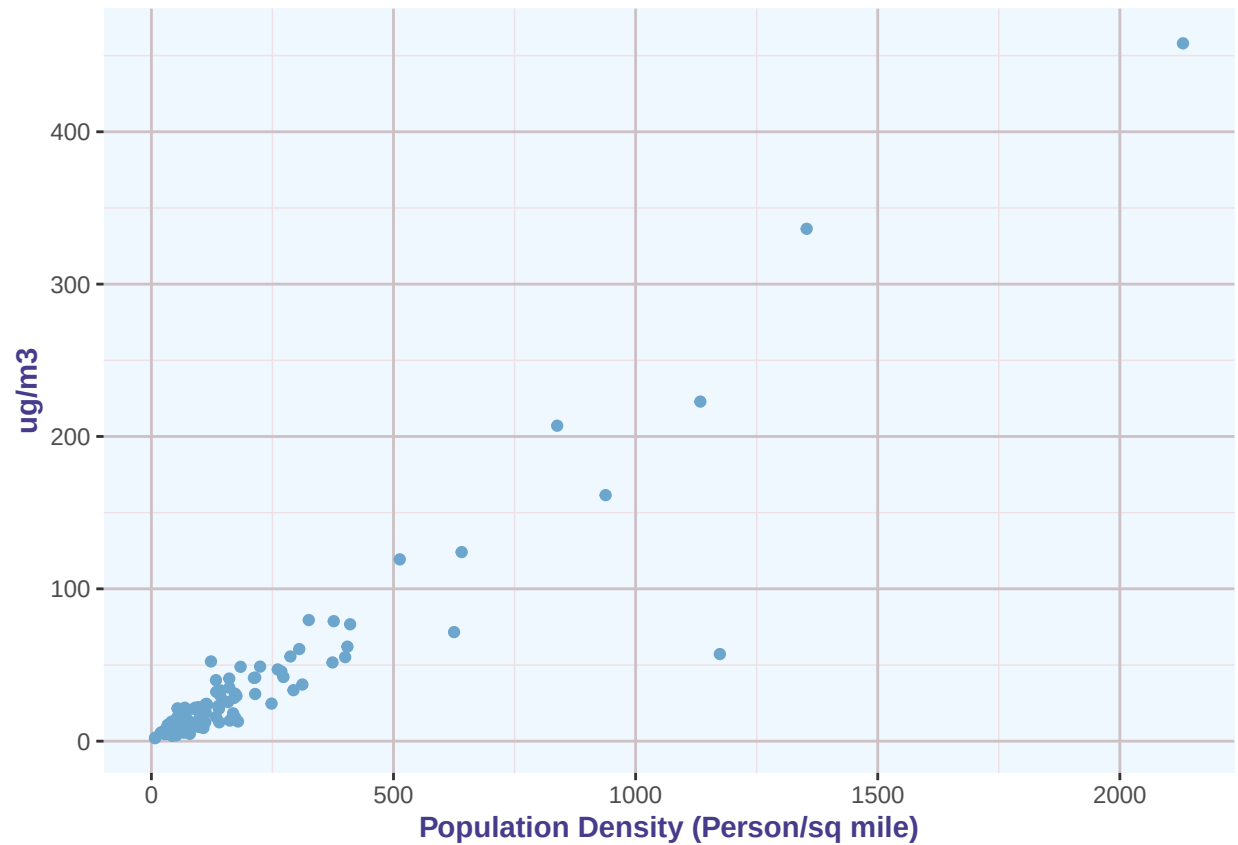
```
(scatter_income <- ggplot(formaldehyde_complete,
  aes(x = `Median Family Income (Dollars)`,
    y = Total.Exposure.Conc)) +
  geom_point(color = "skyblue3") +
  scale_x_continuous(labels = function(x) format(x, scientific = FALSE)) +
  theme(axis.title.x = element_text(face = "bold", color = "slateblue4"),
    axis.title.y = element_text(face = "bold", color = "slateblue4"),
    panel.background = element_rect(fill = "aliceblue"),
    panel.grid.major = element_line(color = "lavenderblush3"),
    panel.grid.minor = element_line(color = "lavenderblush2")) +
  labs(x = "Median Family Income ($)",
    y = "ug/m3"))
```



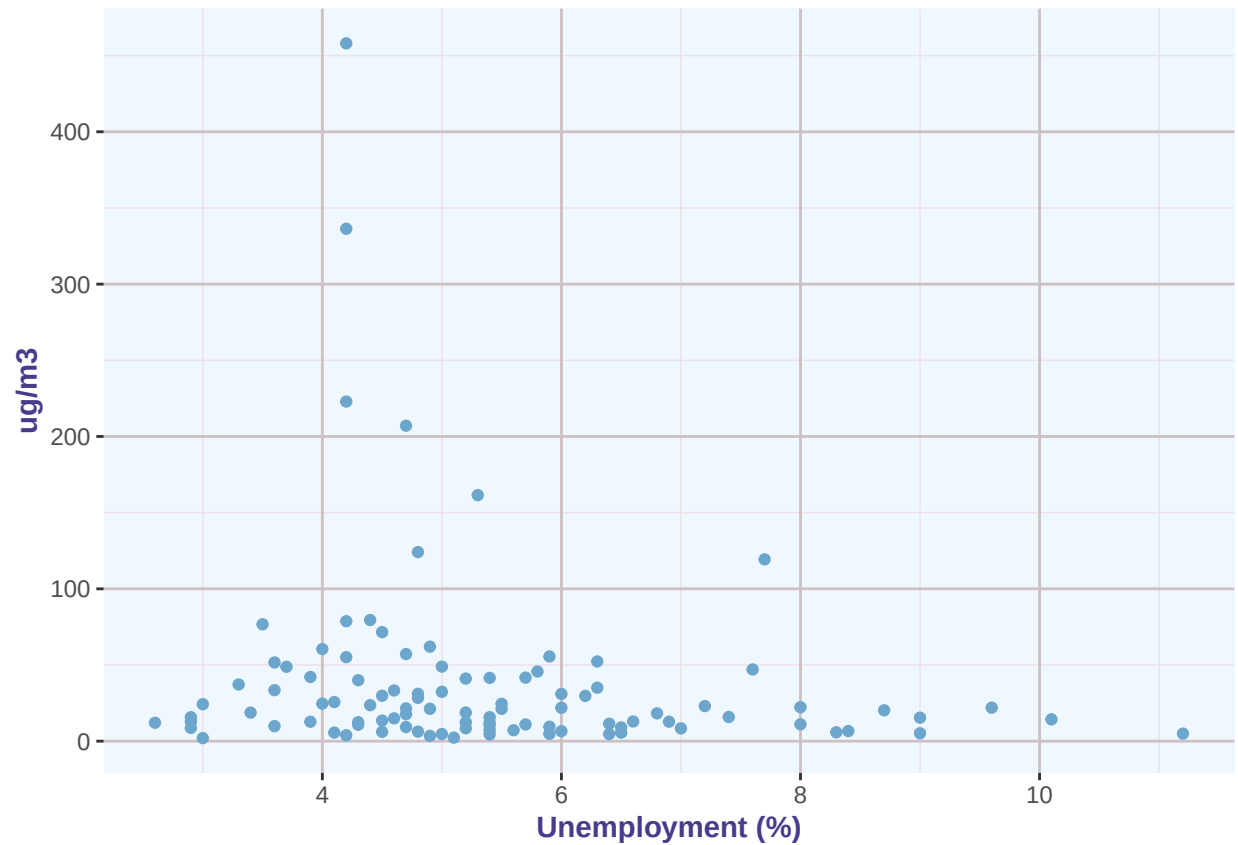
```
(scatter_urban <- ggplot(formaldehyde_complete,
  aes(x = URBAN_RURAL, y = Total.Exposure.Conc)) +
  geom_point(color = "skyblue3") +
  theme(axis.title.x = element_text(face = "bold", color = "slateblue4"),
    axis.title.y = element_text(face = "bold", color = "slateblue4"),
    panel.background = element_rect(fill = "aliceblue"),
    panel.grid.major = element_line(color = "lavenderblush3"),
    panel.grid.minor = element_line(color = "lavenderblush2")) +
  labs(x = "Urban or Rural County",
    y = "ug/m3"))
```



```
(scatter_density <- ggplot(formaldehyde_complete,
  aes(x = `Population Density (Persons Per SqMile)`,
    y = Total.Exposure.Conc)) +
  geom_point(color = "skyblue3") +
  theme(axis.title.x = element_text(face = "bold", color = "slateblue4"),
    axis.title.y = element_text(face = "bold", color = "slateblue4"),
    panel.background = element_rect(fill = "aliceblue"),
    panel.grid.major = element_line(color = "lavenderblush3"),
    panel.grid.minor = element_line(color = "lavenderblush2")) +
  labs(x = "Population Density (Person/sq mile)",
    y = "ug/m3"))
```

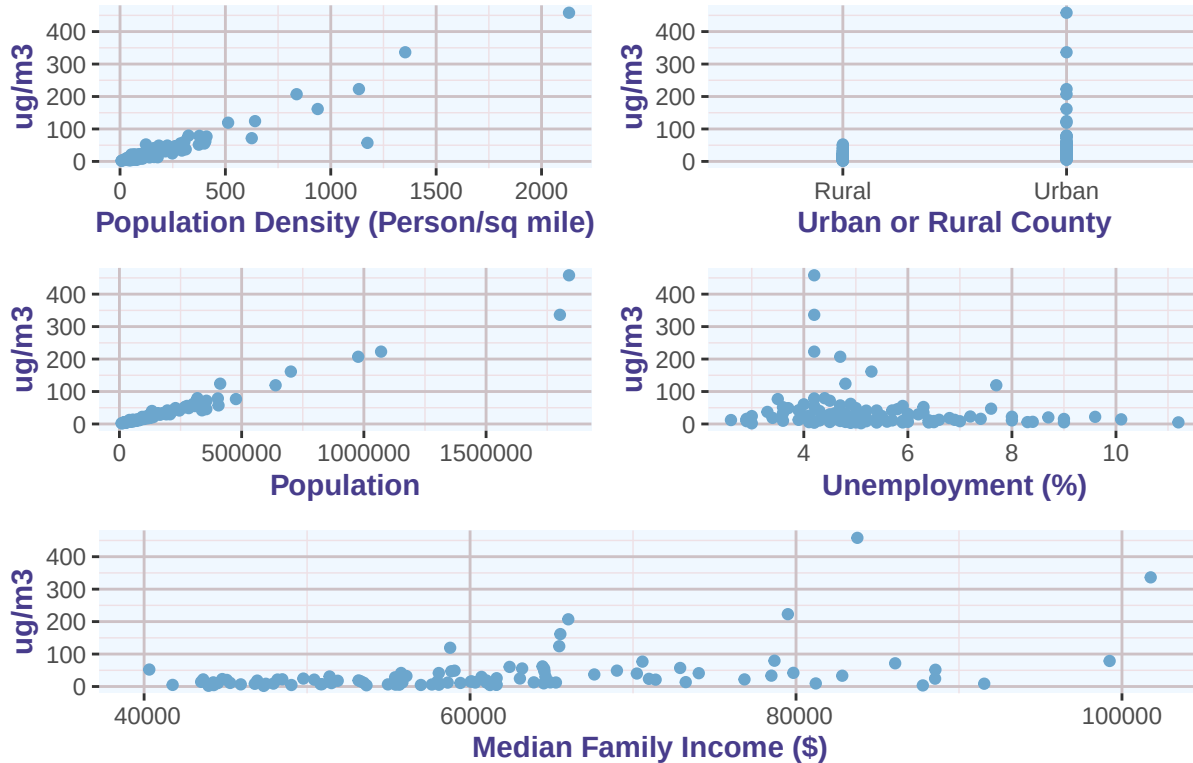


```
(scatter_unemployment <- ggplot(formaldehyde_complete,
                                aes(x = `Unemployment(Percent)`,
                                    y = Total.Exposure.Conc)) +
  geom_point(color = "skyblue3") +
  theme(axis.title.x = element_text(face = "bold", color = "slateblue4"),
        axis.title.y = element_text(face = "bold", color = "slateblue4"),
        panel.background = element_rect(fill = "aliceblue"),
        panel.grid.major = element_line(color = "lavenderblush3"),
        panel.grid.minor = element_line(color = "lavenderblush2")) +
  labs(x = "Unemployment (%)",
       y = "ug/m3"))
```



```
final_plot <- ((scatter_density + scatter_urban) /
  (scatter_population + scatter_unemployment) / scatter_income) +
  plot_annotation(title = "Formaldehyde Concentrations vs. County Demographics",
    theme = theme(
      plot.title = element_text(hjust = 0.5, face = "bold",
        color = "slateblue4")))
plot(final_plot)
```

Formaldehyde Concentrations vs. County Demographics



```
# Perform correlation tests to test for collinearity
```

```
cor.test(formaldehyde_complete$Population,
         formaldehyde_complete$`Median Family Income (Dollars)`)
```

```
##
## Pearson's product-moment correlation
##
## data: formaldehyde_complete$Population and formaldehyde_complete$`Median Family Income (Dollars)`
## t = 5.4197, df = 98, p-value = 4.281e-07
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.3133539 0.6183125
## sample estimates:
##      cor
## 0.4802155
```

```
# p < 0.01, cor = 0.4802
```

```
cor.test(formaldehyde_complete$Population,
         formaldehyde_complete$`Population Density (Persons Per SqMile)`)
```

```
##
## Pearson's product-moment correlation
##
```



```
## data: formaldehyde_complete$Population and formaldehyde_complete$'Population Density (Persons Per S
## t = 25.141, df = 98, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.8982012 0.9527564
## sample estimates:
## cor
## 0.9304641
```

```
# p < 0.01, cor = 0.9144
```

```
cor.test(formaldehyde_complete$`Unemployment(Percent)` ,
          formaldehyde_complete$`Median Family Income (Dollars)`)
```

```
##
## Pearson's product-moment correlation
##
## data: formaldehyde_complete$`Unemployment(Percent)` and formaldehyde_complete$`Median Family Income
## t = -5.5819, df = 98, p-value = 2.12e-07
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.6270834 -0.3262148
## sample estimates:
## cor
## -0.491162
```

```
# p < 0.01, cor = -0.4911
```

```
cor.test(formaldehyde_complete$`Median Family Income (Dollars)` ,
          formaldehyde_complete$`Population Density (Persons Per SqMile)`)
```

```
##
## Pearson's product-moment correlation
##
## data: formaldehyde_complete$`Median Family Income (Dollars)` and formaldehyde_complete$`Population I
## t = 5.6216, df = 98, p-value = 1.783e-07
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.3293295 0.6291961
## sample estimates:
## cor
## 0.4938052
```

```
# p < 0.01, cor = 0.402
```

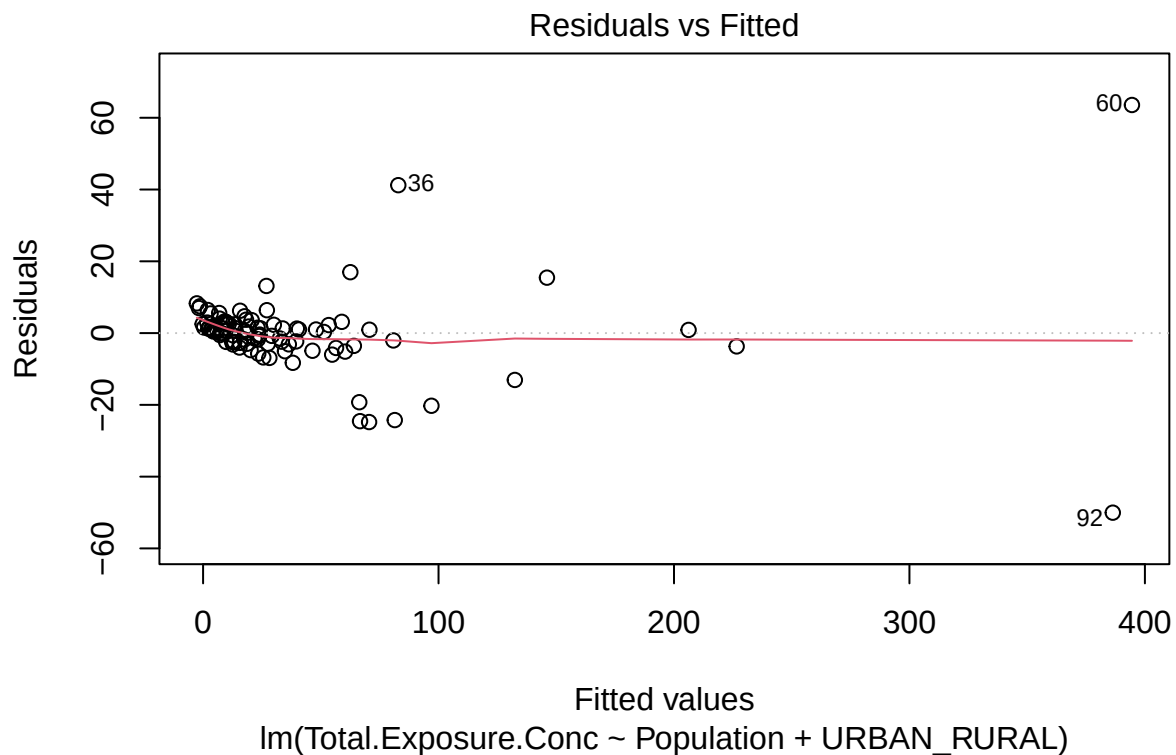
```
# Generate regression model(s)
```

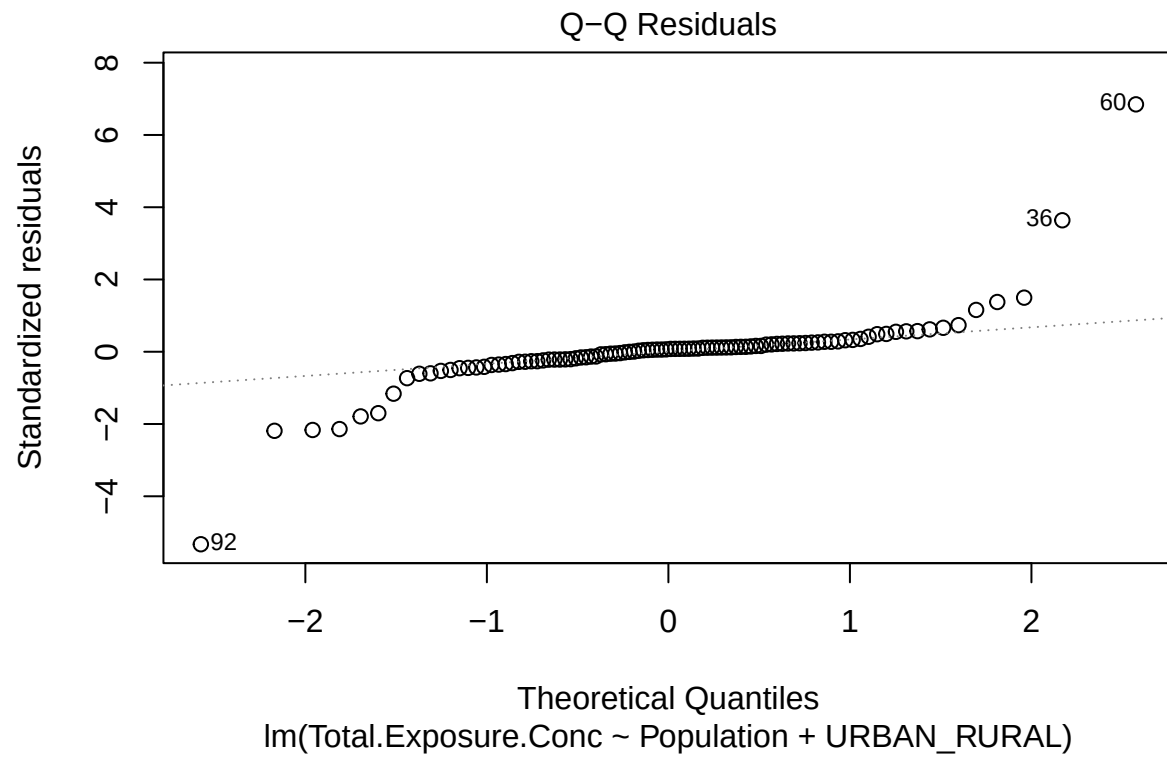
```
F_model1 <- lm(Total.Exposure.Conc ~ Population + URBAN_RURAL,
               formaldehyde_complete)
summary(F_model1)
```

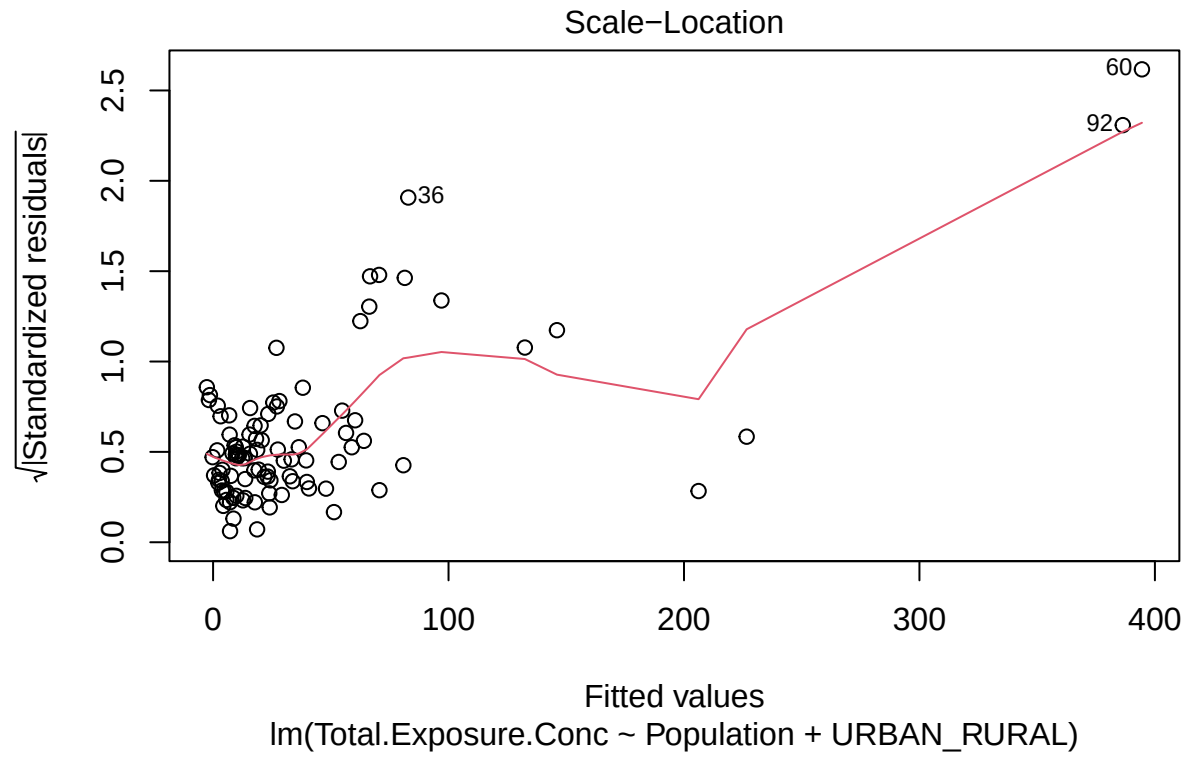
```
##
## Call:
```

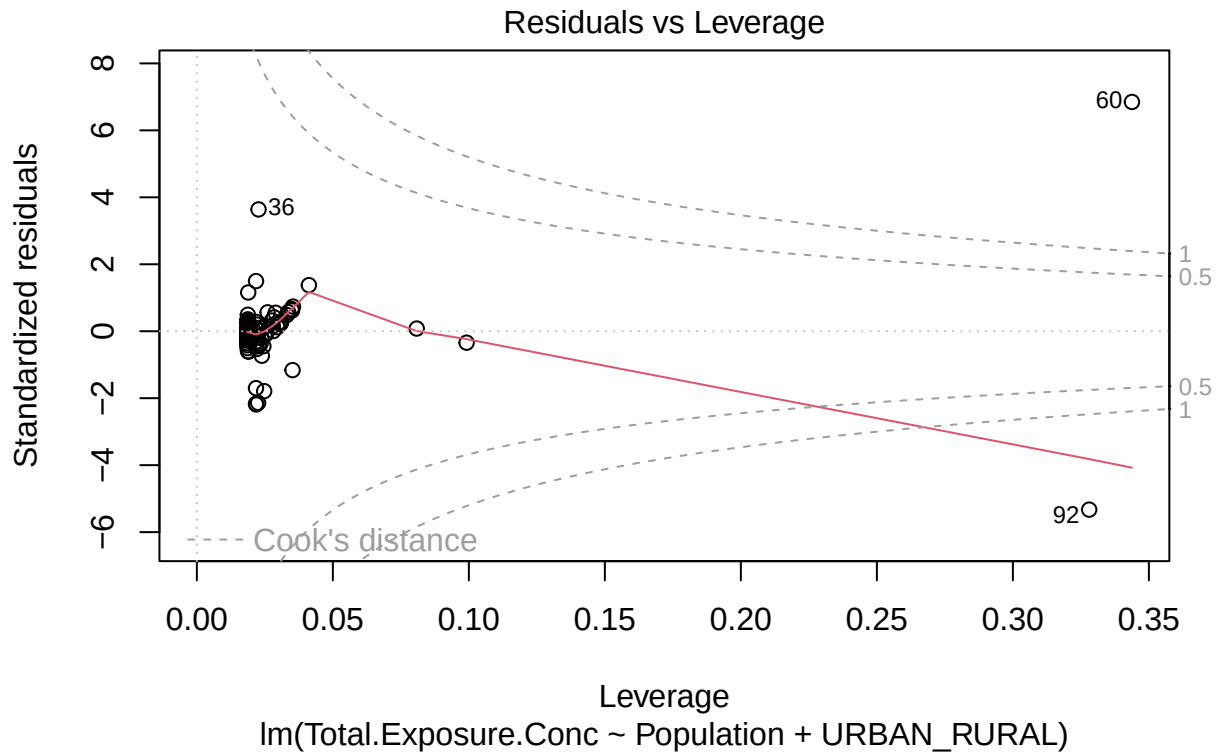
```
## lm(formula = Total.Exposure.Conc ~ Population + URBAN_RURAL,
##     data = formaldehyde_complete)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -50.038  -2.538   0.815   2.597  63.555
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -2.116e+00  1.598e+00  -1.324   0.1887
## Population      2.183e-04  4.309e-06  50.666 <2e-16 ***
## URBAN_RURALUrban -4.967e+00  2.537e+00  -1.958   0.0531 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.46 on 97 degrees of freedom
## Multiple R-squared:  0.969, Adjusted R-squared:  0.9683
## F-statistic: 1514 on 2 and 97 DF, p-value: < 2.2e-16
```

```
plot(F_model1)
```







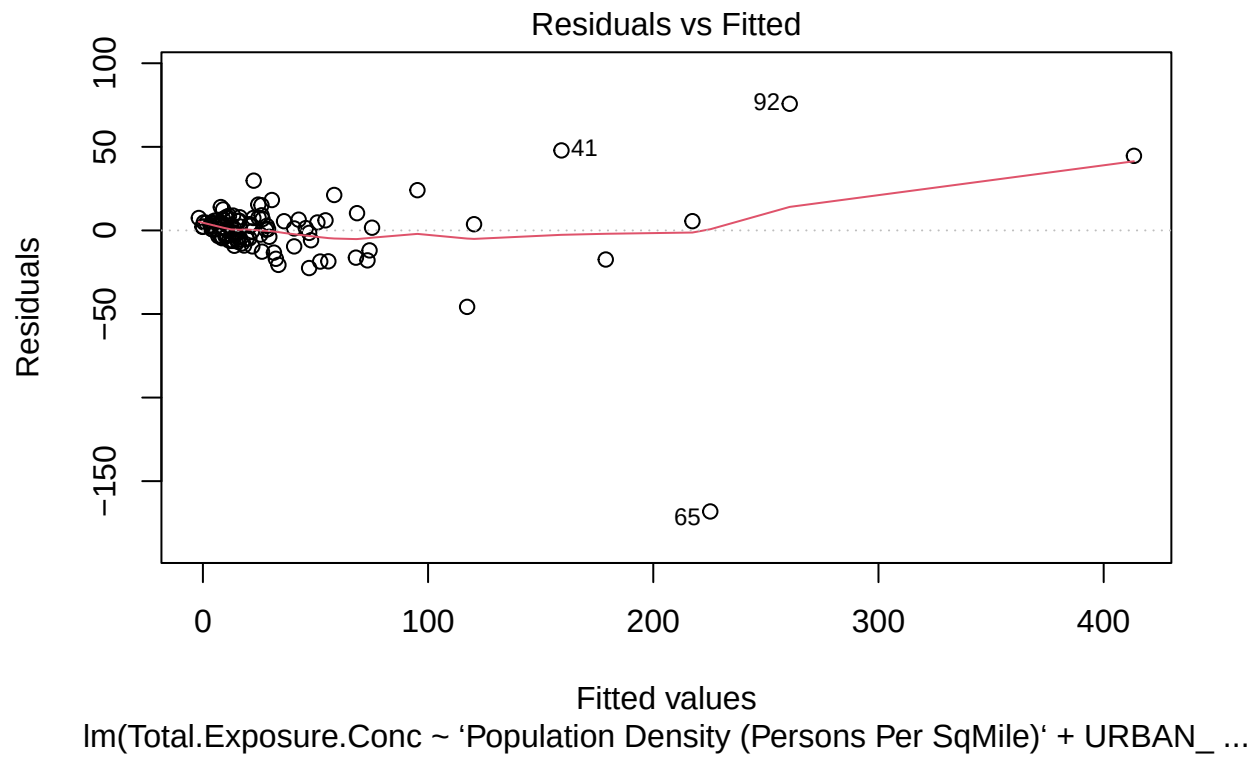


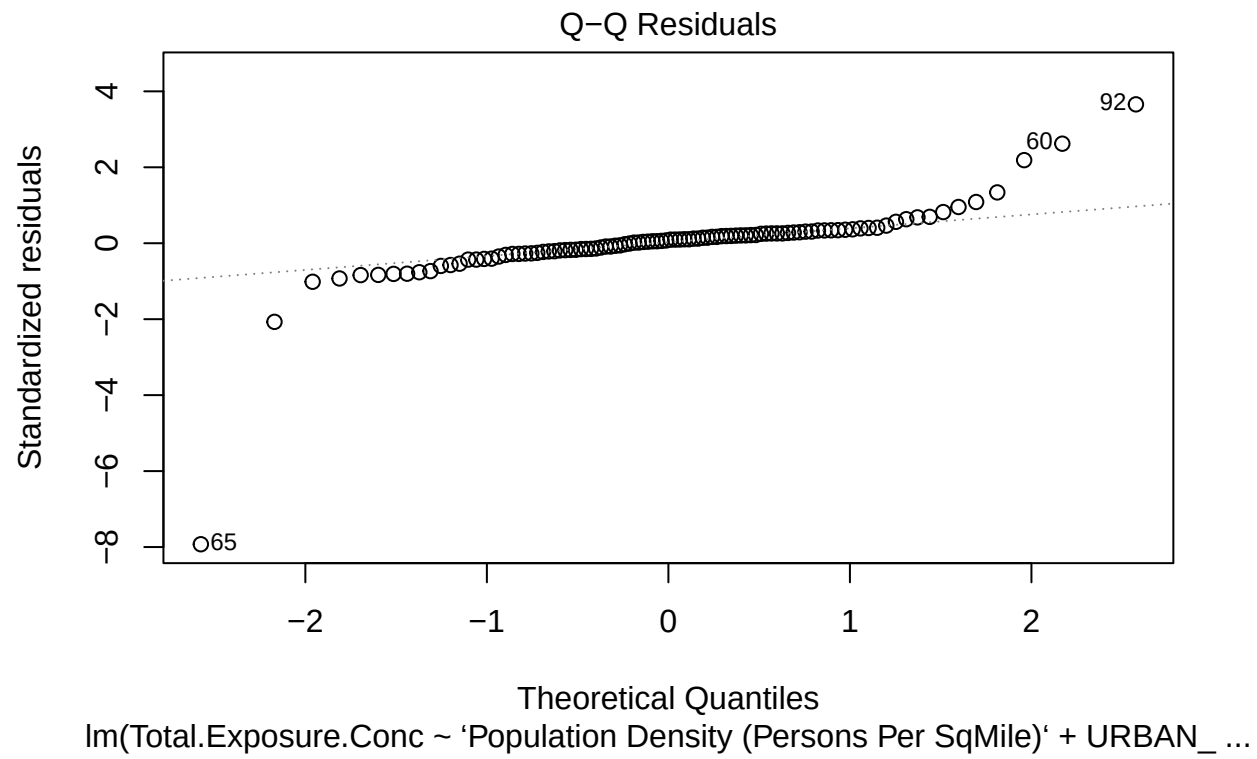
```
F_model2 <- lm(Total.Exposure.Conc ~ `Population Density (Persons Per SqMile)` +
               URBAN_RURAL, formaldehyde_complete)
summary(F_model2)
```

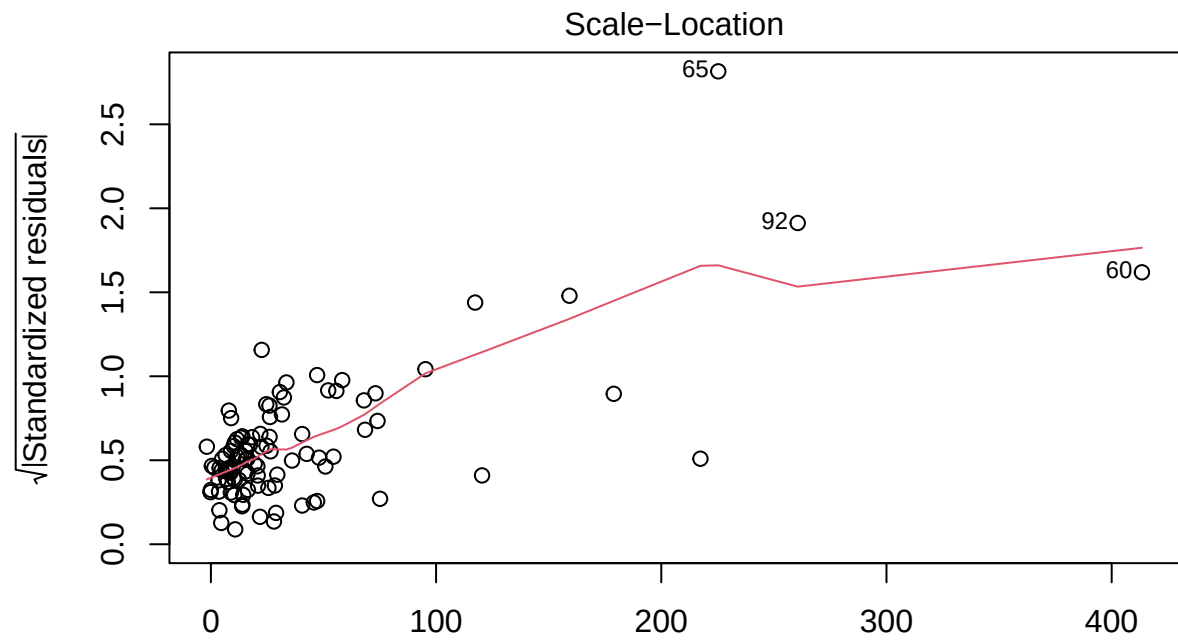
```
##
## Call:
## lm(formula = Total.Exposure.Conc ~ `Population Density (Persons Per SqMile)` +
##     URBAN_RURAL, data = formaldehyde_complete)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -168.131   -4.846    1.865    6.089   75.752
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.633169   3.135897  -0.521   0.604
## `Population Density (Persons Per SqMile)`  0.196709   0.008051  24.434 <2e-16
## URBAN_RURALUrban  -3.993179   5.005943  -0.798   0.427
##
## (Intercept)
## `Population Density (Persons Per SqMile)` ***
## URBAN_RURALUrban
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 22.45 on 97 degrees of freedom
## Multiple R-squared:  0.8809, Adjusted R-squared:  0.8784
## F-statistic: 358.6 on 2 and 97 DF,  p-value: < 2.2e-16
```

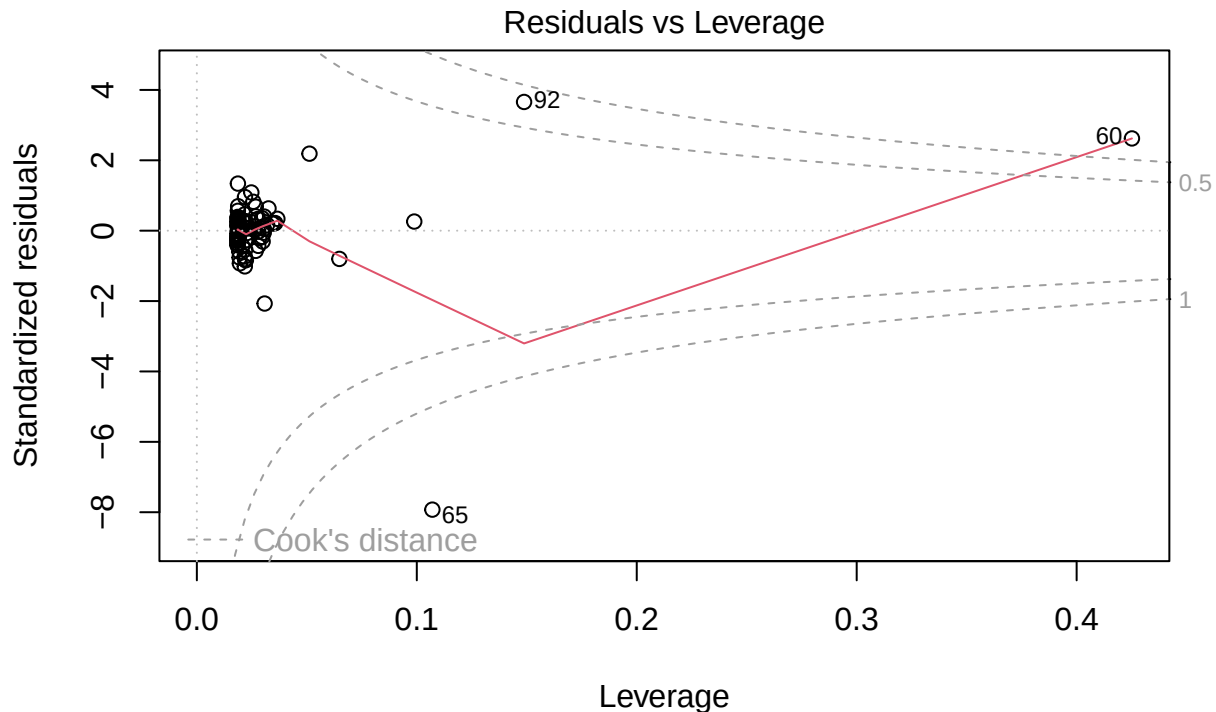
```
plot(F_model2)
```







Fitted values
 $\text{lm}(\text{Total.Exposure.Conc} \sim \text{'Population Density (Persons Per SqMile)'} + \text{URBAN_} \dots$

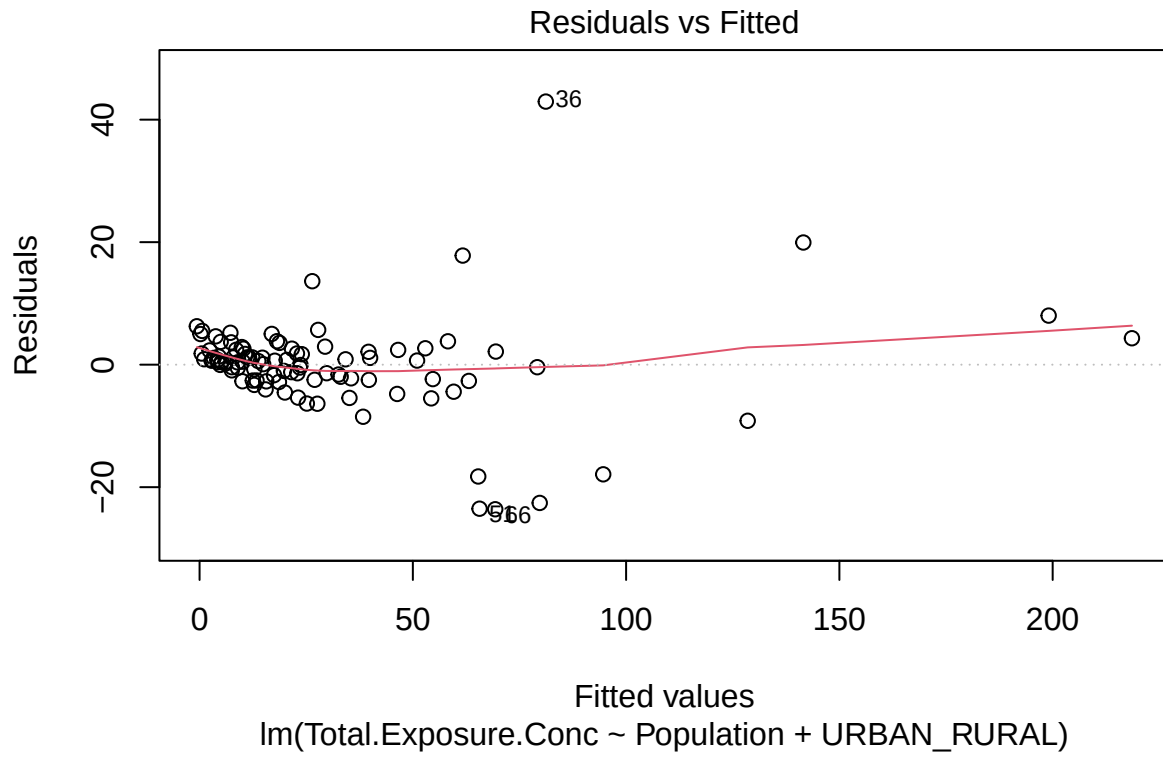


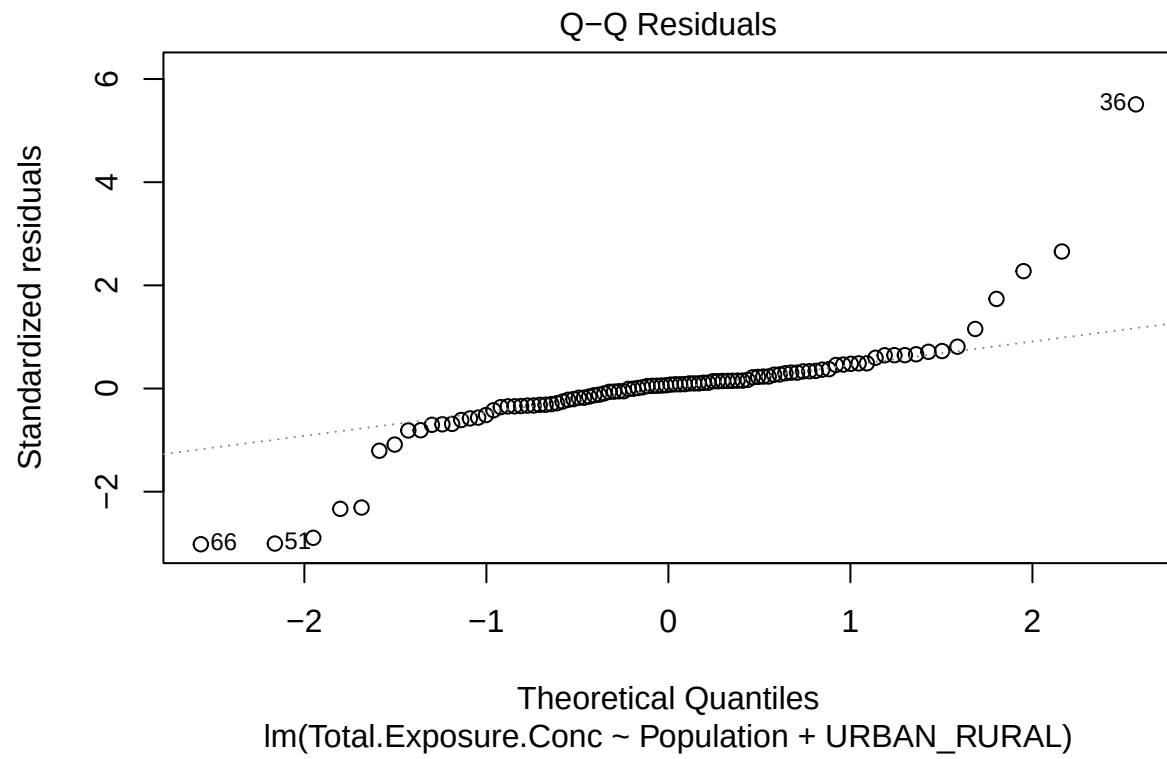
lm(Total.Exposure.Conc ~ 'Population Density (Persons Per SqMile)' + URBAN_ ...

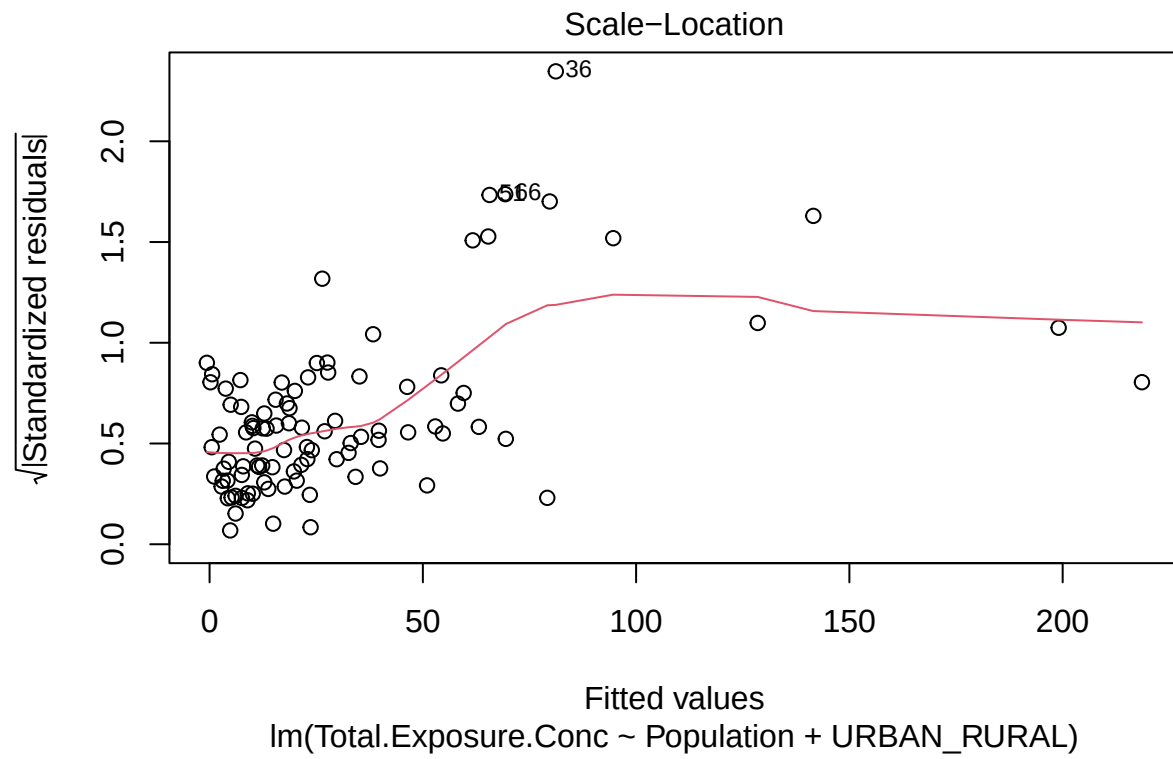
```
formaldehyde_nooutlier <- formaldehyde_complete[-92,]
formaldehyde_nooutlier <- formaldehyde_nooutlier[-60,]
modell1_nooutlier <- lm(Total.Exposure.Conc ~ Population + URBAN_RURAL,
                        formaldehyde_nooutlier)
summary(modell1_nooutlier)
```

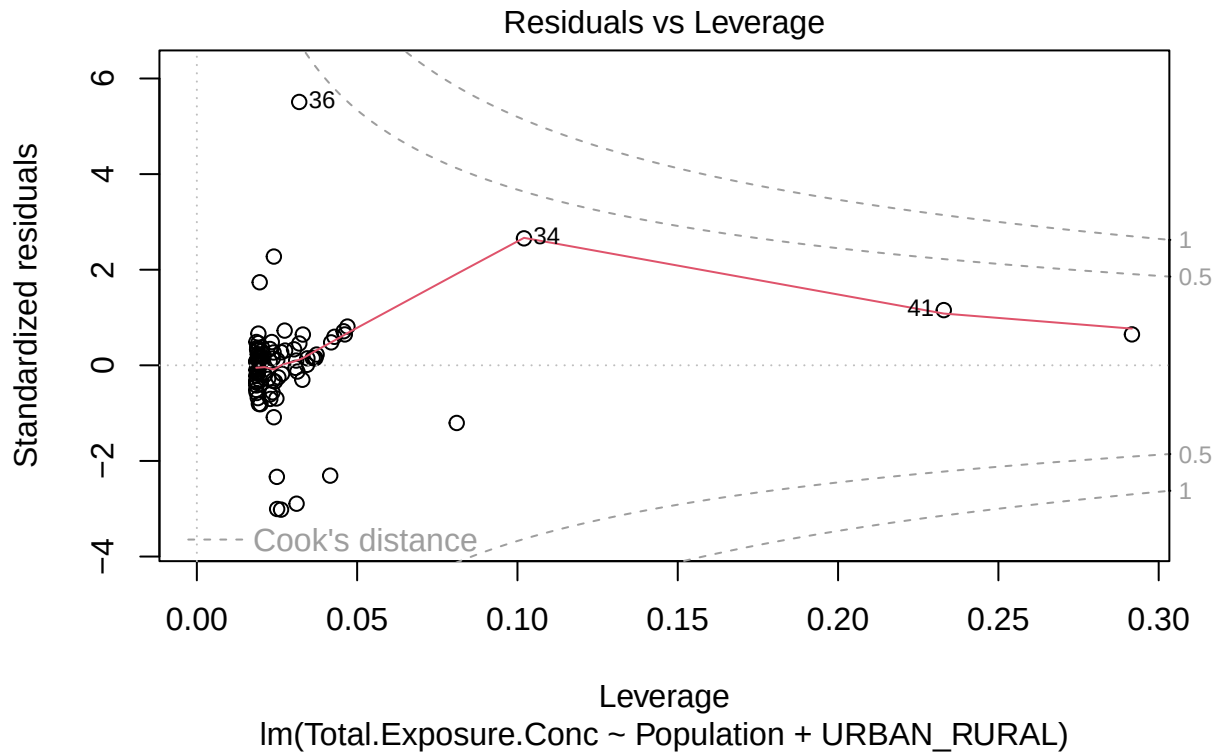
```
##
## Call:
## lm(formula = Total.Exposure.Conc ~ Population + URBAN_RURAL,
##     data = formaldehyde_nooutlier)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -23.602  -2.439   0.543   2.388  42.948
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.336e+00  1.155e+00  -1.157  0.2503
## Population      2.088e-04  5.088e-06  41.034 <2e-16 ***
## URBAN_RURALUrban -3.543e+00  1.855e+00  -1.910  0.0591 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.923 on 95 degrees of freedom
## Multiple R-squared:  0.9575, Adjusted R-squared:  0.9566
## F-statistic: 1069 on 2 and 95 DF, p-value: < 2.2e-16
```

```
plot(model1_nooutlier)
```







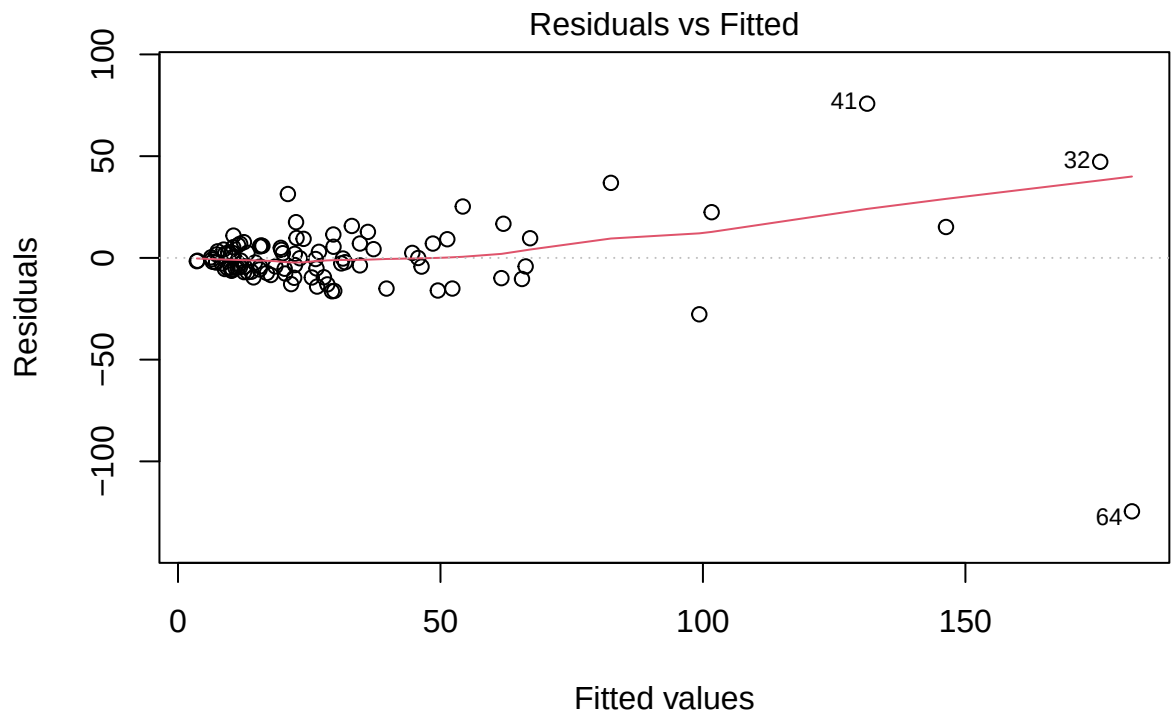


```
model2_nooutlier <- lm(Total.Exposure.Conc ~
  `Population Density (Persons Per SqMile)` +
  URBAN_RURAL, formaldehyde_nooutlier)
summary(model2_nooutlier)
```

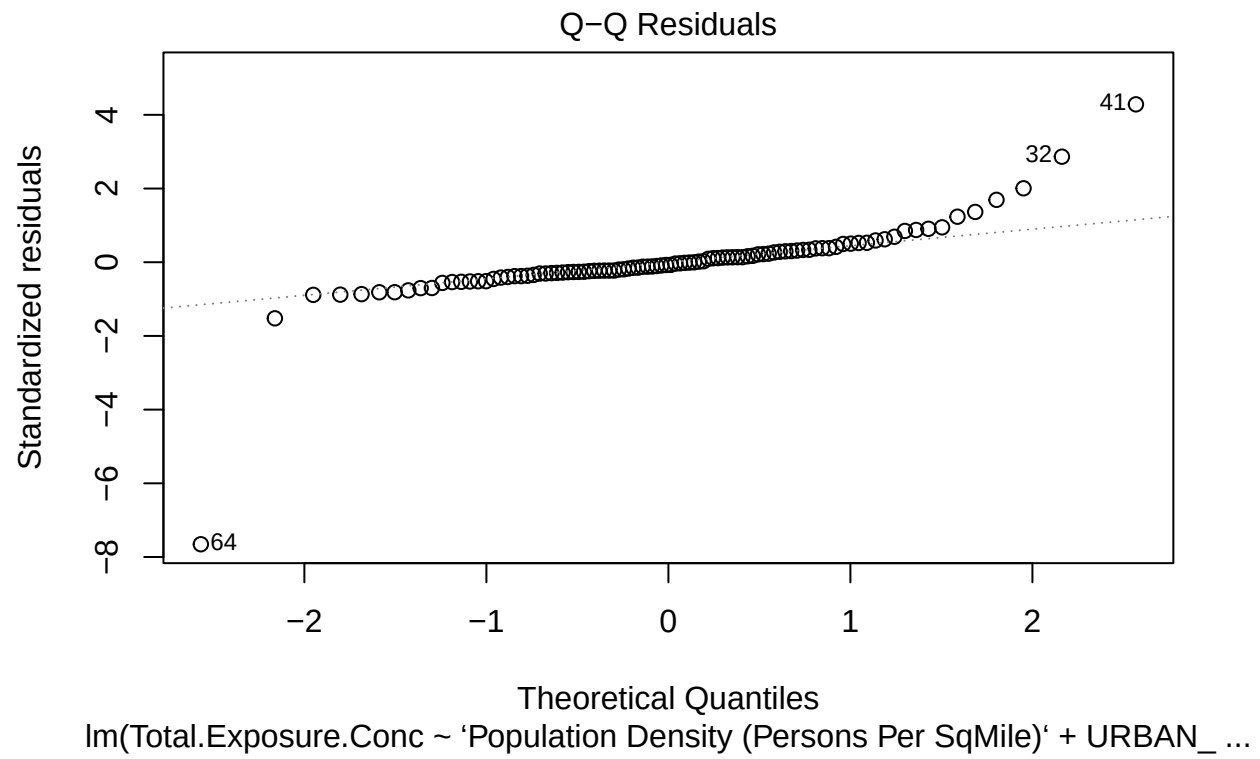
```
##
## Call:
## lm(formula = Total.Exposure.Conc ~ `Population Density (Persons Per SqMile)` +
##     URBAN_RURAL, data = formaldehyde_nooutlier)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -124.551   -5.619   -1.361    5.591   75.794
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      2.46572    2.69620   0.915   0.363
## `Population Density (Persons Per SqMile)`  0.15016    0.01004  14.958 <2e-16
## URBAN_RURALUrban    2.98029    4.33987   0.687   0.494
##
## (Intercept)
## `Population Density (Persons Per SqMile)` ***
## URBAN_RURALUrban
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

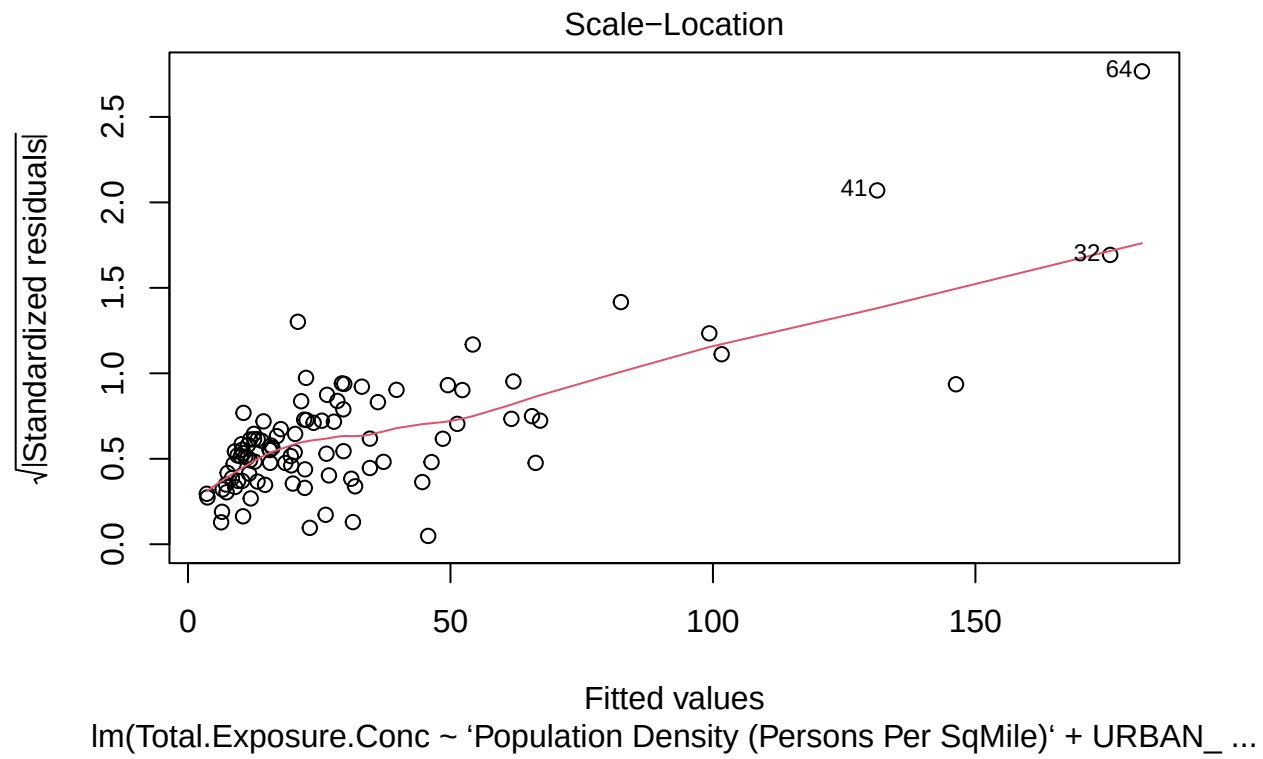
```
##  
## Residual standard error: 18.72 on 95 degrees of freedom  
## Multiple R-squared:  0.7626, Adjusted R-squared:  0.7576  
## F-statistic: 152.6 on 2 and 95 DF,  p-value: < 2.2e-16
```

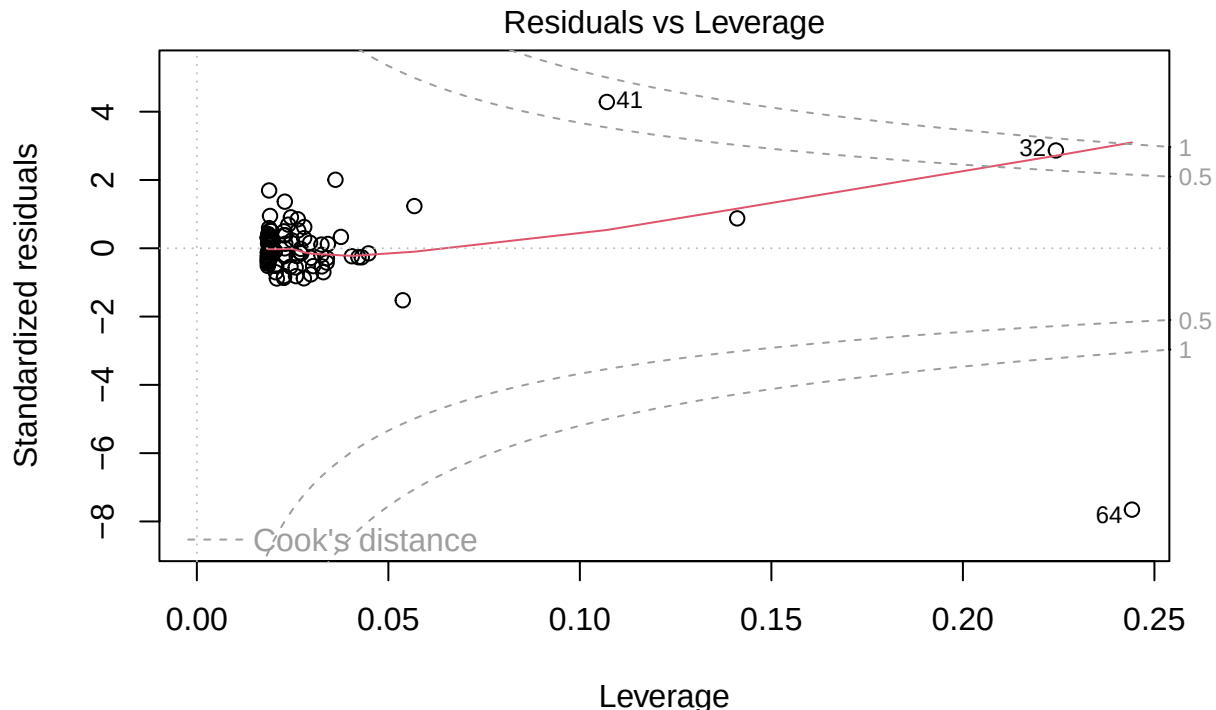
```
plot(model2_nooutlier)
```



lm(Total.Exposure.Conc ~ 'Population Density (Persons Per SqMile)' + URBAN_ ...





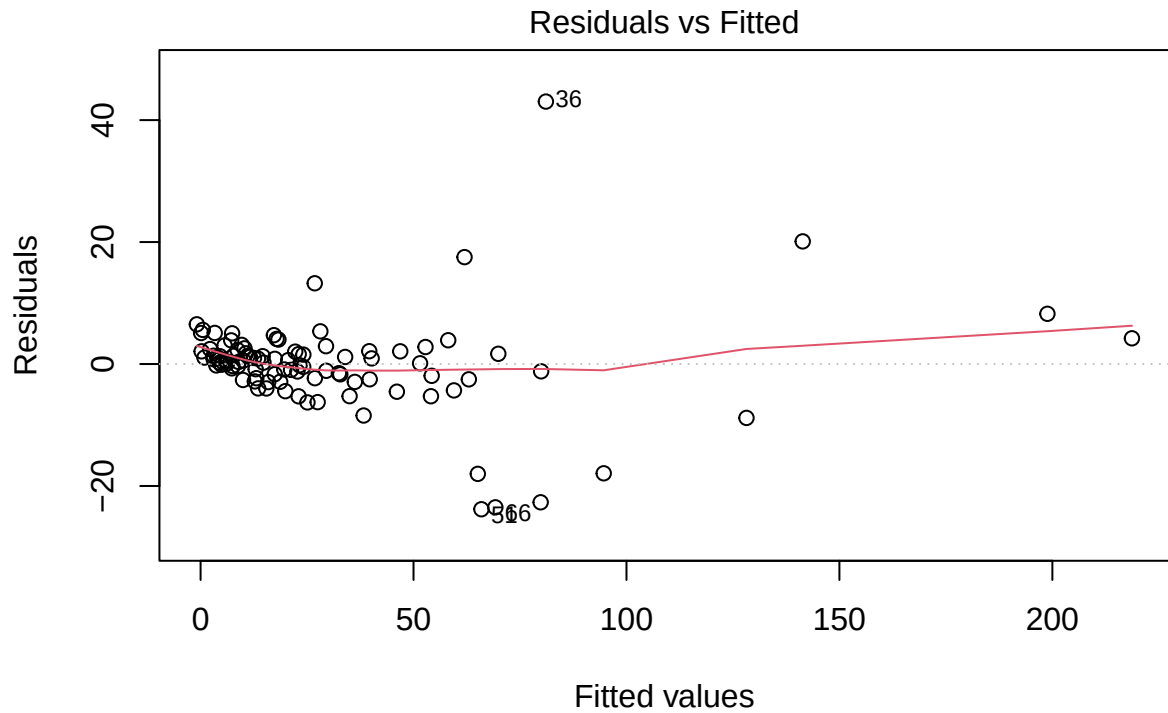


lm(Total.Exposure.Conc ~ 'Population Density (Persons Per SqMile)' + URBAN_ ...

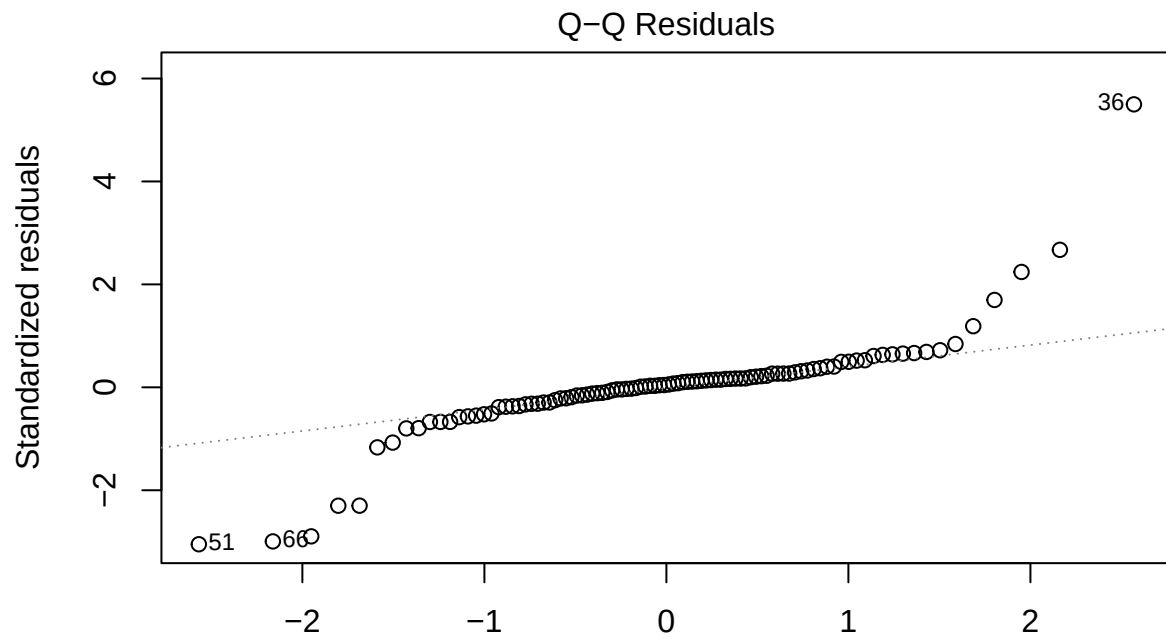
```
model3 <- lm(Total.Exposure.Conc ~ URBAN_RURAL +
  `Median Family Income (Dollars)` +
  Population, formaldehyde_nooutlier)
summary(model3)
```

```
##
## Call:
## lm(formula = Total.Exposure.Conc ~ URBAN_RURAL + 'Median Family Income (Dollars)' +
##   Population, data = formaldehyde_nooutlier)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -23.816  -2.323   0.389   2.097  43.039
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -2.734e+00  4.255e+00  -0.642  0.5221
## URBAN_RURALUrban    -3.822e+00  2.036e+00  -1.878  0.0635 .
## 'Median Family Income (Dollars)'  2.593e-05  7.596e-05   0.341  0.7336
## Population       2.085e-04  5.172e-06  40.314 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.961 on 94 degrees of freedom
## Multiple R-squared:  0.9575, Adjusted R-squared:  0.9562
## F-statistic: 706.1 on 3 and 94 DF, p-value: < 2.2e-16
```

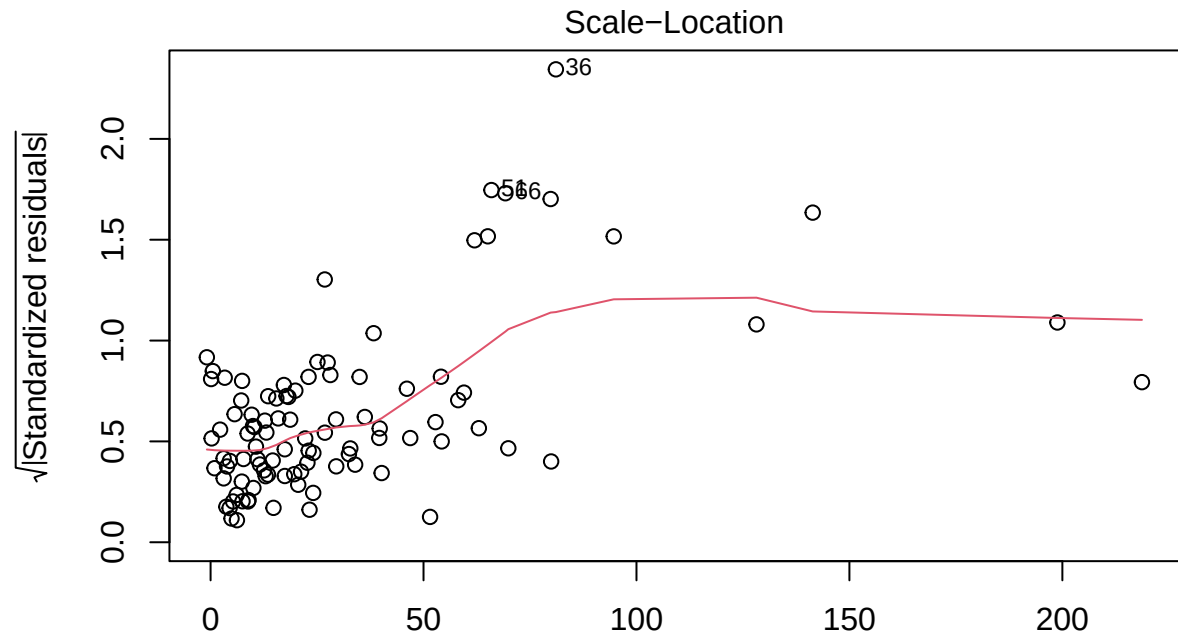
```
plot(model13)
```



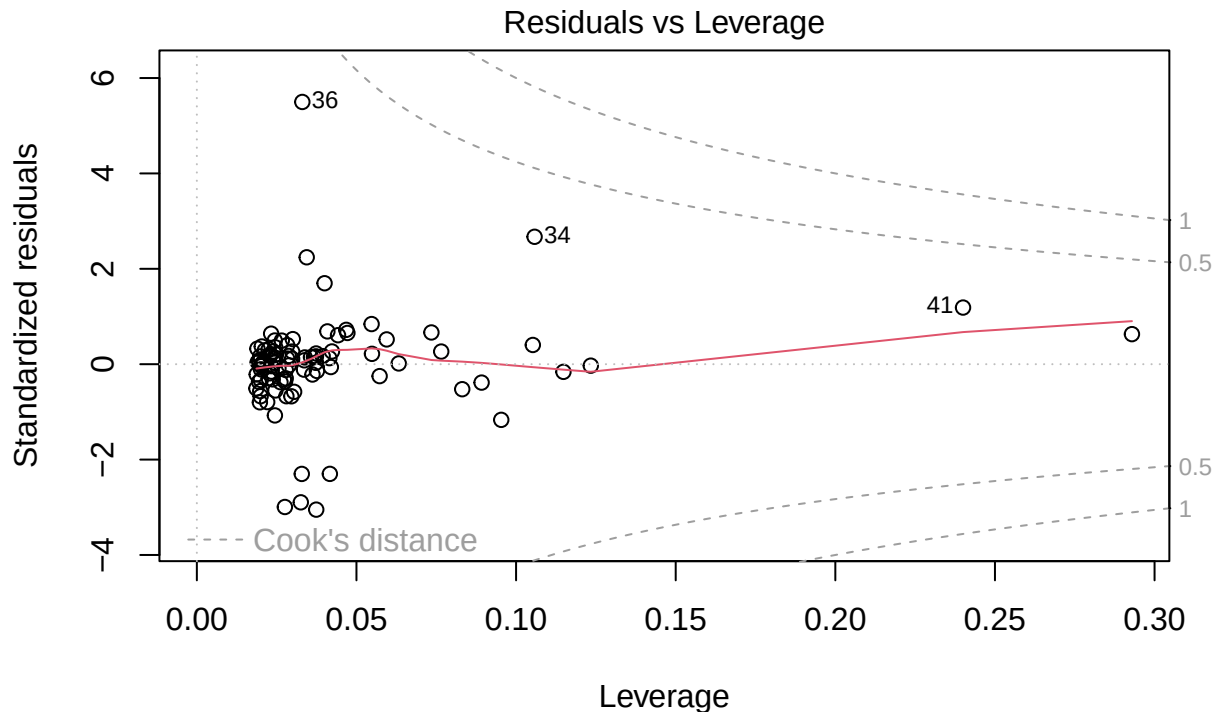
lm(Total.Exposure.Conc ~ URBAN_RURAL + 'Median Family Income (Dollars)' + P .



Im(Total.Exposure.Conc ~ URBAN_RURAL + 'Median Family Income (Dollars)' + P .



Fitted values
 $\text{lm}(\text{Total.Exposure.Conc} \sim \text{URBAN_RURAL} + \text{'Median Family Income (Dollars)'} + \text{P} .$

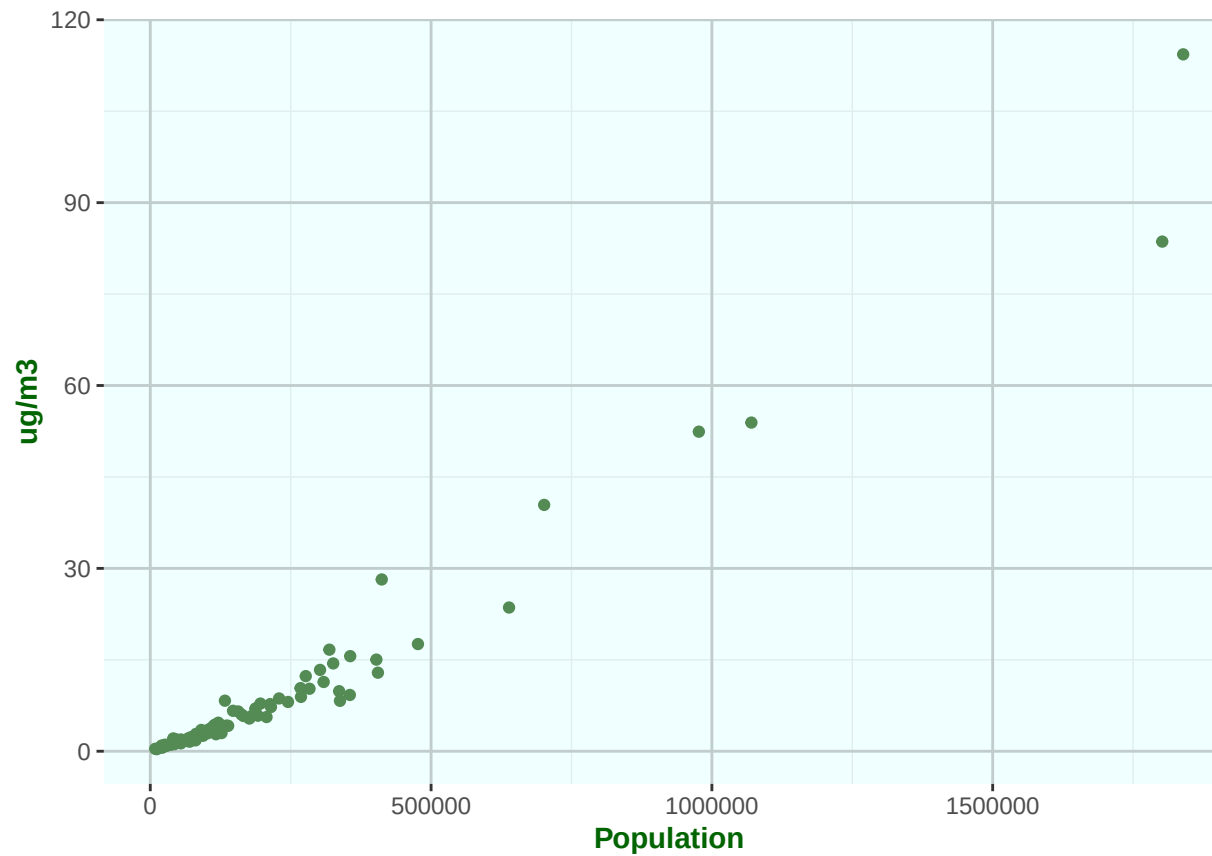


lm(Total.Exposure.Conc ~ URBAN_RURAL + 'Median Family Income (Dollars)' + P .

```
AIC1 <- AIC(model1_nooutlier)
AIC2 <- AIC(model2_nooutlier)
AIC3 <- AIC(model3)
```

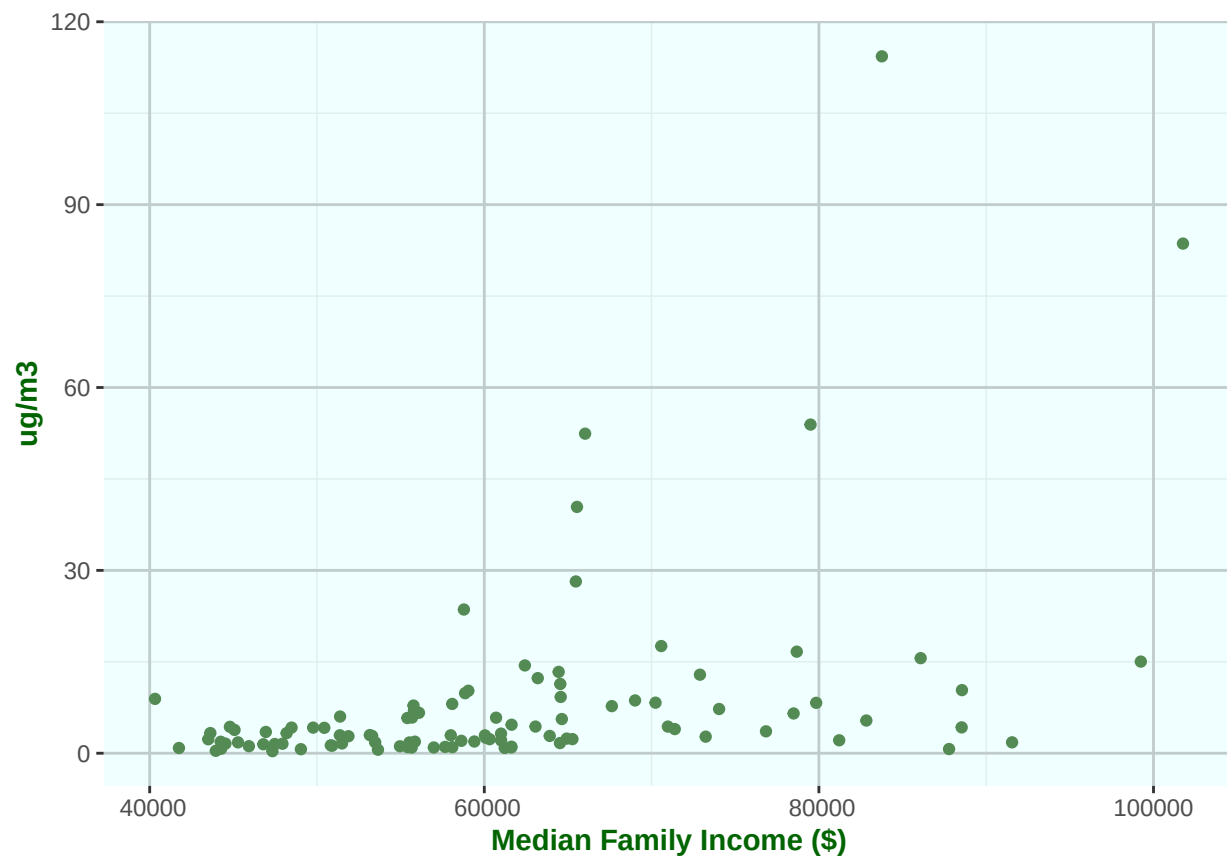
Model 1 without outliers has the lowest AIC value and is the one that I will write the results about.

```
(scatter_populationB <- ggplot(benzene_complete,
                              aes(x = Population, y = Total.Exposure.Conc)) +
  geom_point(color = "palegreen4") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold",
                                  color = "darkgreen"),
        axis.title.x = element_text(face = "bold", color = "darkgreen"),
        axis.title.y = element_text(face = "bold", color = "darkgreen"),
        panel.background = element_rect(fill = "azure"),
        panel.grid.major = element_line(color = "azure3"),
        panel.grid.minor = element_line(color = "azure2")) +
  labs(x = "Population",
       y = "ug/m3"))
```

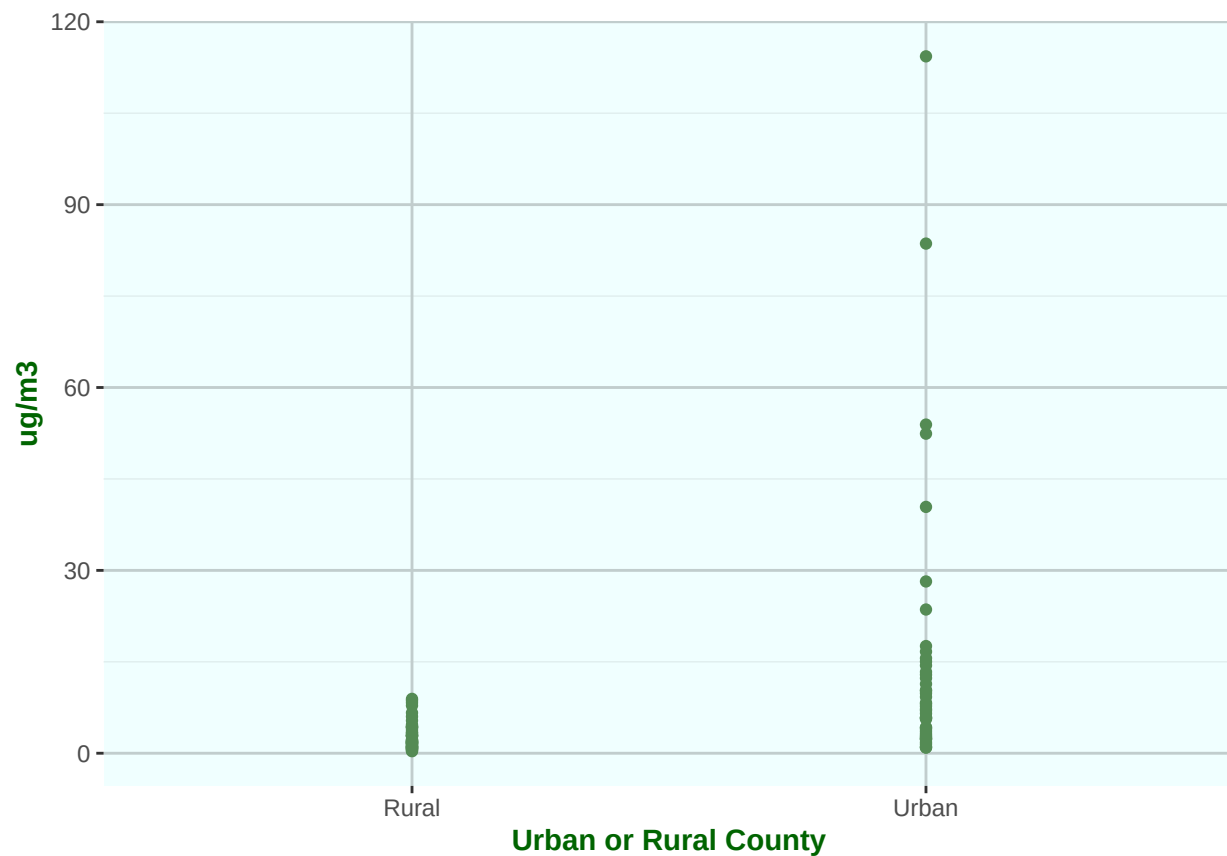


Benzene Model

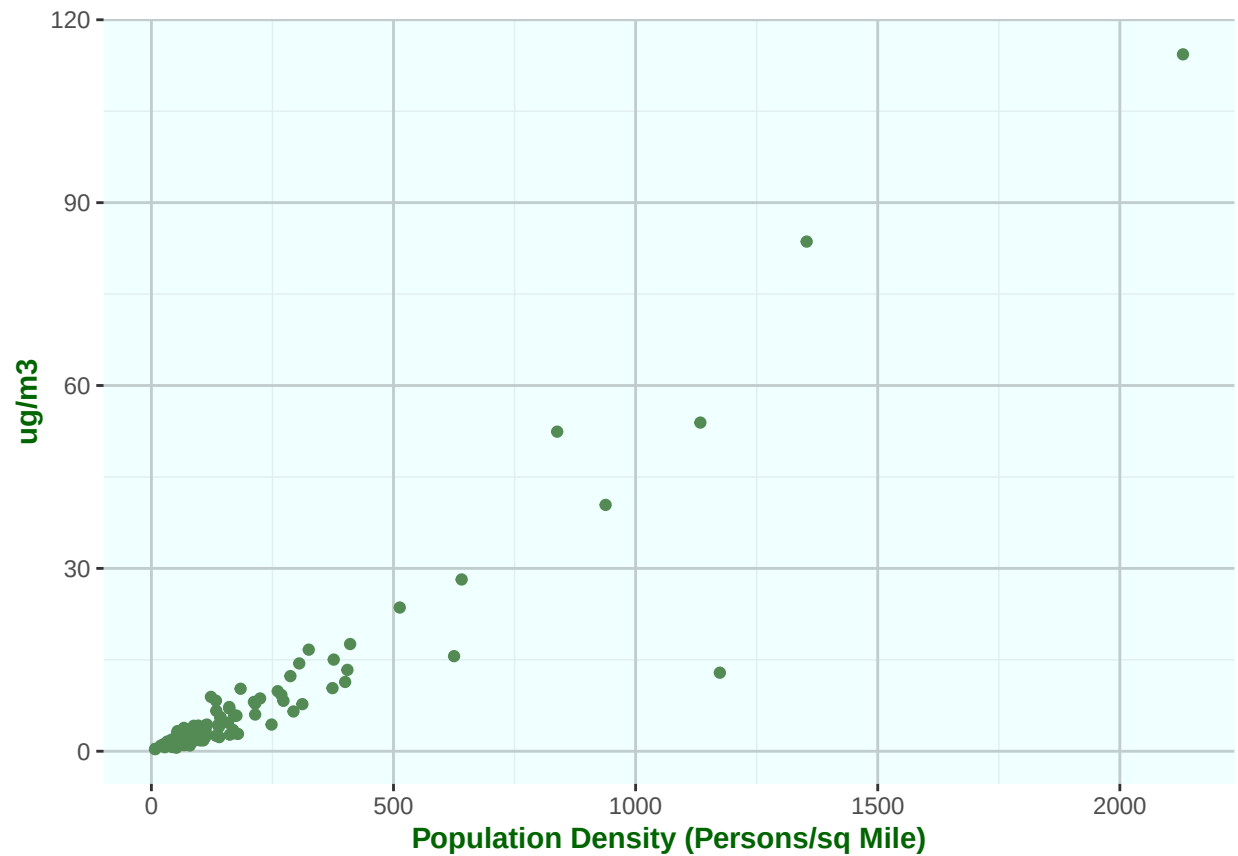
```
(scatter_incomeB <- ggplot(benzene_complete,
  aes(x = `Median Family Income (Dollars)`,
    y = Total.Exposure.Conc)) +
  geom_point(color = "palegreen4") +
  scale_x_continuous(labels = function(x) format(x, scientific = FALSE)) +
  theme(plot.title = element_text(hjust = 0.5, face = "bold",
    color = "darkgreen"),
    axis.title.x = element_text(face = "bold", color = "darkgreen"),
    axis.title.y = element_text(face = "bold", color = "darkgreen"),
    panel.background = element_rect(fill = "azure"),
    panel.grid.major = element_line(color = "azure3"),
    panel.grid.minor = element_line(color = "azure2")) +
  labs(x = "Median Family Income ($)",
    y = "ug/m3"))
```



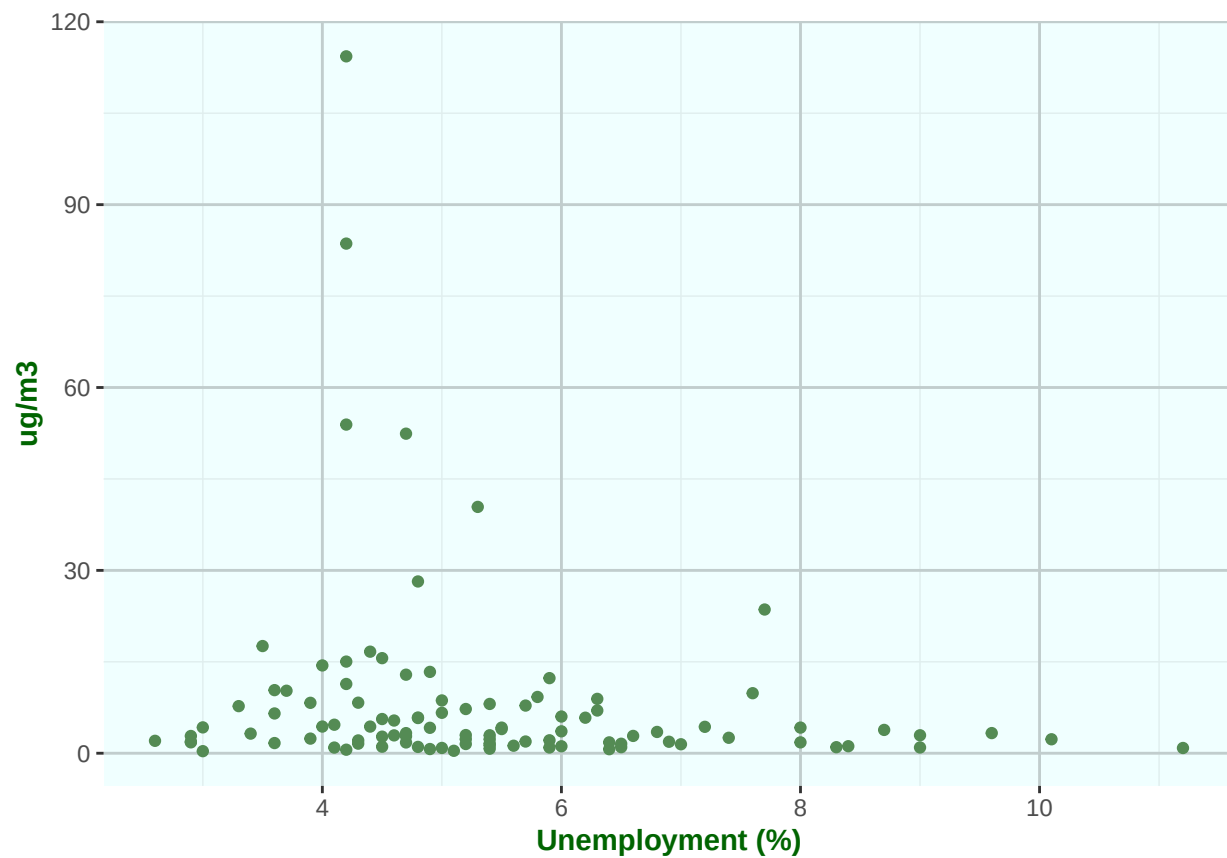
```
(scatter_urbanB <- ggplot(benzene_complete,
  aes(x = URBAN_RURAL, y = Total.Exposure.Conc)) +
  geom_point(color = "palegreen4") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold",
    color = "darkgreen"),
    axis.title.x = element_text(face = "bold", color = "darkgreen"),
    axis.title.y = element_text(face = "bold", color = "darkgreen"),
    panel.background = element_rect(fill = "azure"),
    panel.grid.major = element_line(color = "azure3"),
    panel.grid.minor = element_line(color = "azure2")) +
  labs(x = "Urban or Rural County",
    y = "ug/m3"))
```



```
(scatter_densityB <- ggplot(benzene_complete,
                             aes(x = `Population Density (Persons Per SqMile)`,
                                 y = Total.Exposure.Conc)) +
  geom_point(color = "palegreen4") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold",
                                   color = "darkgreen"),
        axis.title.x = element_text(face = "bold", color = "darkgreen"),
        axis.title.y = element_text(face = "bold", color = "darkgreen"),
        panel.background = element_rect(fill = "azure"),
        panel.grid.major = element_line(color = "azure3"),
        panel.grid.minor = element_line(color = "azure2")) +
  labs(x = "Population Density (Persons/sq Mile)",
       y = "ug/m3"))
```

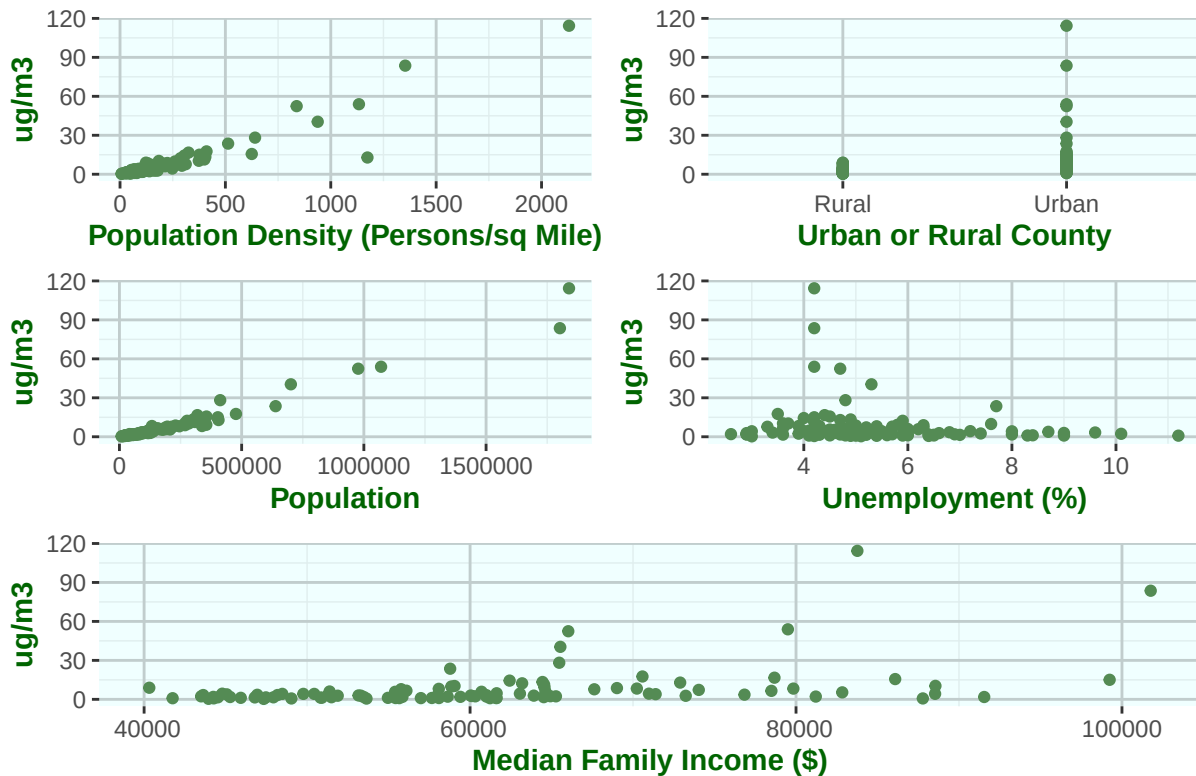



```
(scatter_unemploymentB <- ggplot(benzene_complete,
                                aes(x = `Unemployment(Percent)`,
                                    y = Total.Exposure.Conc)) +
  geom_point(color = "palegreen4") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold",
                                  color = "darkgreen"),
        axis.title.x = element_text(face = "bold", color = "darkgreen"),
        axis.title.y = element_text(face = "bold", color = "darkgreen"),
        panel.background = element_rect(fill = "azure"),
        panel.grid.major = element_line(color = "azure3"),
        panel.grid.minor = element_line(color = "azure2")) +
  labs(x = "Unemployment (%)",
       y = "ug/m3"))
```



```
final_plotB <- ((scatter_densityB + scatter_urbanB) /
  (scatter_populationB + scatter_unemploymentB) / scatter_incomeB) +
  plot_annotation(title = "Benzene Concentrations vs. County Demographics",
    theme = theme(
      plot.title = element_text(hjust = 0.5, face = "bold",
        color = "darkgreen")))
plot(final_plotB)
```

Benzene Concentrations vs. County Demographics



```
# Perform correlation tests to test for collinearity
```

```
cor.test(benzene_complete$Population,
         benzene_complete$`Median Family Income (Dollars)`)
```

```
##
## Pearson's product-moment correlation
##
## data: benzene_complete$Population and benzene_complete$`Median Family Income (Dollars)`
## t = 5.4197, df = 98, p-value = 4.281e-07
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.3133539 0.6183125
## sample estimates:
## cor
## 0.4802155
```

```
# p < 0.01, cor = 0.4802
```

```
cor.test(benzene_complete$Population,
         benzene_complete$`Population Density (Persons Per SqMile)`)
```

```
##
## Pearson's product-moment correlation
##
## data: benzene_complete$Population and benzene_complete$`Population Density (Persons Per SqMile)`
```

```
## t = 25.141, df = 98, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.8982012 0.9527564
## sample estimates:
##      cor
## 0.9304641
```

```
# p < 0.01, cor = 0.9144
```

```
cor.test(benzene_complete$`Unemployment(Percent)` ,
         benzene_complete$`Median Family Income (Dollars)`)
```

```
##
## Pearson's product-moment correlation
##
## data: benzene_complete$`Unemployment(Percent)` and benzene_complete$`Median Family Income (Dollars)`
## t = -5.5819, df = 98, p-value = 2.12e-07
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.6270834 -0.3262148
## sample estimates:
##      cor
## -0.491162
```

```
# p < 0.01, cor = -0.4911
```

```
cor.test(benzene_complete$`Median Family Income (Dollars)` ,
         benzene_complete$`Population Density (Persons Per SqMile)`)
```

```
##
## Pearson's product-moment correlation
##
## data: benzene_complete$`Median Family Income (Dollars)` and benzene_complete$`Population Density (Persons Per SqMile)`
## t = 5.6216, df = 98, p-value = 1.783e-07
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.3293295 0.6291961
## sample estimates:
##      cor
## 0.4938052
```

```
# p < 0.01, cor = 0.402
```

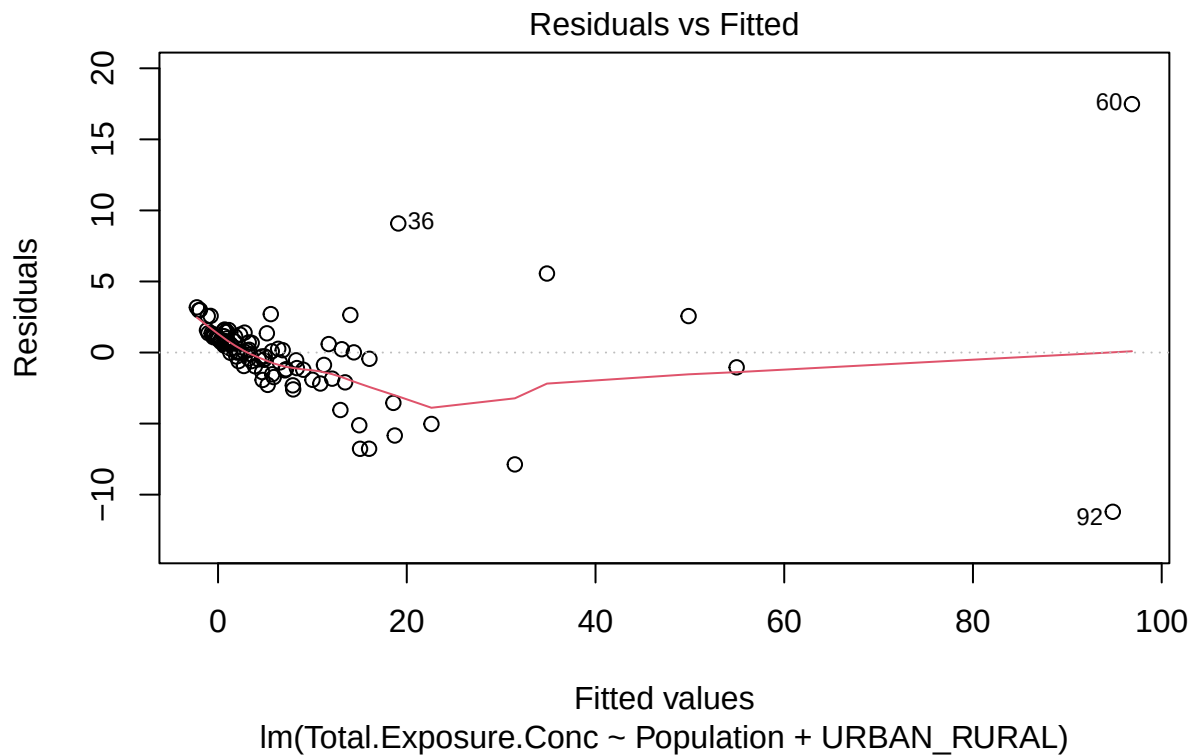
```
# Generate regression models
```

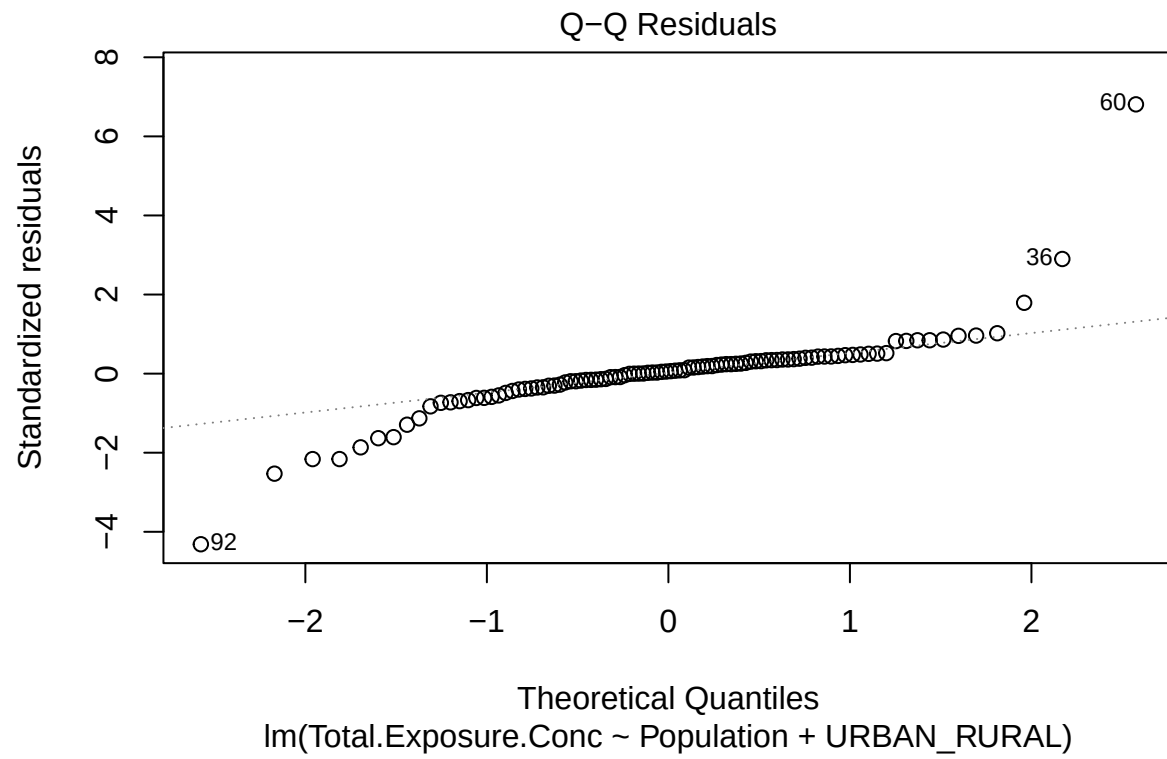
```
B_model1 <- lm(Total.Exposure.Conc ~ Population + URBAN_RURAL,
               benzene_complete)
summary(B_model1)
```

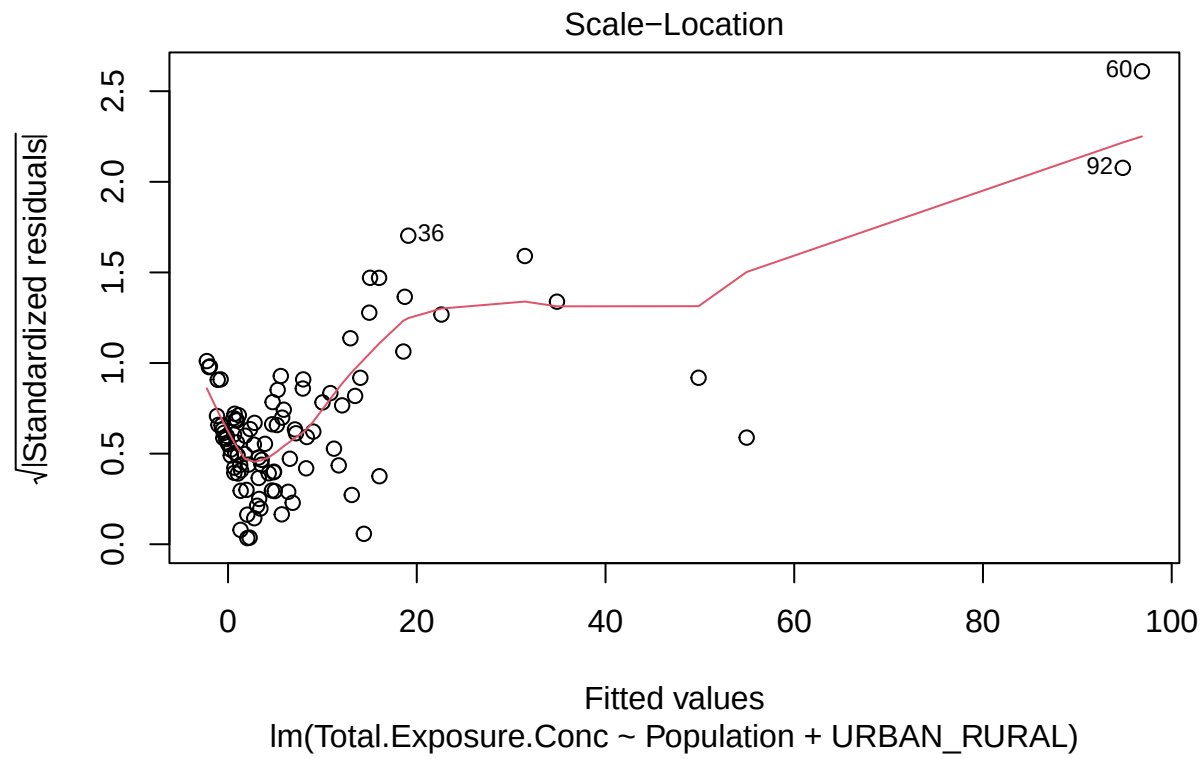
```
##
## Call:
## lm(formula = Total.Exposure.Conc ~ Population + URBAN_RURAL,
```

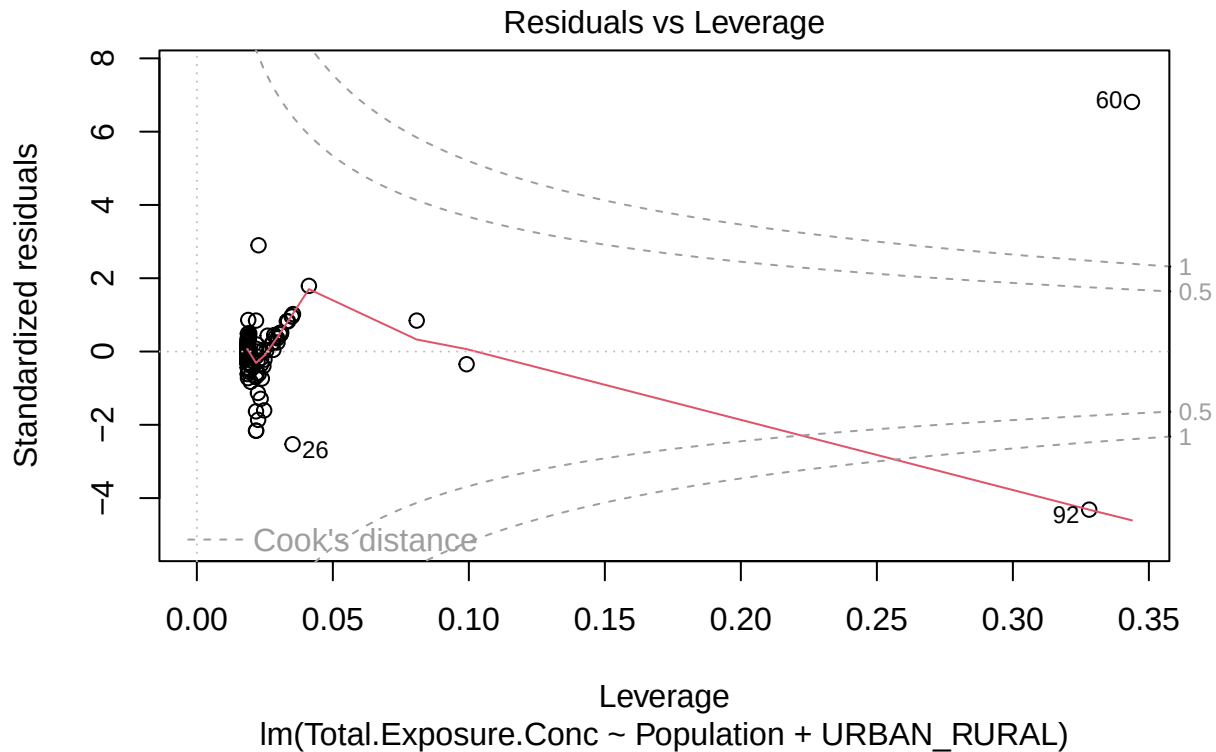
```
##      data = benzene_complete)
##
## Residuals:
##      Min        1Q      Median        3Q        Max
## -11.2065   -0.9831    0.1799    1.1280   17.4801
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.653e+00  4.420e-01  -3.739 0.000312 ***
## Population      5.448e-05  1.192e-06  45.712 < 2e-16 ***
## URBAN_RURALUrban -1.701e+00  7.016e-01  -2.424 0.017197 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.169 on 97 degrees of freedom
## Multiple R-squared:  0.9617, Adjusted R-squared:  0.9609
## F-statistic: 1219 on 2 and 97 DF,  p-value: < 2.2e-16
```

```
plot(B_model1)
```









```
AIC(B_model1) # = 519.3985
```

```
## [1] 519.3985
```

```
benzene_nooutlier <- benzene_complete[-60,]
benzene_nooutlier <- benzene_nooutlier[-92,]

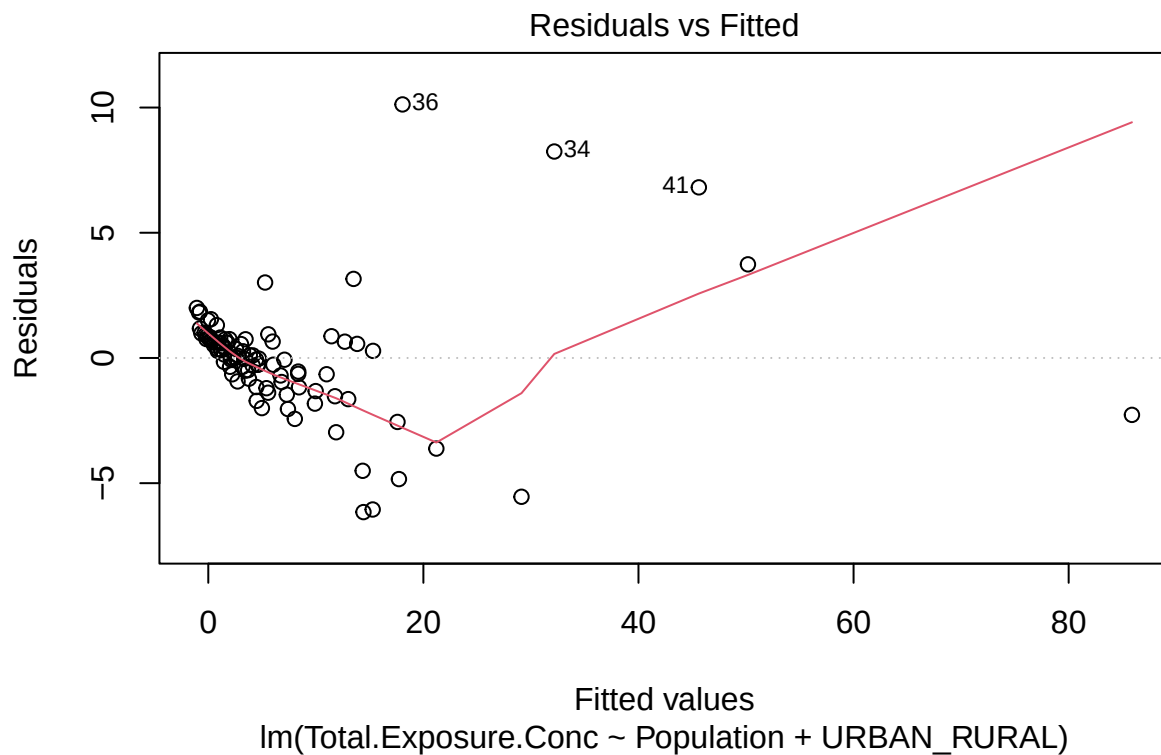
Bmodel1_nooutlier <- lm(Total.Exposure.Conc ~ Population + URBAN_RURAL,
                        benzene_nooutlier)
summary(Bmodel1_nooutlier)
```

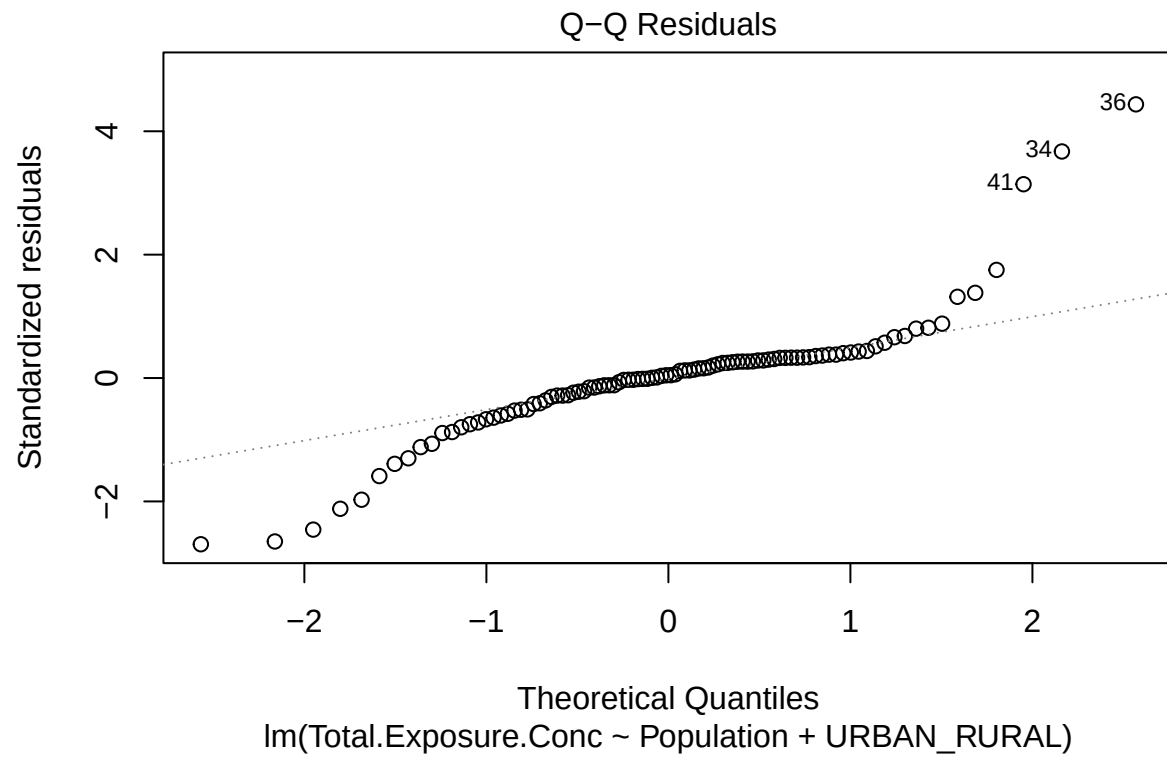
```
##
## Call:
## lm(formula = Total.Exposure.Conc ~ Population + URBAN_RURAL,
##     data = benzene_nooutlier)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.1550 -0.7966  0.1034  0.7496 10.1239
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.207e+00  3.292e-01  -3.667 0.000405 ***
## Population      4.880e-05  1.061e-06  45.975 < 2e-16 ***
## URBAN_RURALUrban -8.485e-01  5.209e-01  -1.629 0.106672
```

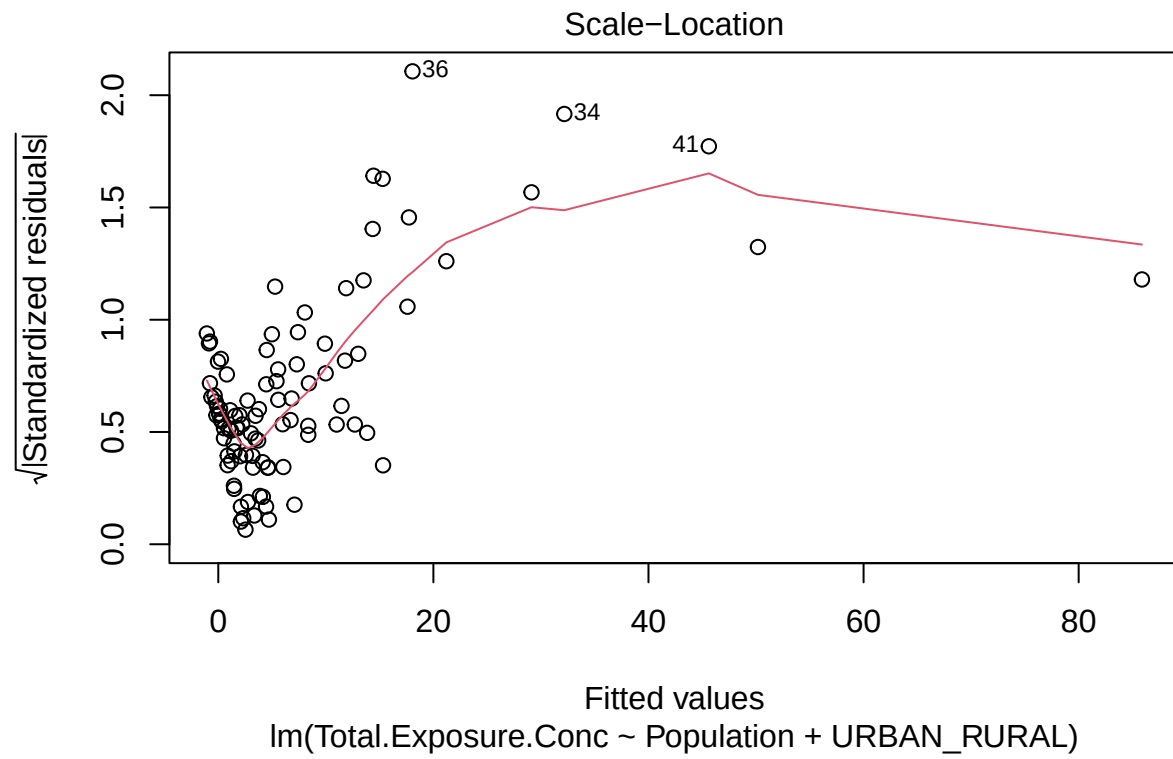


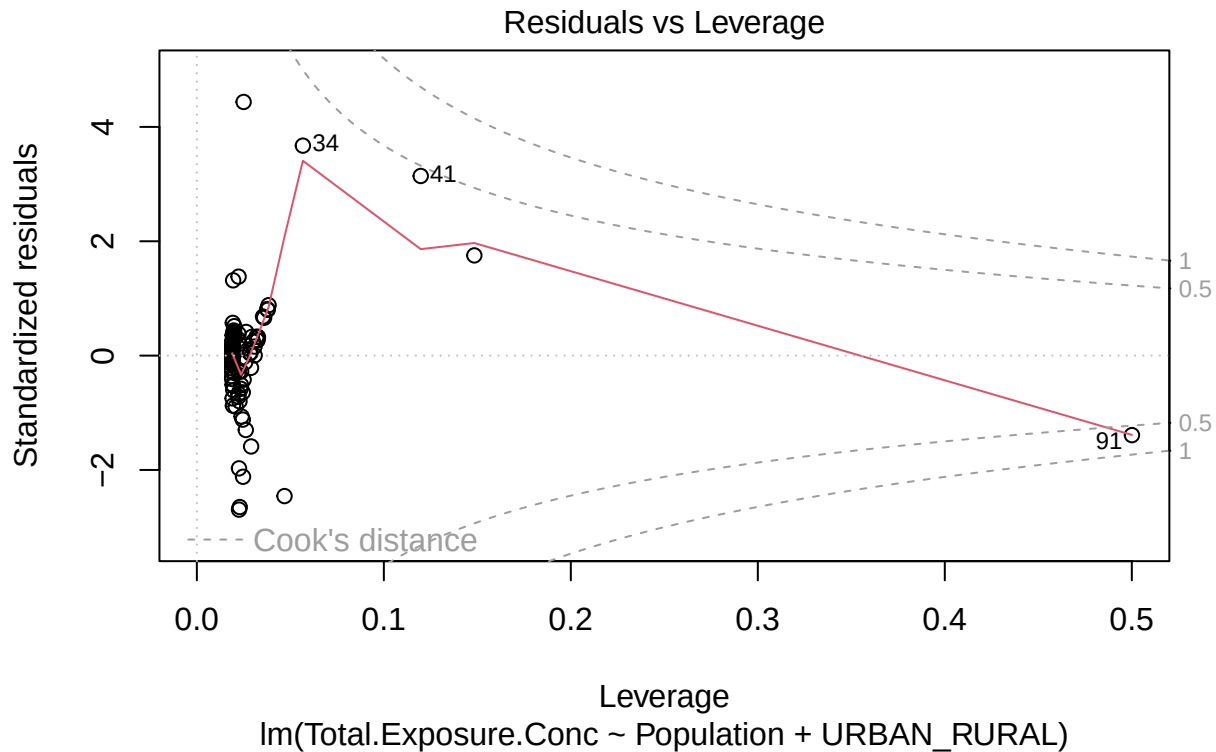
```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.311 on 95 degrees of freedom
## Multiple R-squared:  0.9639, Adjusted R-squared:  0.9631
## F-statistic: 1268 on 2 and 95 DF,  p-value: < 2.2e-16
```

```
plot(Bmodel1_nooutlier)
```









```
AIC(Bmodel1_nooutlier) # = 447.2516
```

```
## [1] 447.2516
```

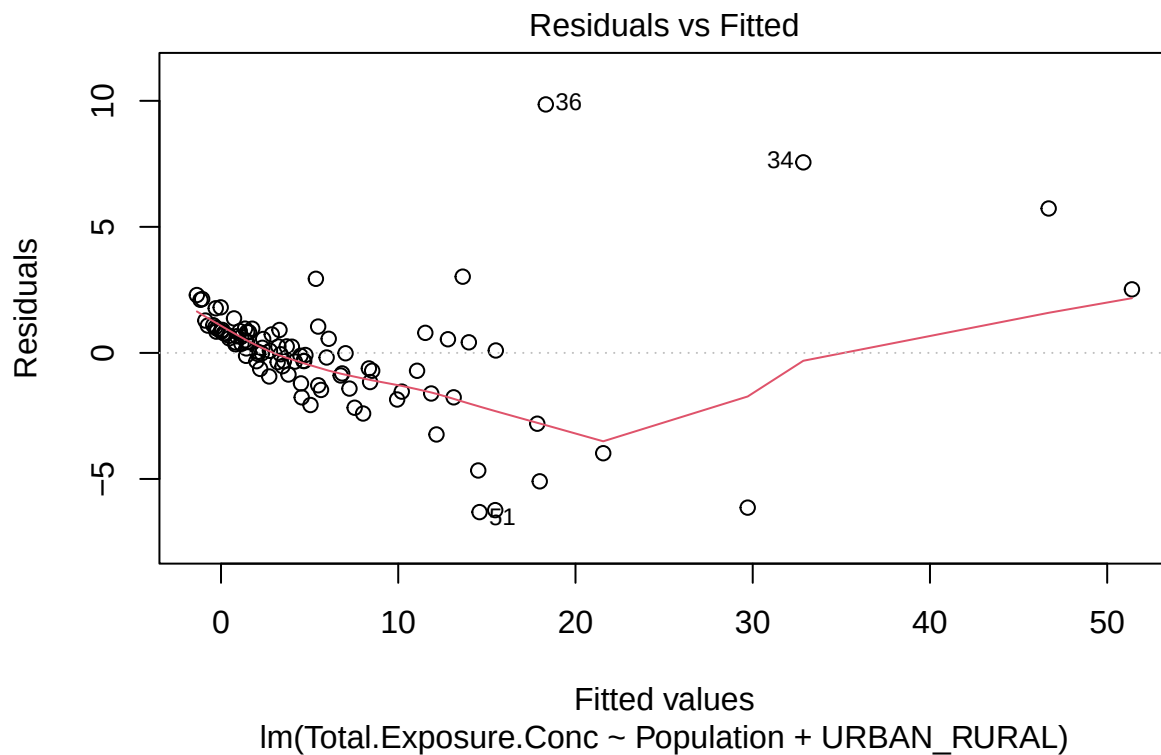
```
benzene_nooutlier <- benzene_nooutlier[-91,]
```

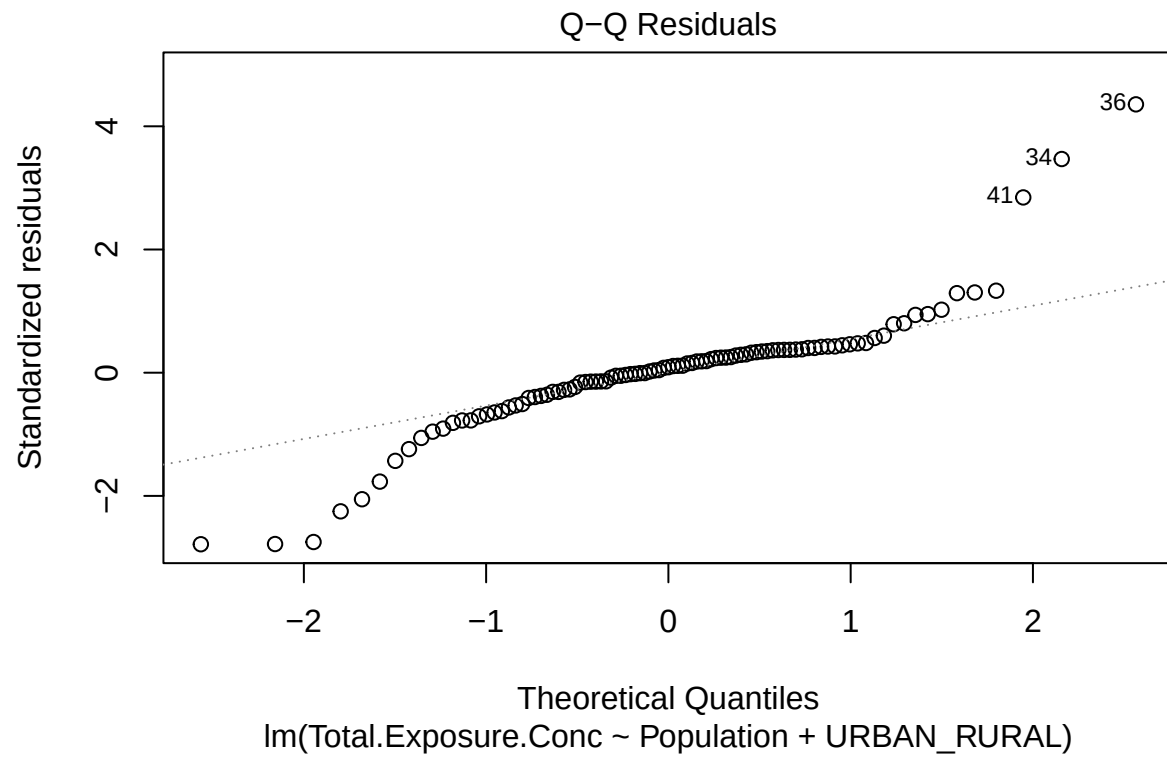
```
B_model2 <- lm(Total.Exposure.Conc ~ Population + URBAN_RURAL,
  benzene_nooutlier)
summary(B_model2)
```

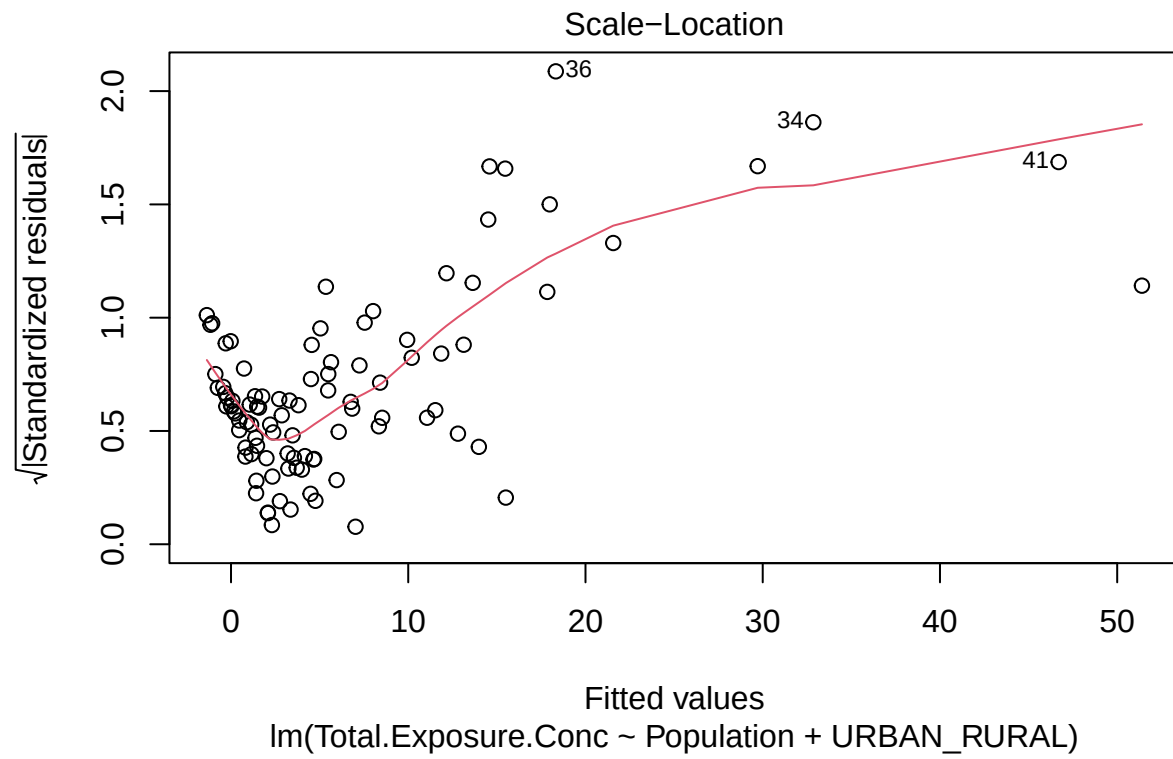
```
##
## Call:
## lm(formula = Total.Exposure.Conc ~ Population + URBAN_RURAL,
##     data = benzene_nooutlier)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.3149 -0.8150  0.2017  0.8447  9.8566
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.326e+00  3.384e-01  -3.918 0.000169 ***
## Population      5.024e-05  1.477e-06  34.016 < 2e-16 ***
## URBAN_RURALUrban -1.057e+00  5.395e-01  -1.960 0.052944 .
##
```

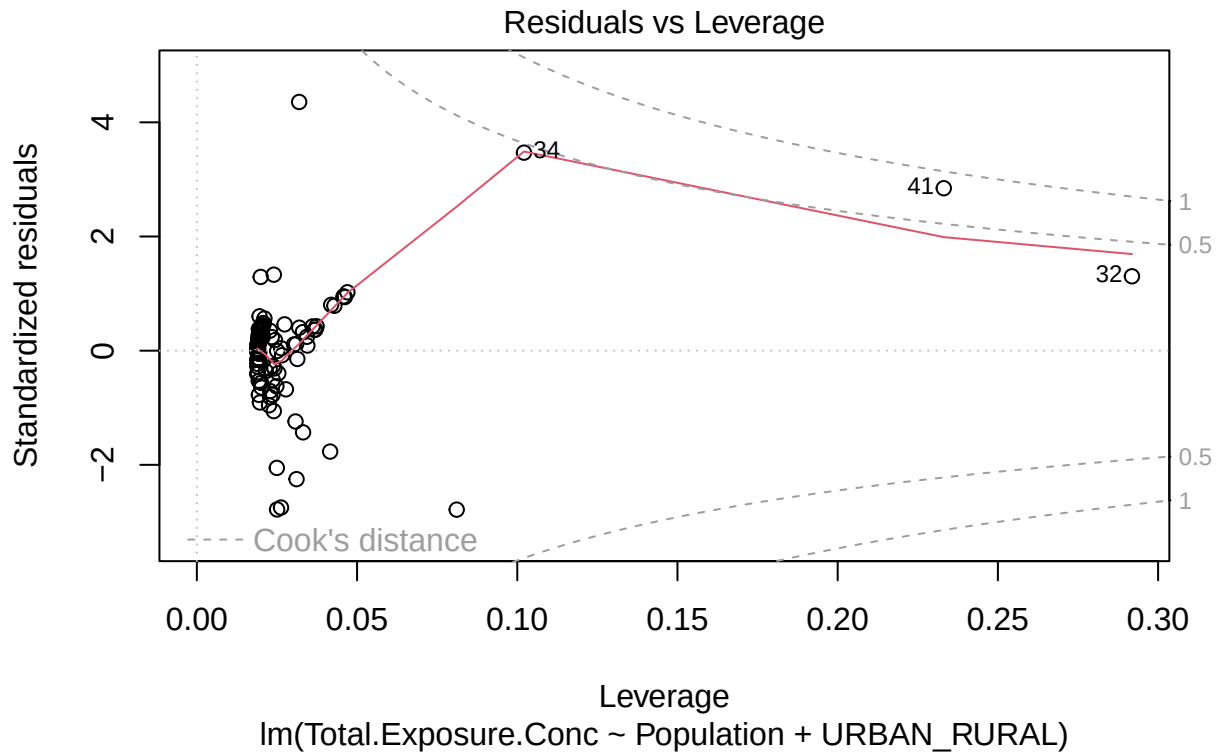
```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.299 on 94 degrees of freedom
## Multiple R-squared:  0.9391, Adjusted R-squared:  0.9378
## F-statistic: 724.5 on 2 and 94 DF,  p-value: < 2.2e-16
```

```
plot(B_model2)
```









```
AIC(B_model2) # = 441.7628
```

```
## [1] 441.7683
```

```
benzene_complete <- benzene_complete %>%
  mutate(log_conc = log10(Total.Exposure.Conc))

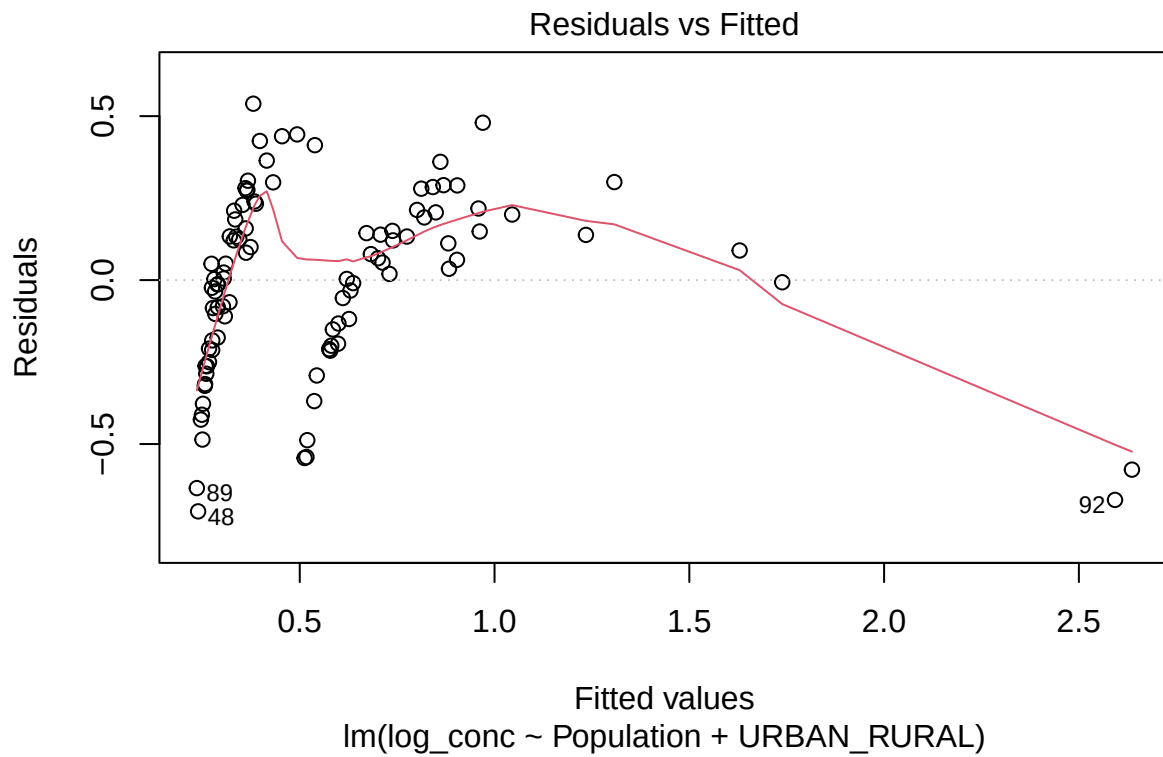
B_model_log <- lm(log_conc ~ Population + URBAN_RURAL,
  benzene_complete)
summary(B_model_log)
```

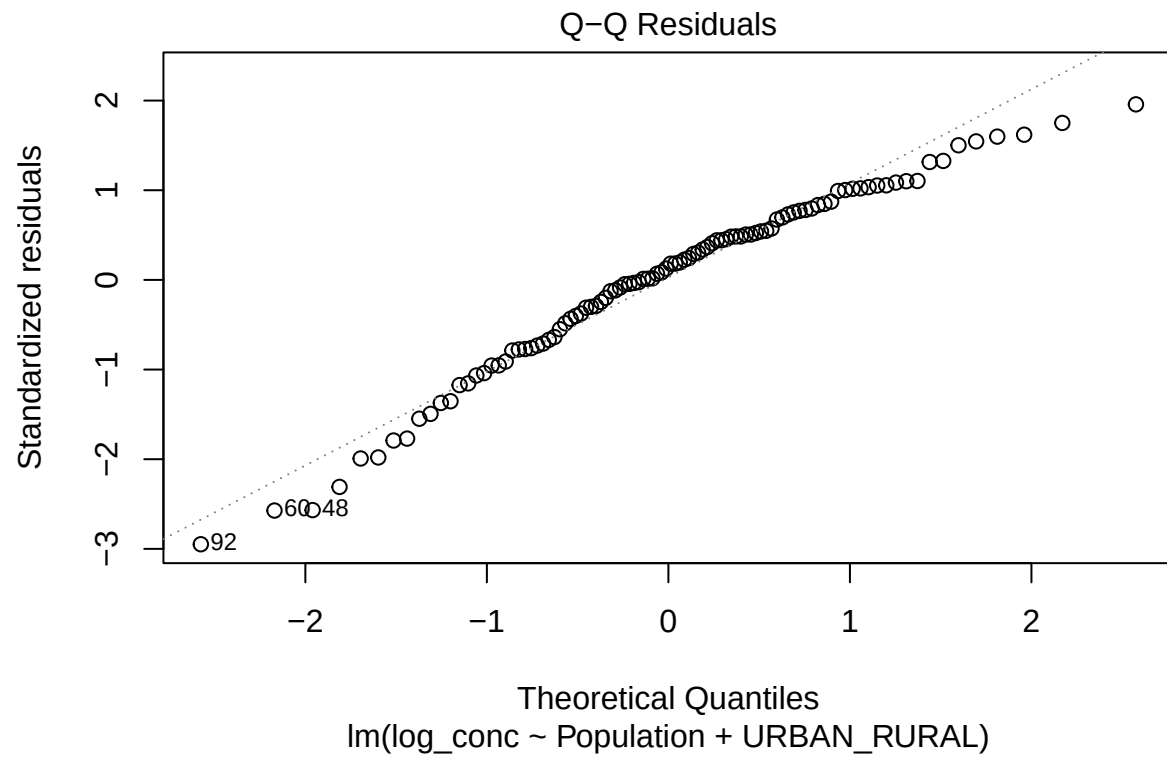
```
##
## Call:
## lm(formula = log_conc ~ Population + URBAN_RURAL, data = benzene_complete)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.70538 -0.18646  0.04213  0.20160  0.53799
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.255e-01  3.870e-02   5.828 7.33e-08 ***
## Population     1.168e-06  1.043e-07  11.190 < 2e-16 ***
## URBAN_RURALUrban 2.631e-01  6.142e-02   4.284 4.32e-05 ***
## ---
```

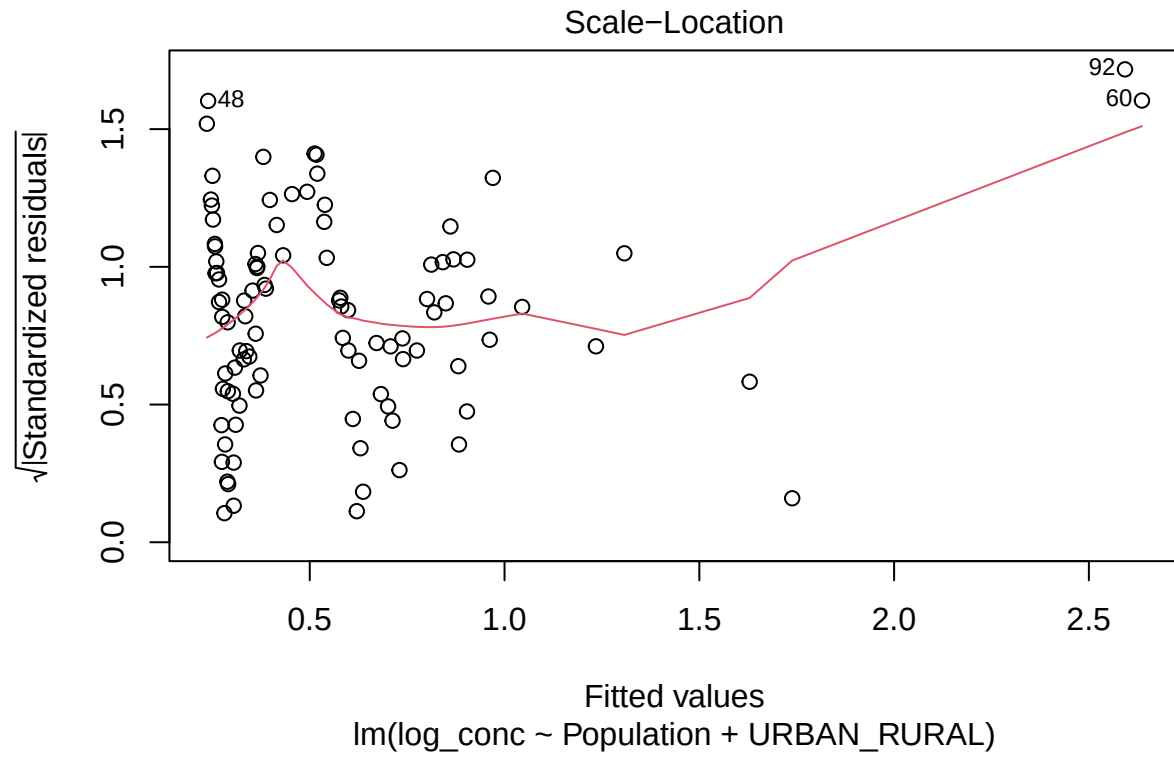


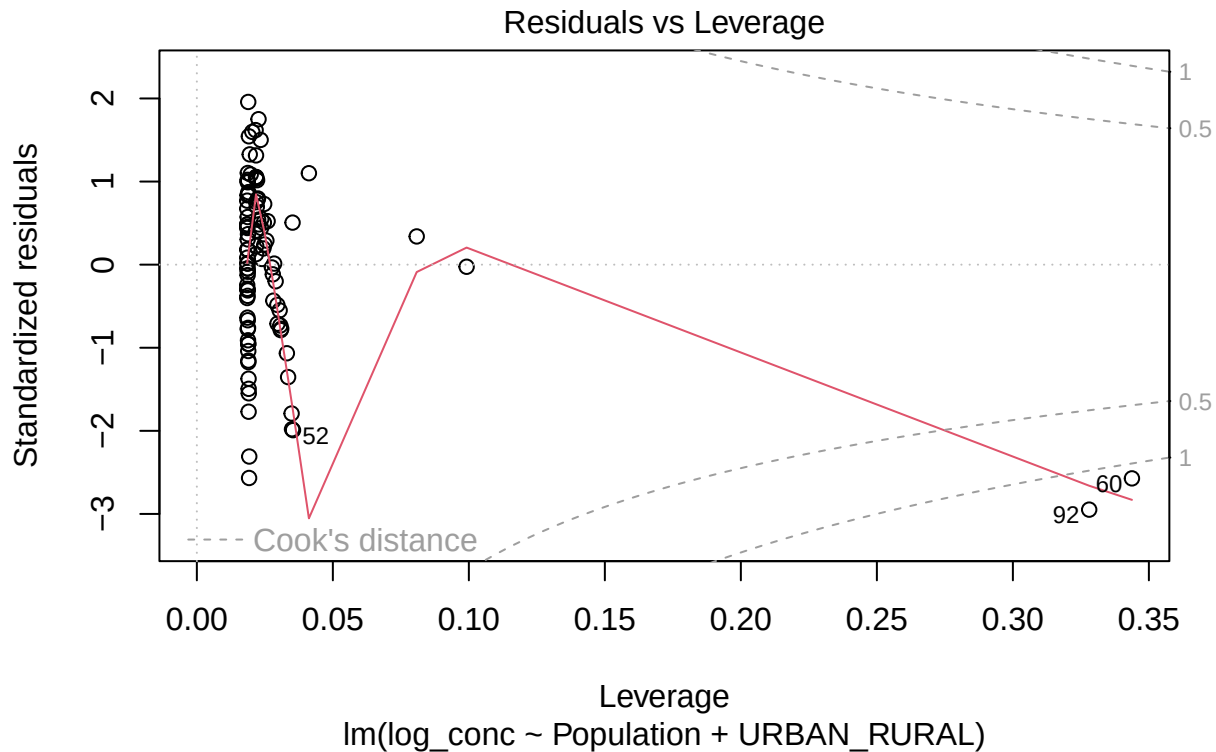
```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2774 on 97 degrees of freedom
## Multiple R-squared:  0.698, Adjusted R-squared:  0.6918
## F-statistic: 112.1 on 2 and 97 DF,  p-value: < 2.2e-16
```

```
plot(B_model_log)
```









```
AIC(B_model_log) #32.28993
```

```
## [1] 32.28993
```

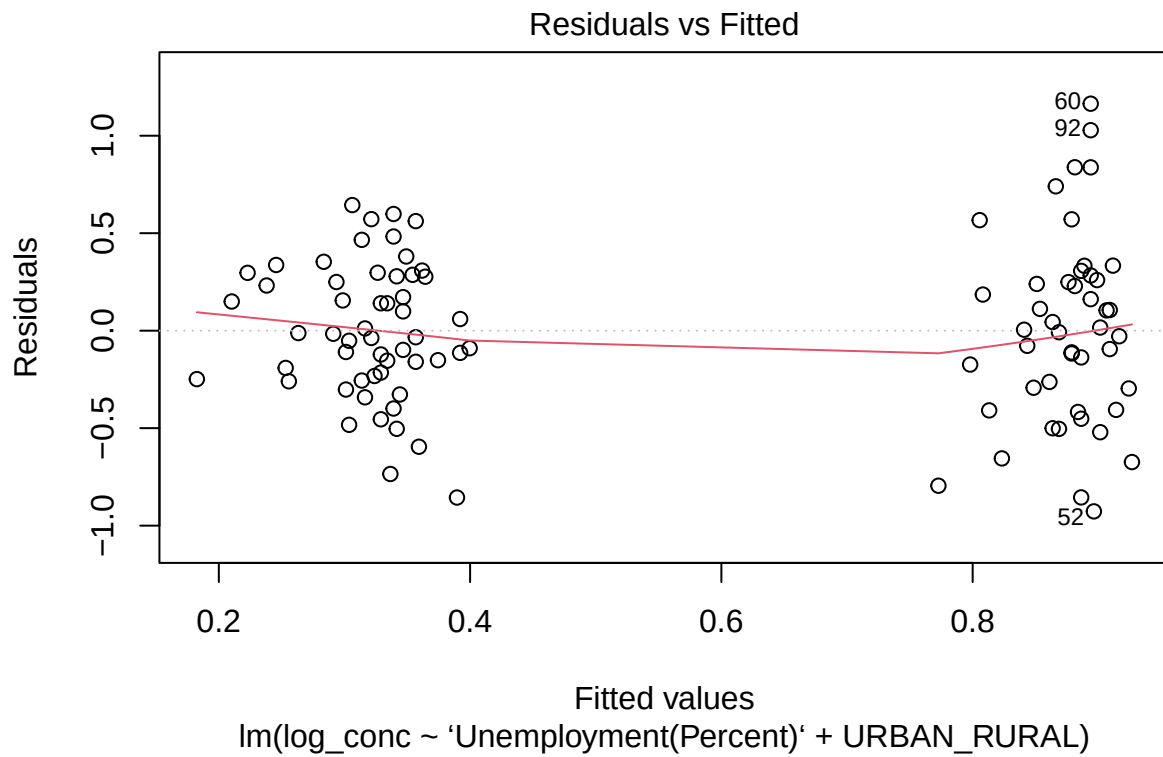
```
B_model_log2 <- lm(log_conc ~ `Unemployment(Percent)` + URBAN_RURAL,
  benzene_complete)
```

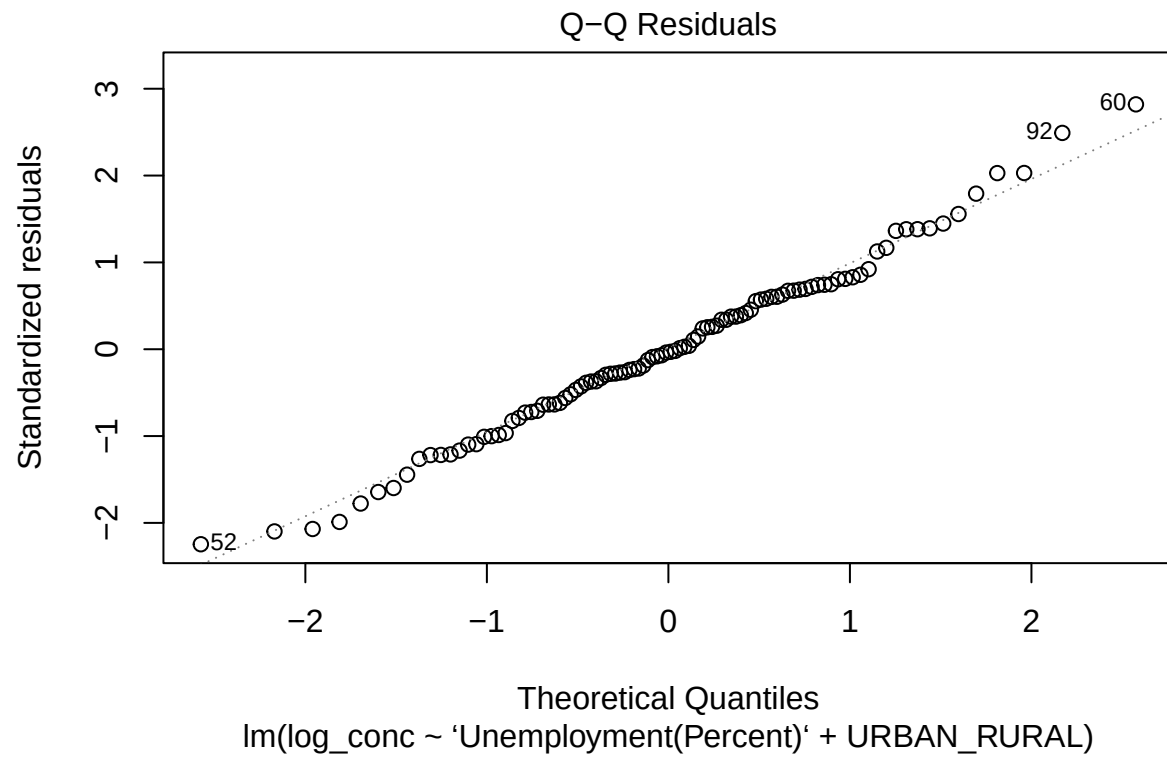
```
summary(B_model_log2)
```

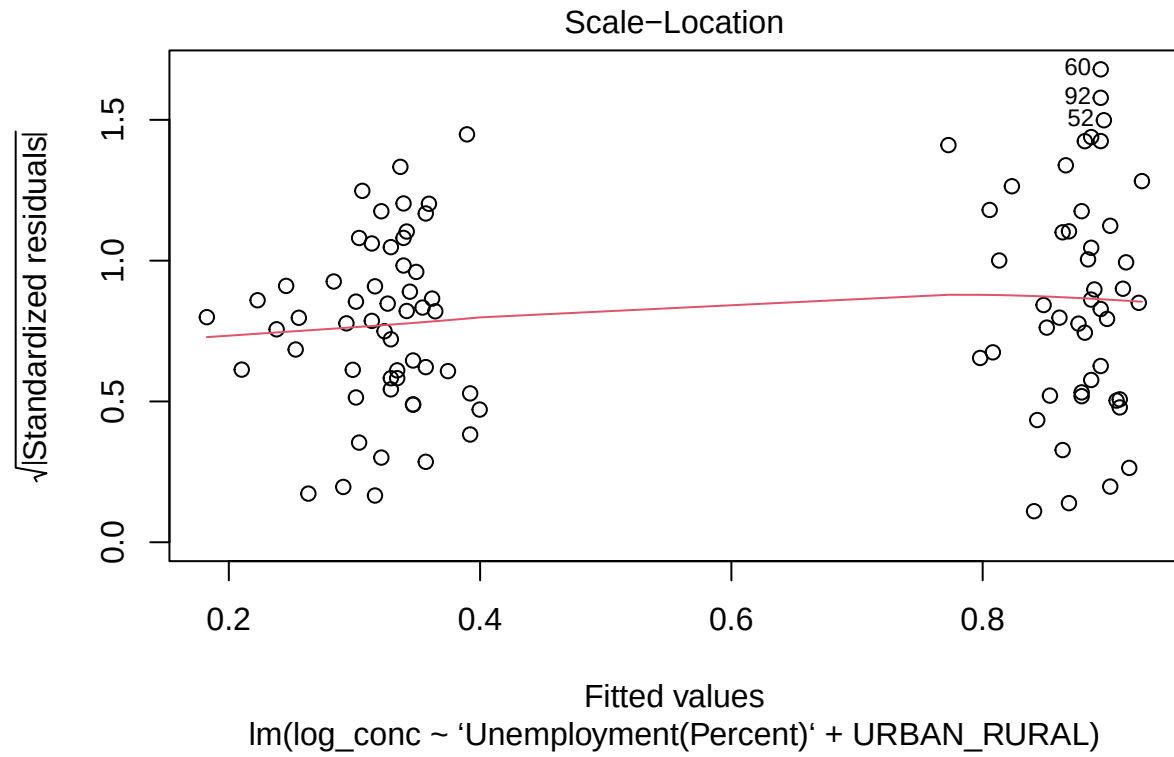
```
##
## Call:
## lm(formula = log_conc ~ `Unemployment(Percent)` + URBAN_RURAL,
##     data = benzene_complete)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.92683 -0.26038 -0.01407  0.27730  1.16421
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.46531    0.15891   2.928  0.00425 **
## `Unemployment(Percent)` -0.02525    0.02592  -0.974  0.33230
## URBAN_RURALUrban    0.53469    0.08611   6.209 1.32e-08 ***
## ---
```

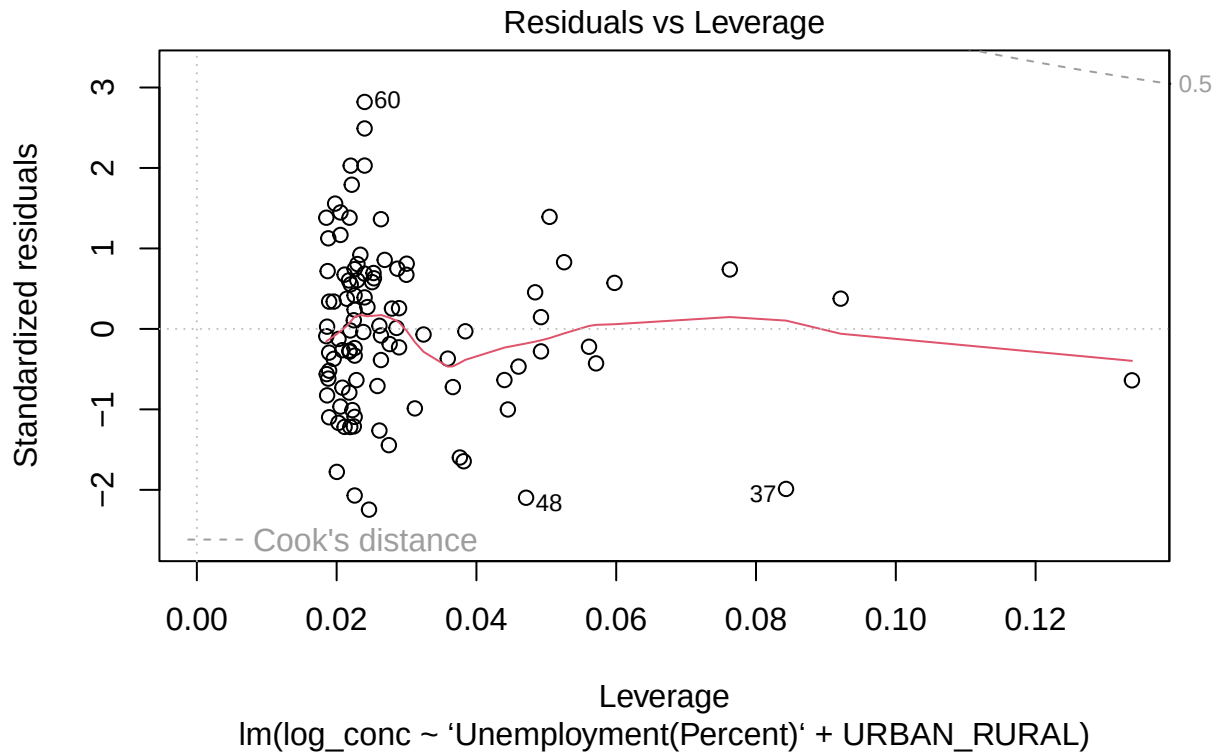
```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4178 on 97 degrees of freedom
## Multiple R-squared:  0.3149, Adjusted R-squared:  0.3008
## F-statistic: 22.29 on 2 and 97 DF,  p-value: 1.08e-08
```

```
plot(B_model_log2)
```









```
AIC(B_model_log2)
```

```
## [1] 114.2148
```

```
log_nooutliers <- benzene_complete[-c(60,48,37),]
```

```
B_model_log3 <- lm(log_conc ~ 'Unemployment(Percent)' + URBAN_RURAL,
  log_nooutliers)
```

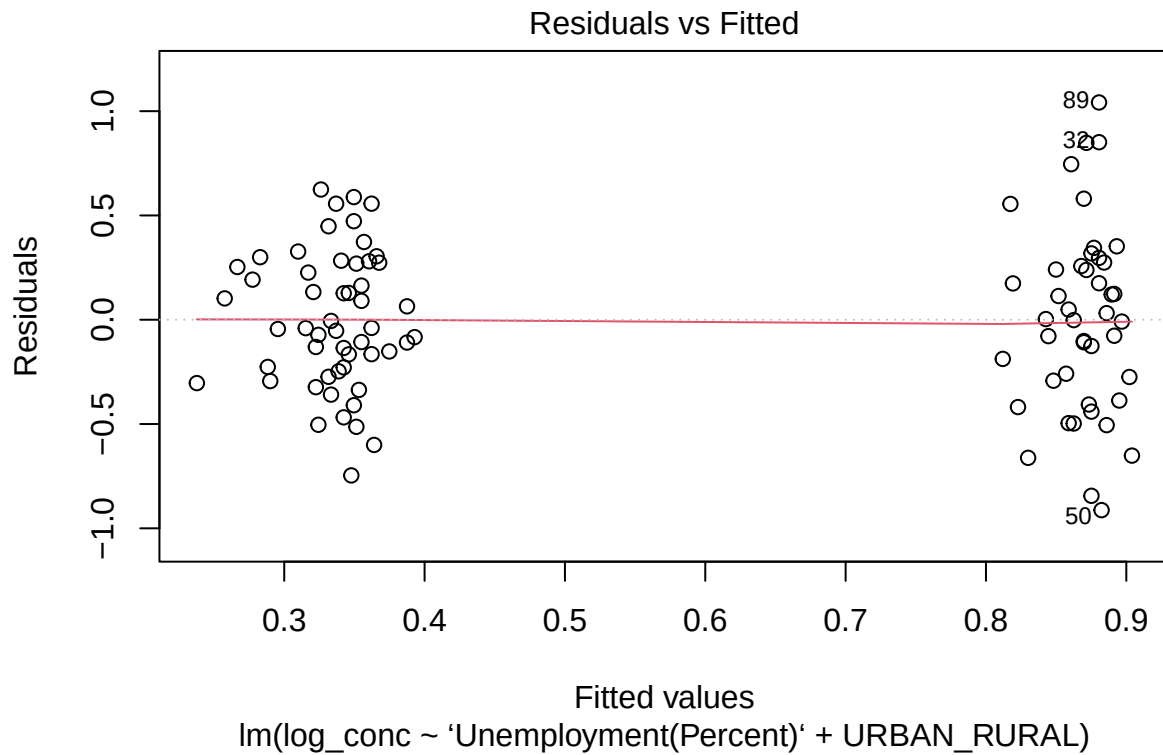
```
summary(B_model_log3)
```

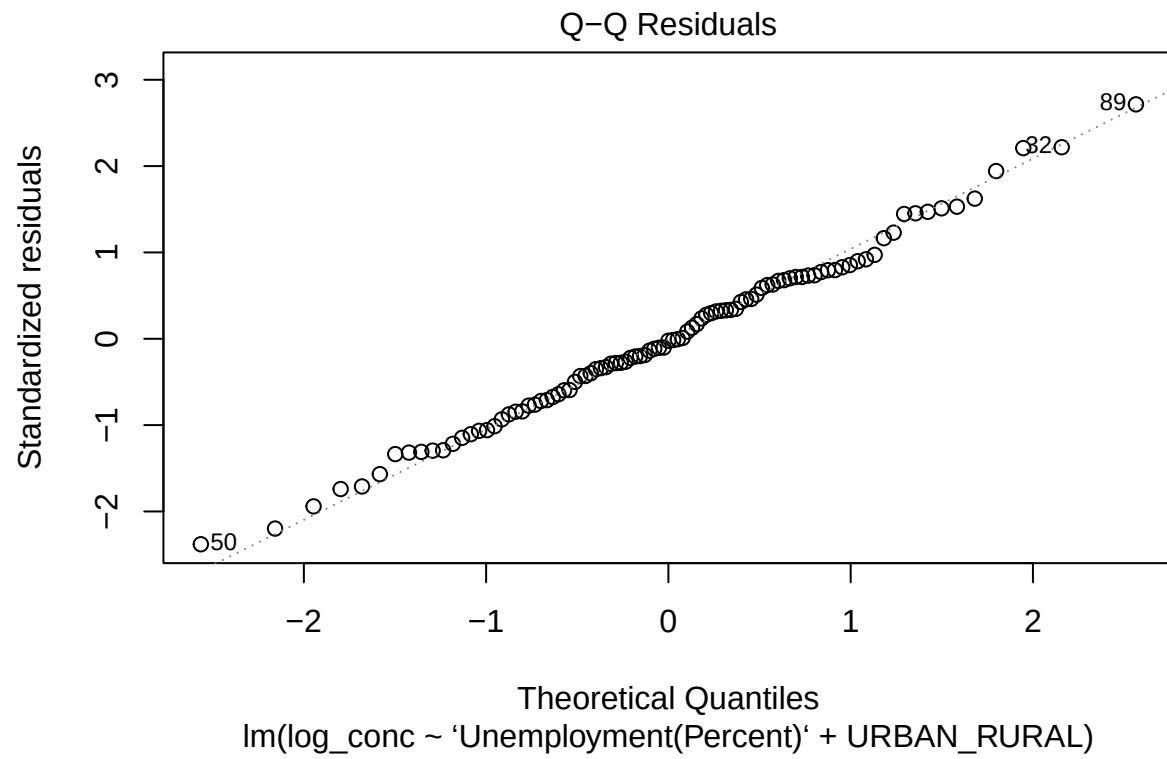
```
##
## Call:
## lm(formula = log_conc ~ 'Unemployment(Percent)' + URBAN_RURAL,
##     data = log_nooutliers)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.91273 -0.27352 -0.00877  0.26900  1.04170
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.43981    0.15574   2.824  0.00579 **
## 'Unemployment(Percent)' -0.01804    0.02533  -0.712  0.47797
```

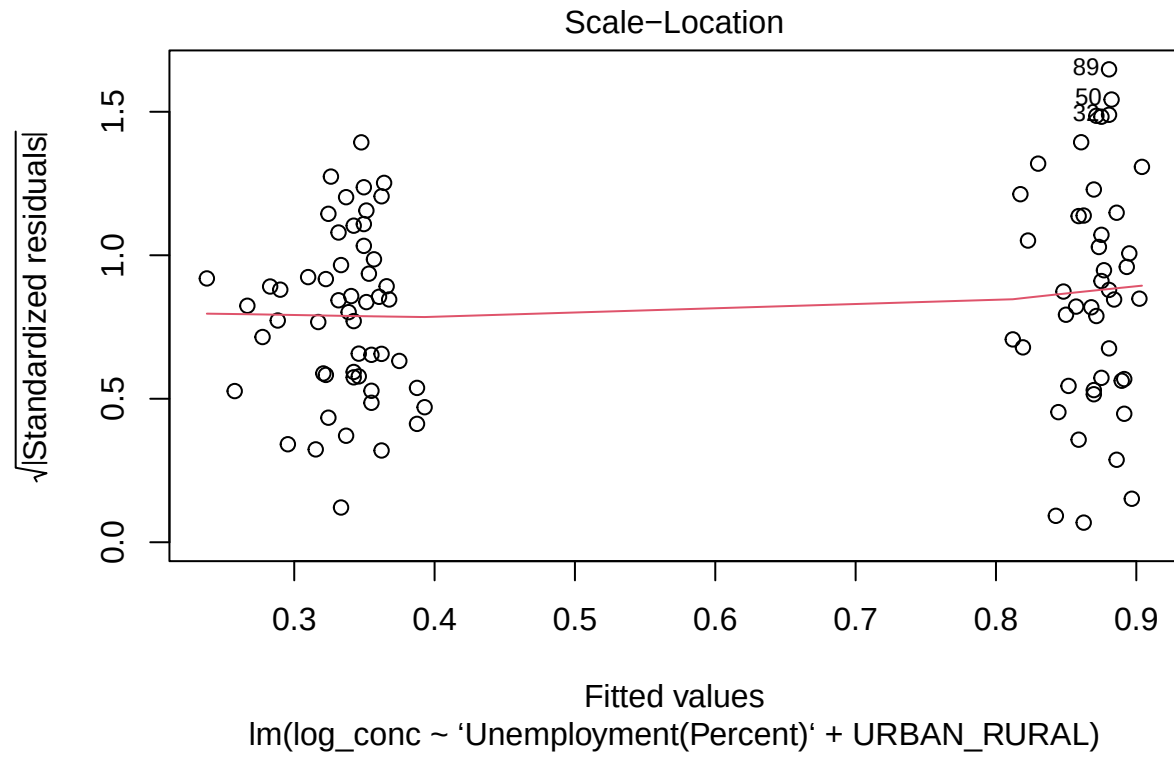


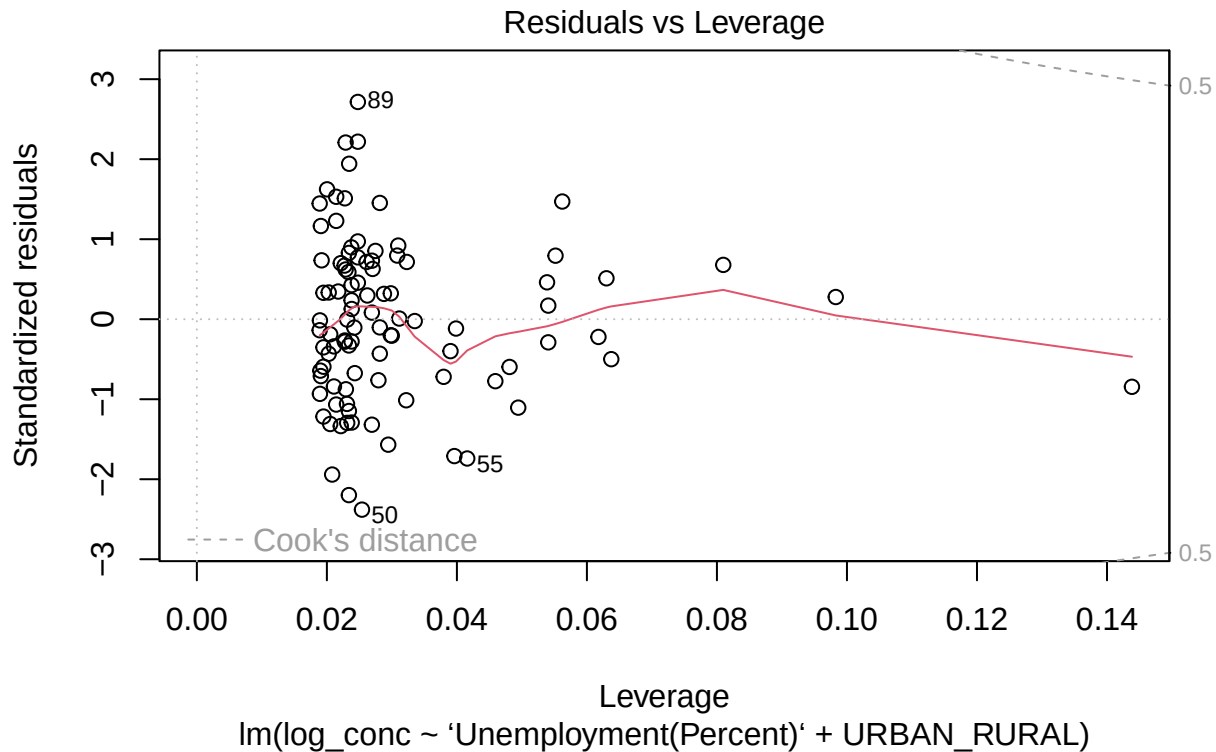
```
## URBAN_RURALUrban      0.51653    0.08233    6.274 1.07e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3885 on 94 degrees of freedom
## Multiple R-squared:  0.327, Adjusted R-squared:  0.3127
## F-statistic: 22.84 on 2 and 94 DF,  p-value: 8.251e-09
```

```
plot(B_model_log3)
```









```
AIC(B_model_log3)
```

```
## [1] 96.79921
```

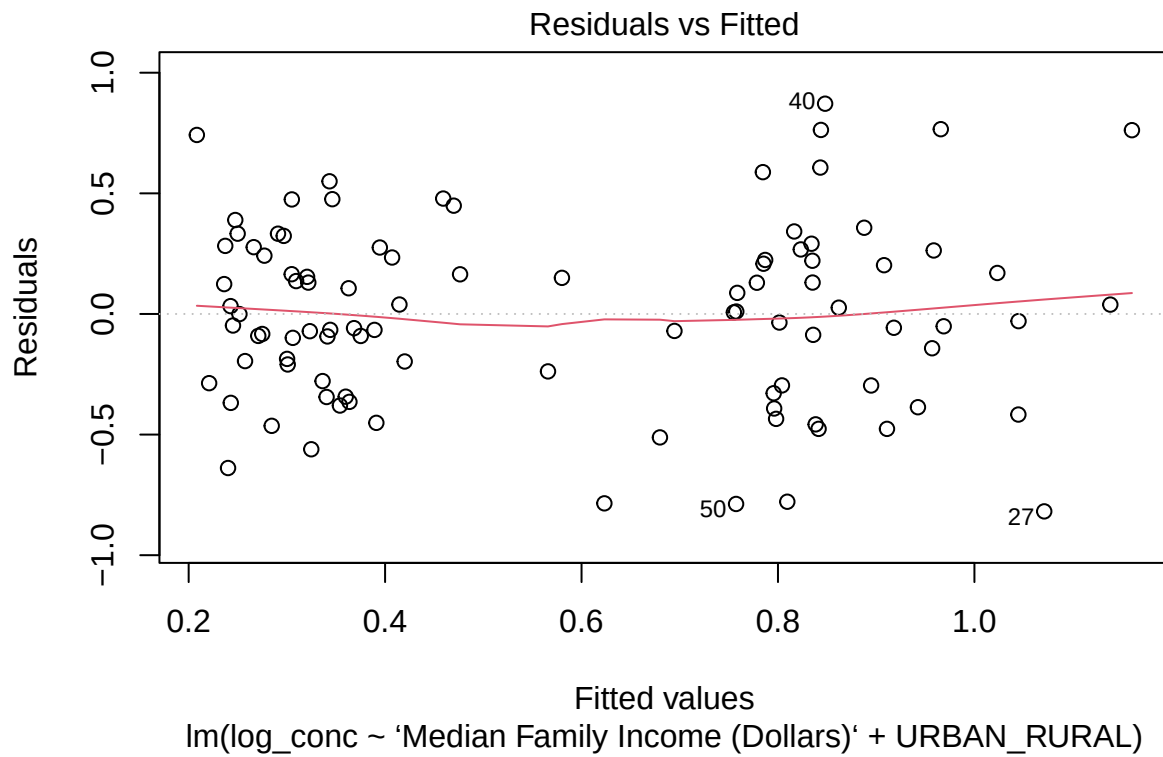
```
B_model_log4 <- lm(log_conc ~ `Median Family Income (Dollars)` + URBAN_RURAL,
  log_nooutliers)
```

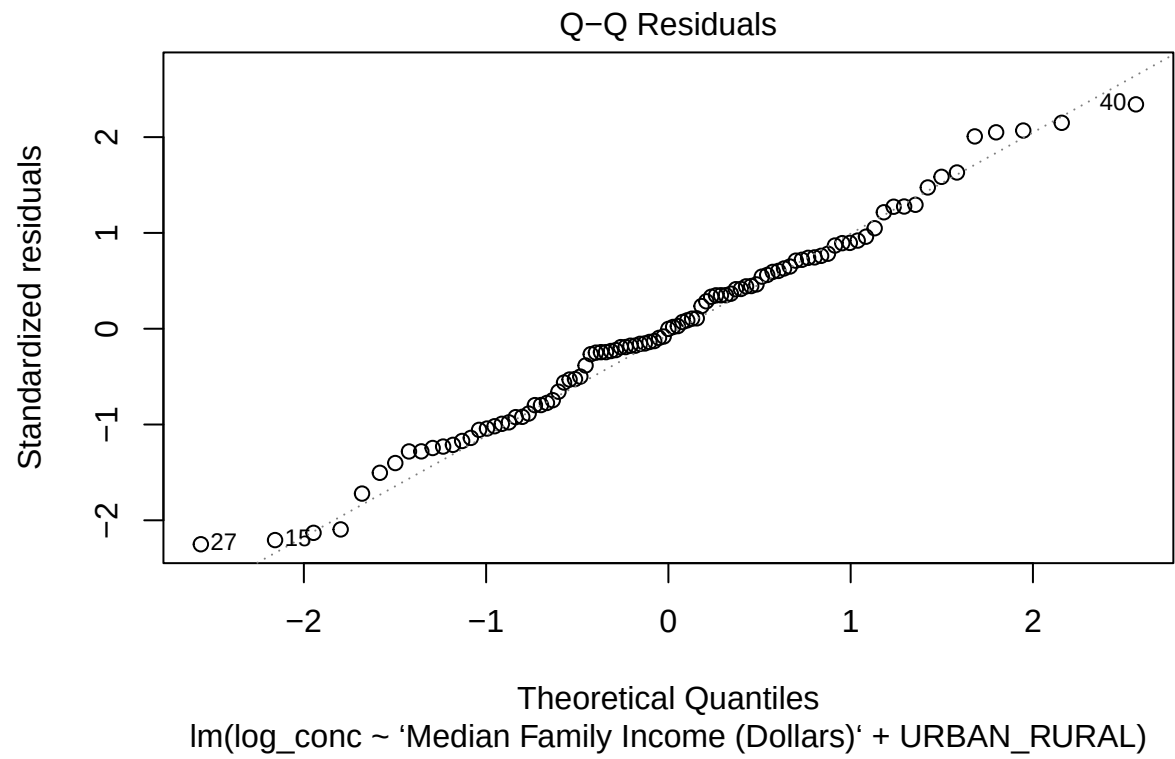
```
summary(B_model_log4)
```

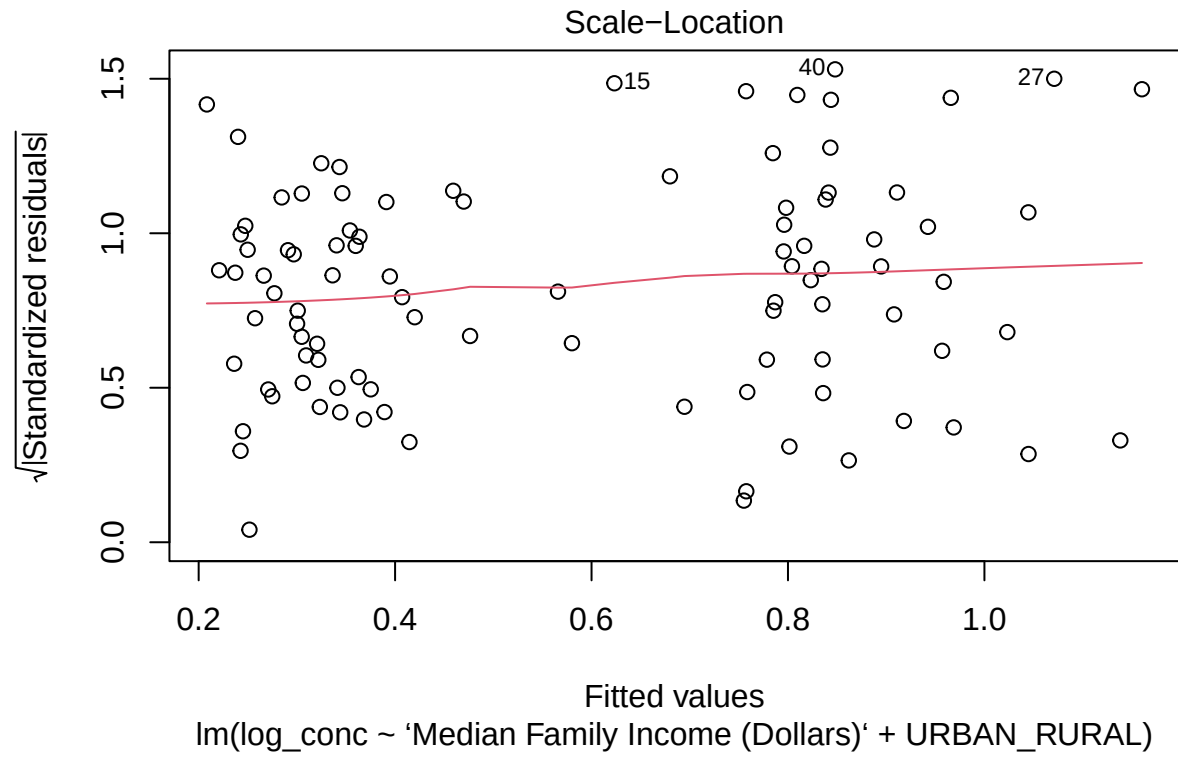
```
##
## Call:
## lm(formula = log_conc ~ `Median Family Income (Dollars)` + URBAN_RURAL,
##     data = log_nooutliers)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.81834 -0.28684 -0.00061  0.24163  0.87157
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.441e-01  1.933e-01  -0.746   0.4578
## `Median Family Income (Dollars)`  8.742e-06  3.395e-06   2.575   0.0116 *
## URBAN_RURALUrban  4.149e-01  8.933e-02   4.644  1.11e-05 ***
## ---
```

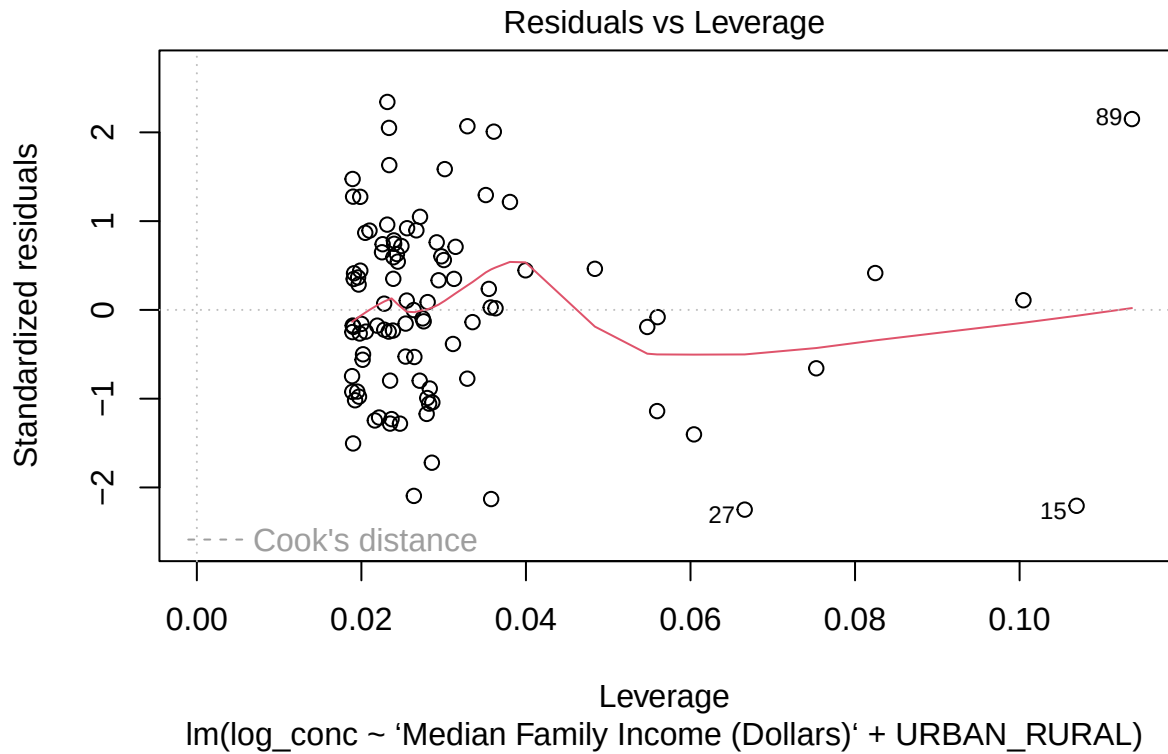
```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3765 on 94 degrees of freedom
## Multiple R-squared:  0.368, Adjusted R-squared:  0.3545
## F-statistic: 27.36 on 2 and 94 DF,  p-value: 4.315e-10
```

```
plot(B_model_log4)
```







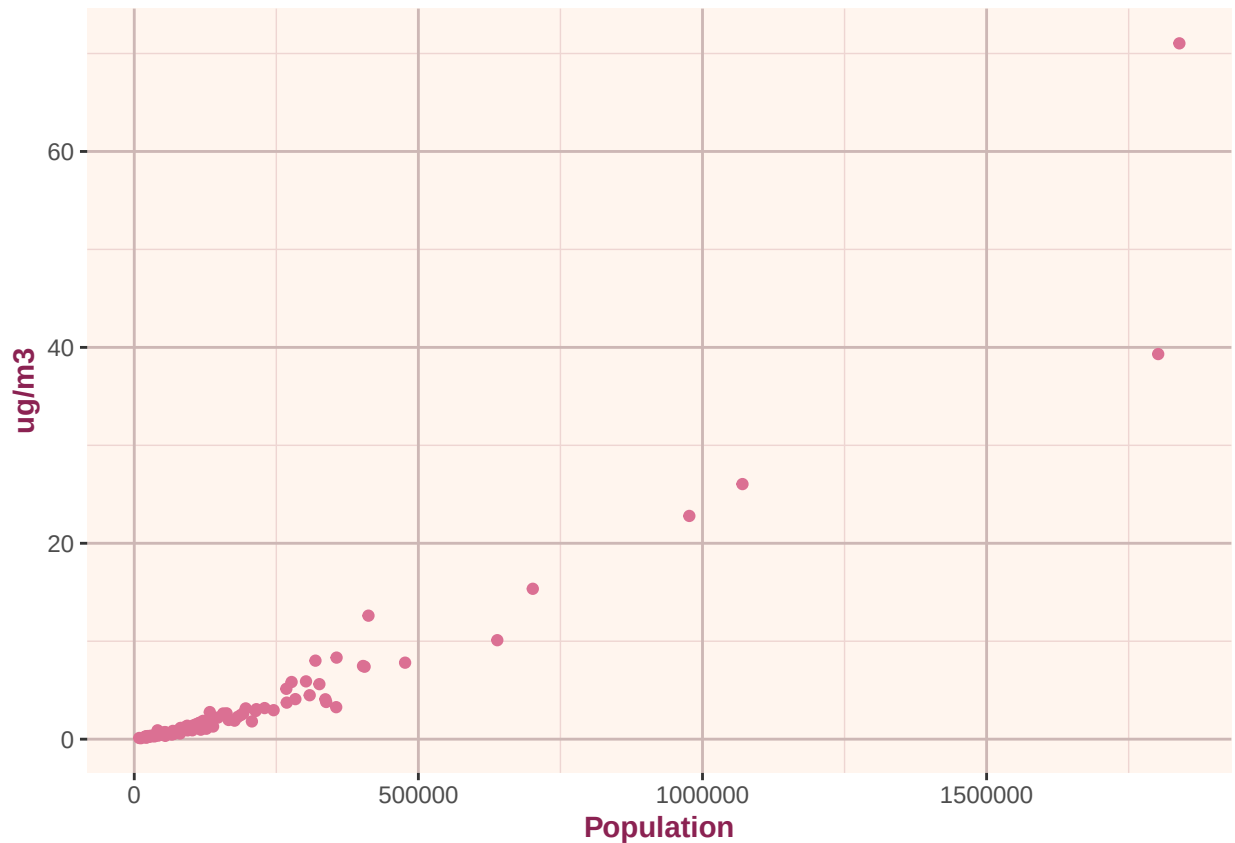


```
AIC(B_model_log4)
```

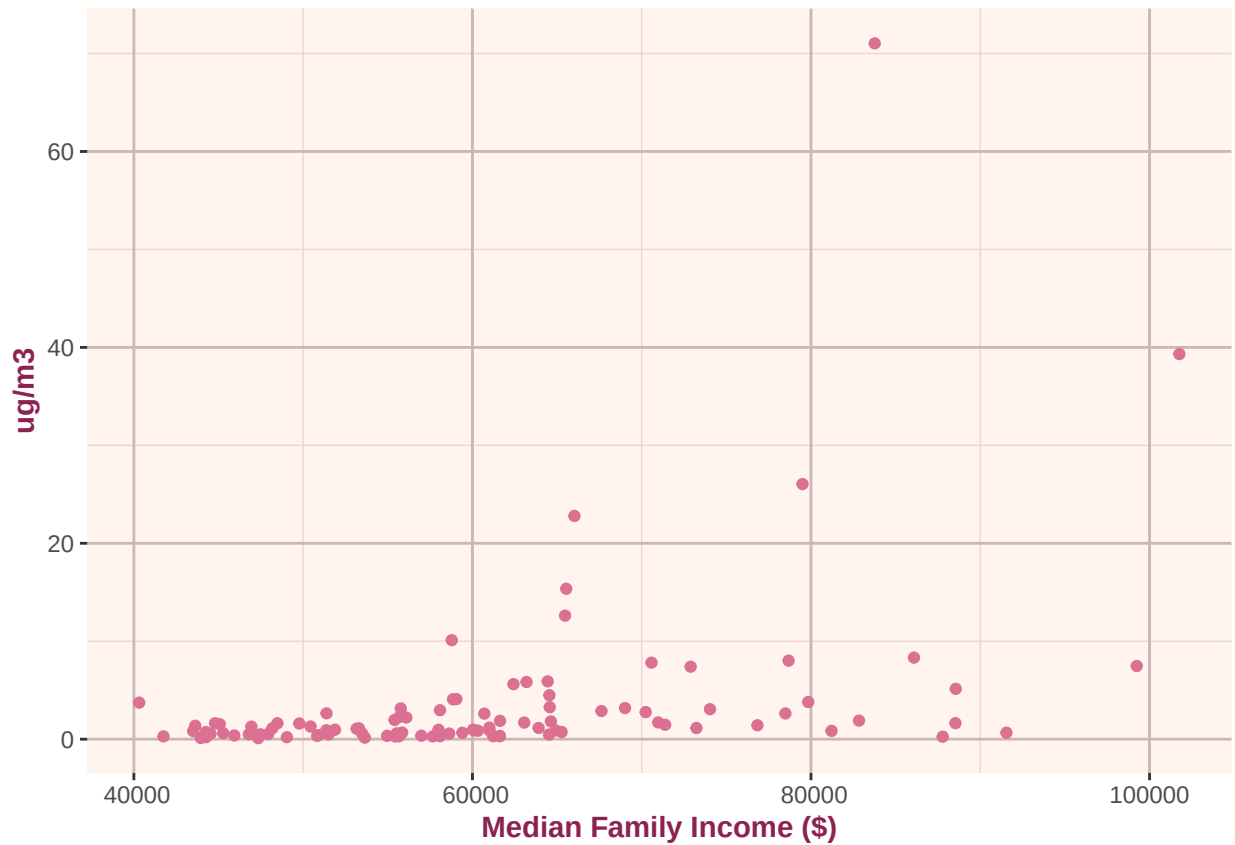
```
## [1] 90.70941
```

Diesel Model

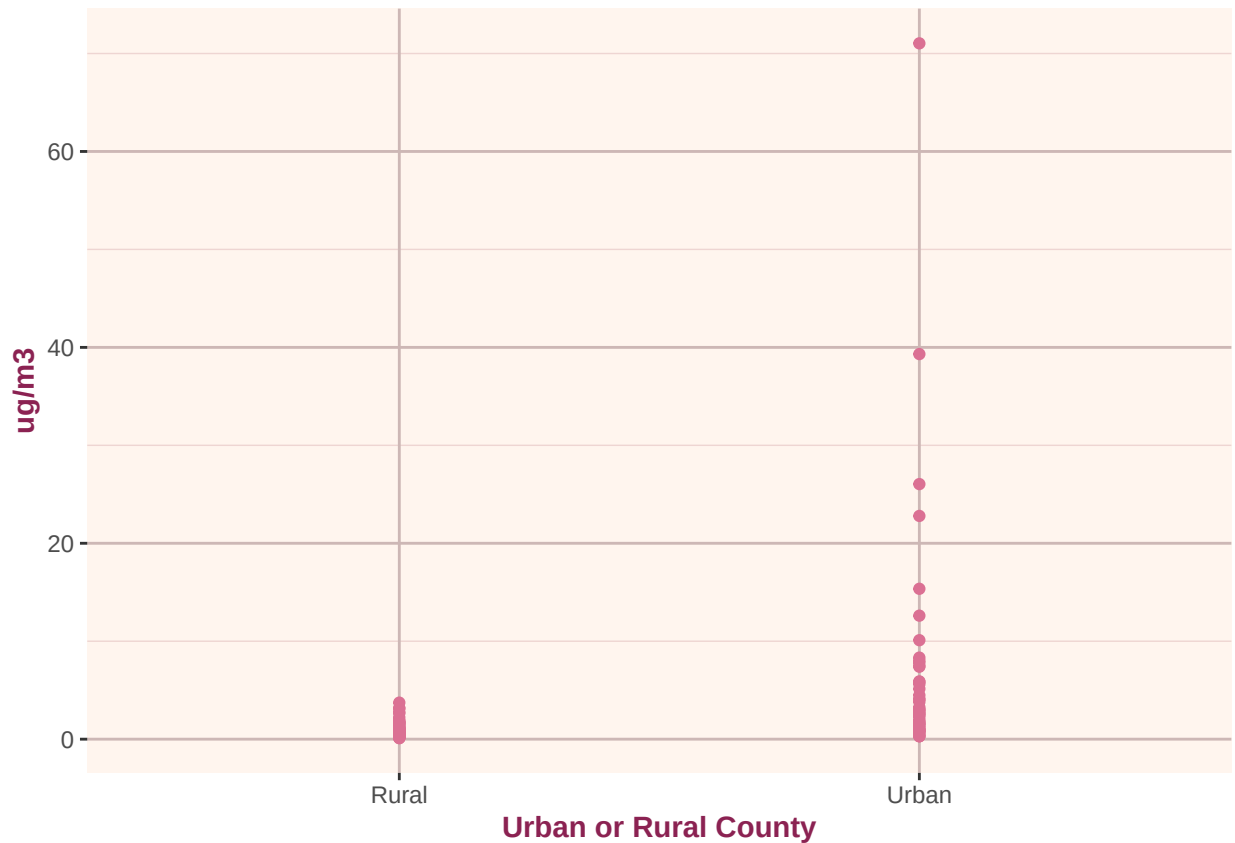
```
(scatter_populationD <- ggplot(diesel_complete,
                              aes(x = Population, y = Total.Exposure.Conc)) +
  geom_point(color = "palevioletred") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold",
                                  color = "violetred4"),
        axis.title.x = element_text(face = "bold", color = "violetred4"),
        axis.title.y = element_text(face = "bold", color = "violetred4"),
        panel.background = element_rect(fill = "seashell"),
        panel.grid.major = element_line(color = "mistyrose3"),
        panel.grid.minor = element_line(color = "mistyrose2")) +
  labs(x = "Population",
       y = "ug/m3"))
```

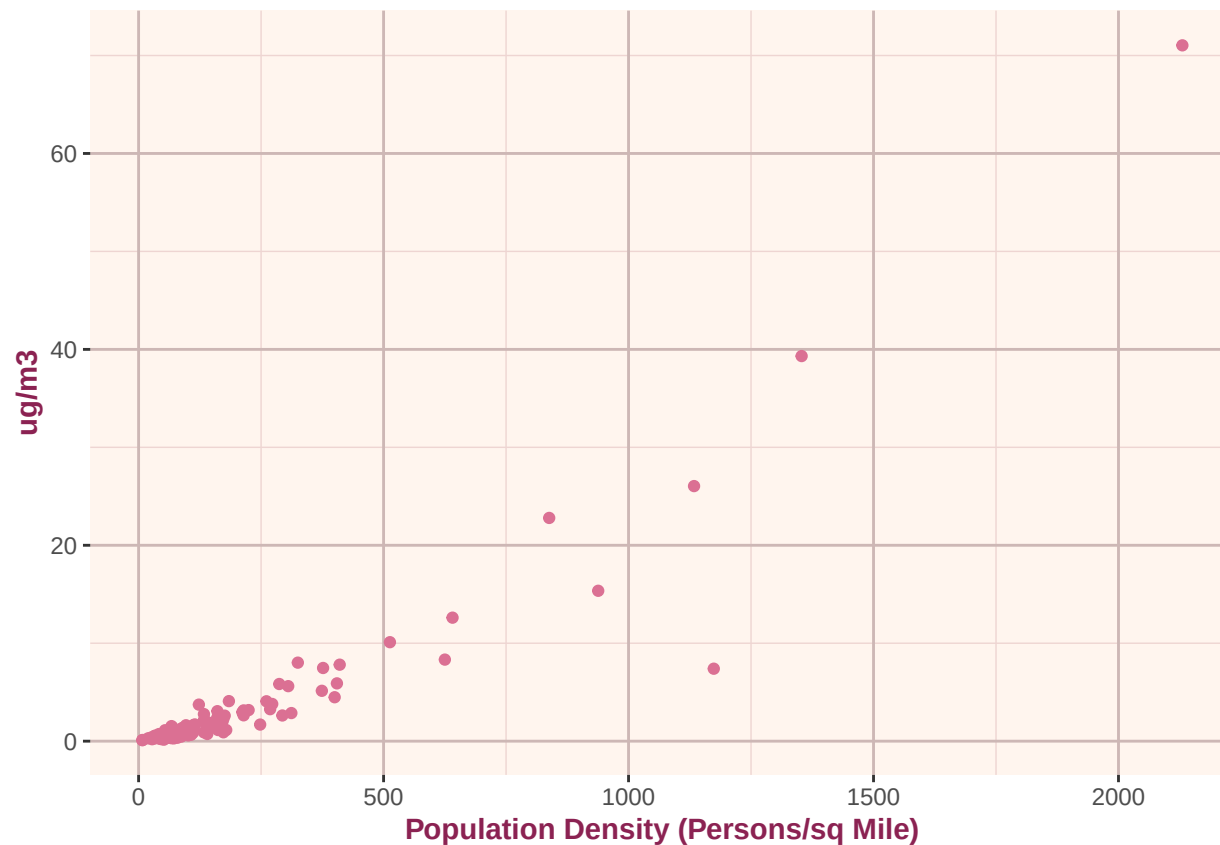
```
(scatter_incomeD <- ggplot(diesel_complete,
  aes(x = `Median Family Income (Dollars)`,
    y = Total.Exposure.Conc)) +
  geom_point(color = "palevioletred") +
  scale_x_continuous(labels = function(x) format(x, scientific = FALSE)) +
  theme(plot.title = element_text(hjust = 0.5, face = "bold",
    color = "violetred4"),
    axis.title.x = element_text(face = "bold", color = "violetred4"),
    axis.title.y = element_text(face = "bold", color = "violetred4"),
    panel.background = element_rect(fill = "seashell"),
    panel.grid.major = element_line(color = "mistyrose3"),
    panel.grid.minor = element_line(color = "mistyrose2")) +
  labs(x = "Median Family Income ($)",
    y = "ug/m3"))
```



```
(scatter_urbanD <- ggplot(diesel_complete,
  aes(x = URBAN_RURAL, y = Total.Exposure.Conc)) +
  geom_point(color = "palevioletred") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold",
    color = "violetred4"),
    axis.title.x = element_text(face = "bold", color = "violetred4"),
    axis.title.y = element_text(face = "bold", color = "violetred4"),
    panel.background = element_rect(fill = "seashell"),
    panel.grid.major = element_line(color = "mistyrose3"),
    panel.grid.minor = element_line(color = "mistyrose2")) +
  labs(x = "Urban or Rural County",
    y = "ug/m3"))
```



```
(scatter_densityD <- ggplot(diesel_complete,
  aes(x = `Population Density (Persons Per SqMile)`,
    y = Total.Exposure.Conc)) +
  geom_point(color = "palevioletred") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold",
    color = "violetred4"),
    axis.title.x = element_text(face = "bold", color = "violetred4"),
    axis.title.y = element_text(face = "bold", color = "violetred4"),
    panel.background = element_rect(fill = "seashell"),
    panel.grid.major = element_line(color = "mistyrose3"),
    panel.grid.minor = element_line(color = "mistyrose2")) +
  labs(x = "Population Density (Persons/sq Mile)",
    y = "ug/m3"))
```

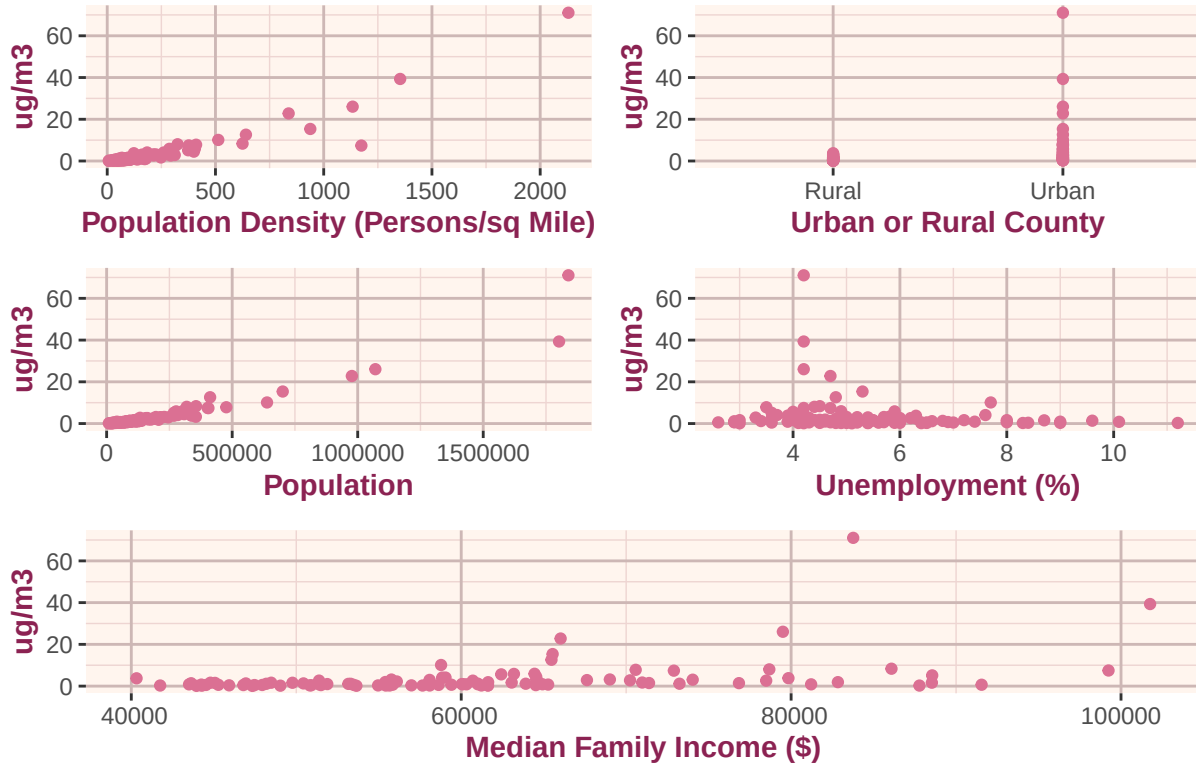


```
(scatter_unemploymentD <- ggplot(diesel_complete,
                                aes(x = `Unemployment(Percent)`,
                                    y = Total.Exposure.Conc)) +
  geom_point(color = "palevioletred") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold",
                                  color = "violetred4"),
        axis.title.x = element_text(face = "bold", color = "violetred4"),
        axis.title.y = element_text(face = "bold", color = "violetred4"),
        panel.background = element_rect(fill = "seashell"),
        panel.grid.major = element_line(color = "mistyrose3"),
        panel.grid.minor = element_line(color = "mistyrose2")) +
  labs(x = "Unemployment (%)",
       y = "ug/m3"))
```



```
final_plot_diesel <- ((scatter_densityD + scatter_urbanD) /
  (scatter_populationD + scatter_unemploymentD) / scatter_incomeD) +
  plot_annotation(title = "Diesel Concentrations vs. County Demographics",
    theme = theme(
      plot.title = element_text(hjust = 0.5, face = "bold",
        color = "violetred4")))
plot(final_plot_diesel)
```

Diesel Concentrations vs. County Demographics



```
# Perform correlation tests to test for collinearity
```

```
cor.test(diesel_complete$Population,
         diesel_complete$`Median Family Income (Dollars)`)
```

```
##
## Pearson's product-moment correlation
##
## data: diesel_complete$Population and diesel_complete$`Median Family Income (Dollars)`
## t = 5.4197, df = 98, p-value = 4.281e-07
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.3133539 0.6183125
## sample estimates:
##      cor
## 0.4802155
```

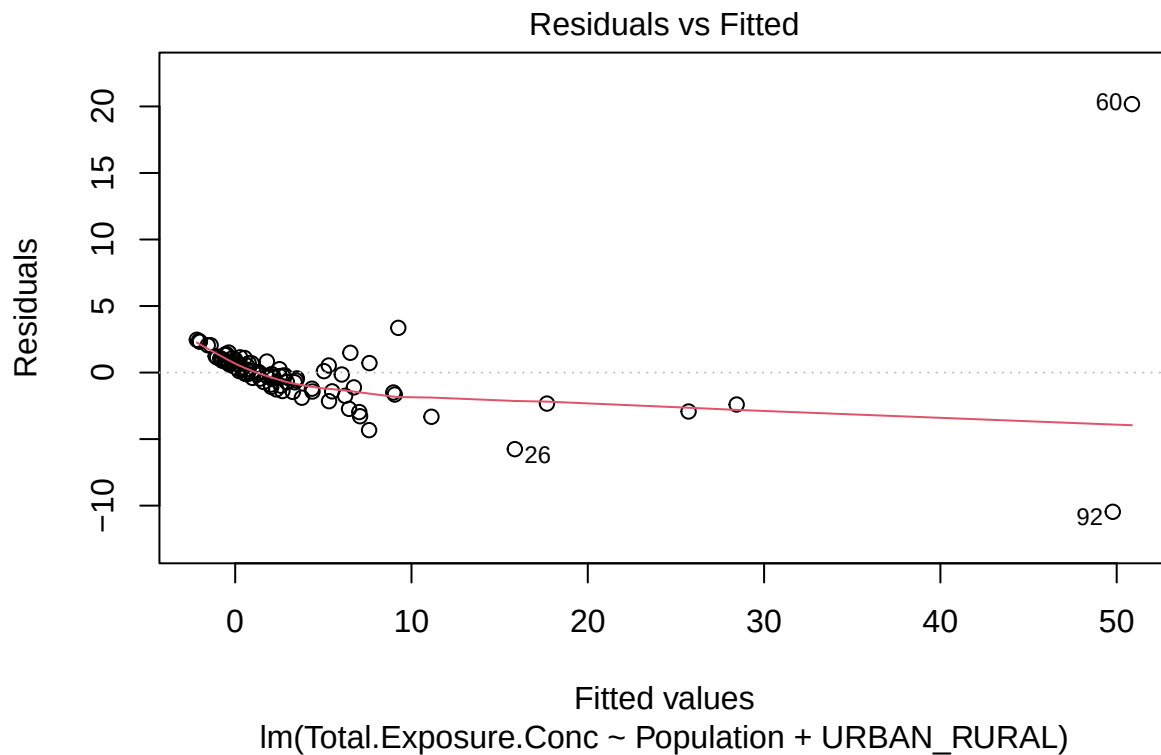
```
# Generate regression models
```

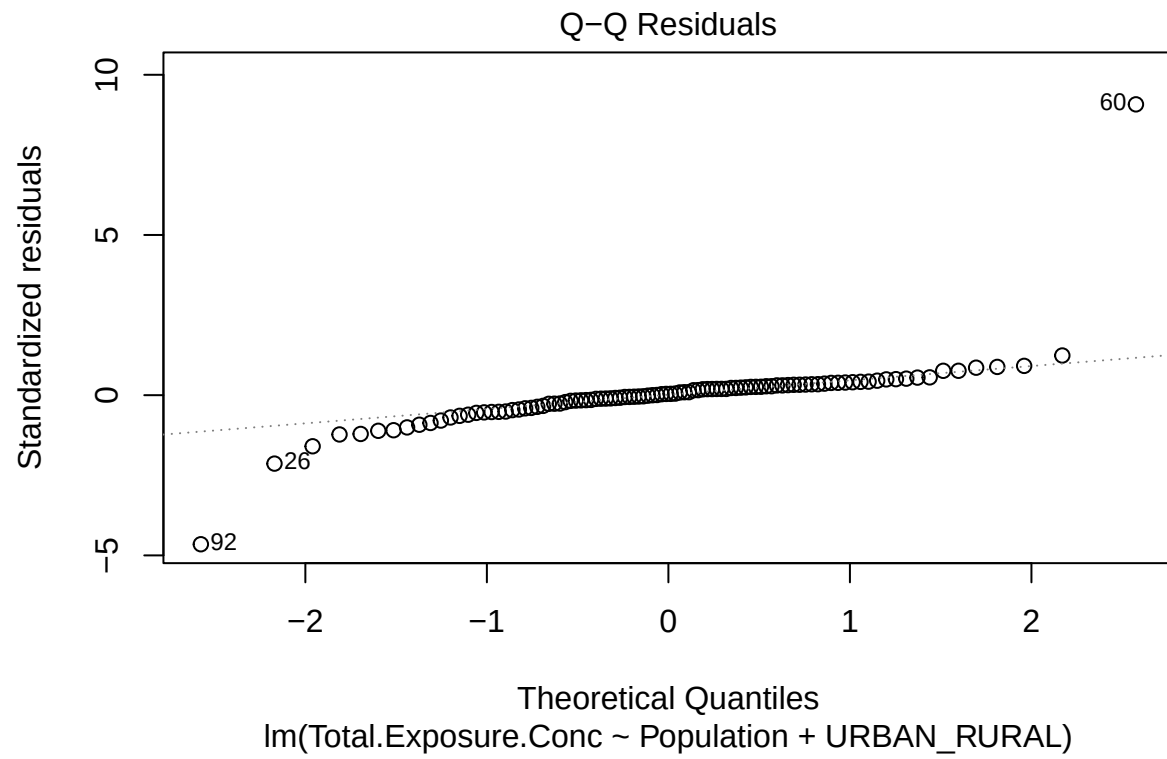
```
D_model1 <- lm(Total.Exposure.Conc ~ Population + URBAN_RURAL,
               diesel_complete)
summary(D_model1)
```

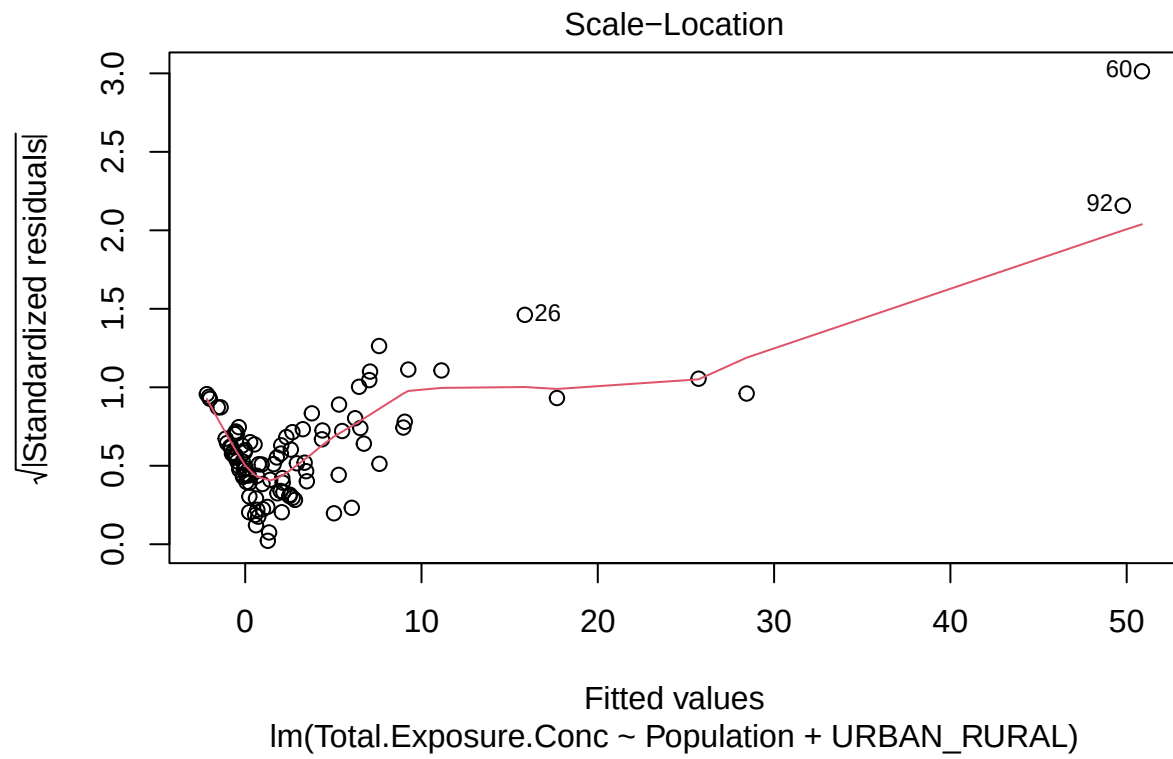
```
##
## Call:
## lm(formula = Total.Exposure.Conc ~ Population + URBAN_RURAL,
```

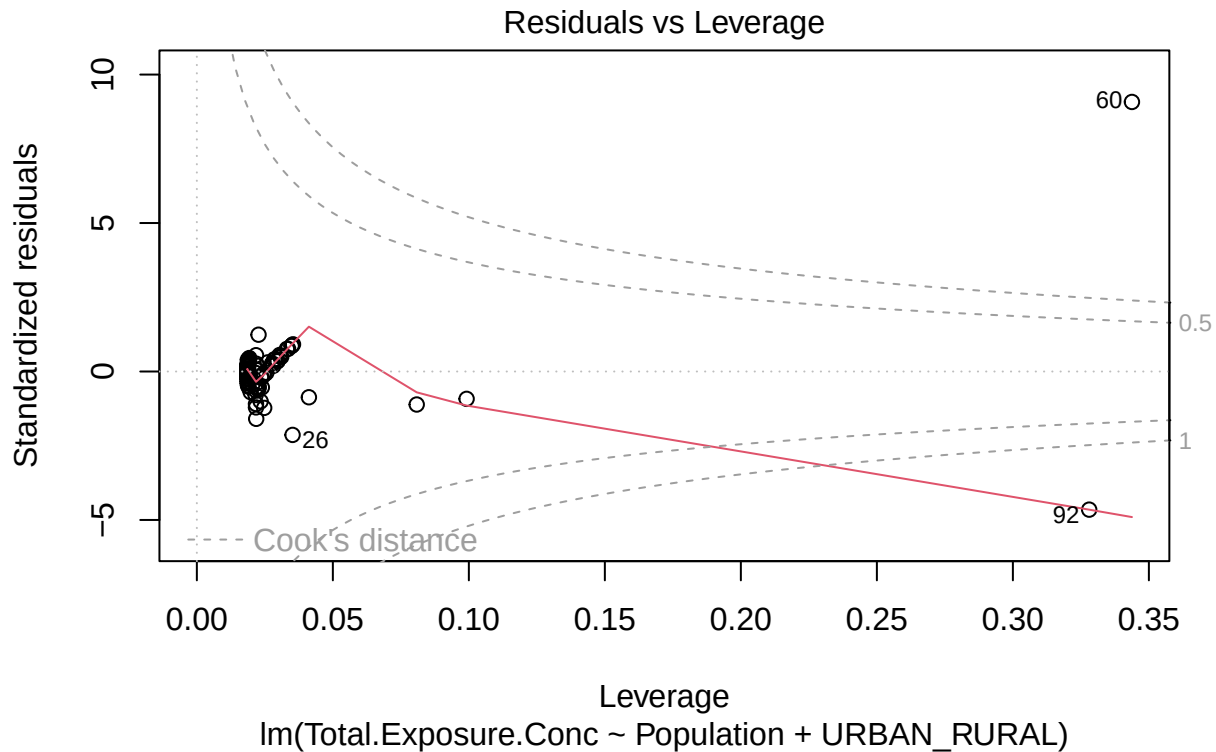
```
##      data = diesel_complete)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.4621  -0.7770   0.1092   0.8616  20.1732
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.365e+00  3.827e-01  -3.567 0.000564 ***
## Population      2.916e-05  1.032e-06  28.259 < 2e-16 ***
## URBAN_RURALUrban -1.402e+00  6.075e-01  -2.308 0.023098 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.743 on 97 degrees of freedom
## Multiple R-squared:  0.9038, Adjusted R-squared:  0.9019
## F-statistic: 455.9 on 2 and 97 DF,  p-value: < 2.2e-16
```

```
plot(D_model1)
```









```
AIC(D_model1) #490.5862
```

```
## [1] 490.5862
```

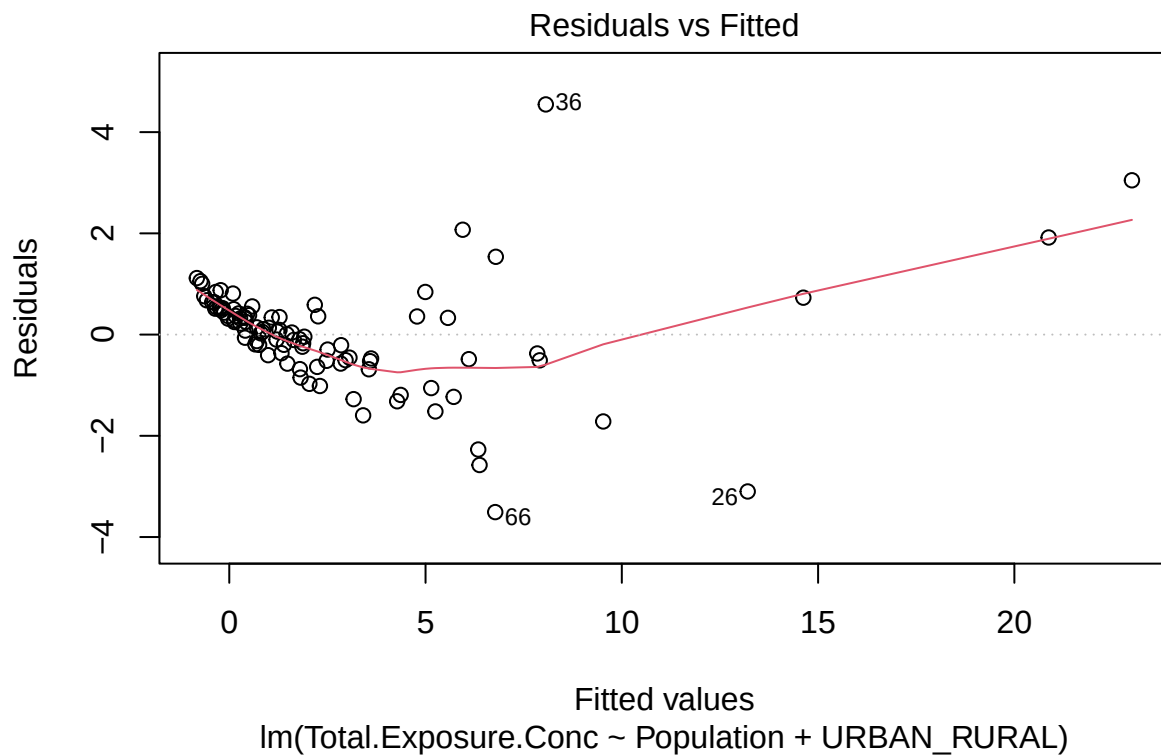
```
diesel_nooutlier <- diesel_complete[-c(60,92),]
```

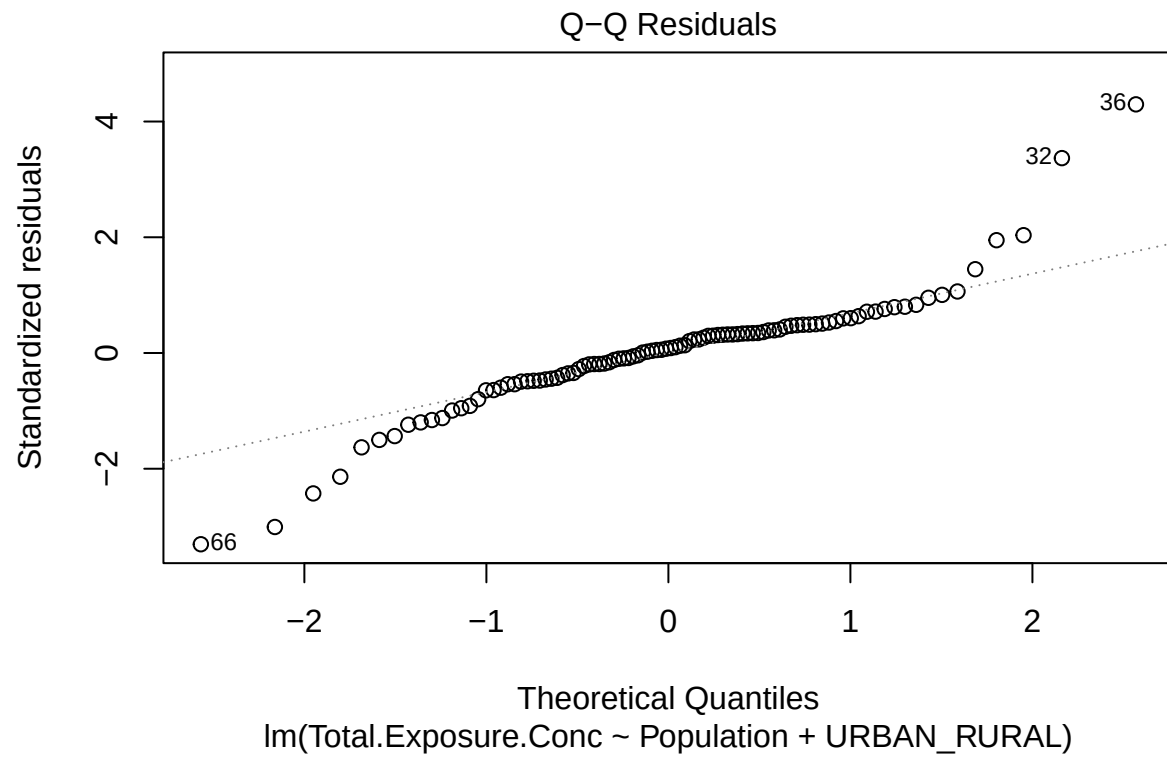
```
D_model1_nooutlier <- lm(Total.Exposure.Conc ~ Population + URBAN_RURAL,
  diesel_nooutlier)
summary(D_model1_nooutlier)
```

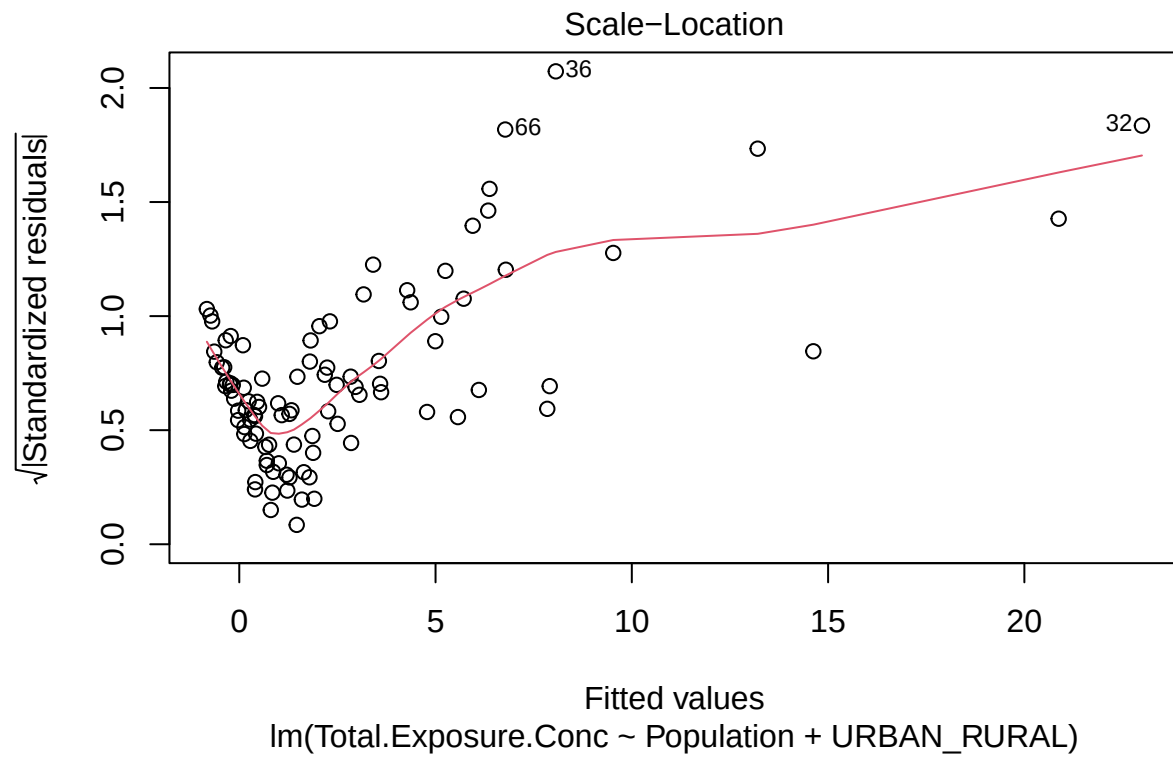
```
##
## Call:
## lm(formula = Total.Exposure.Conc ~ Population + URBAN_RURAL,
##     data = diesel_nooutlier)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.5074 -0.4824  0.0852  0.4966  4.5471
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -8.371e-01  1.568e-01  -5.338 6.38e-07 ***
## Population      2.269e-05  6.906e-07  32.848 < 2e-16 ***
## URBAN_RURALUrban -4.497e-01  2.518e-01  -1.786  0.0772 .
##
```

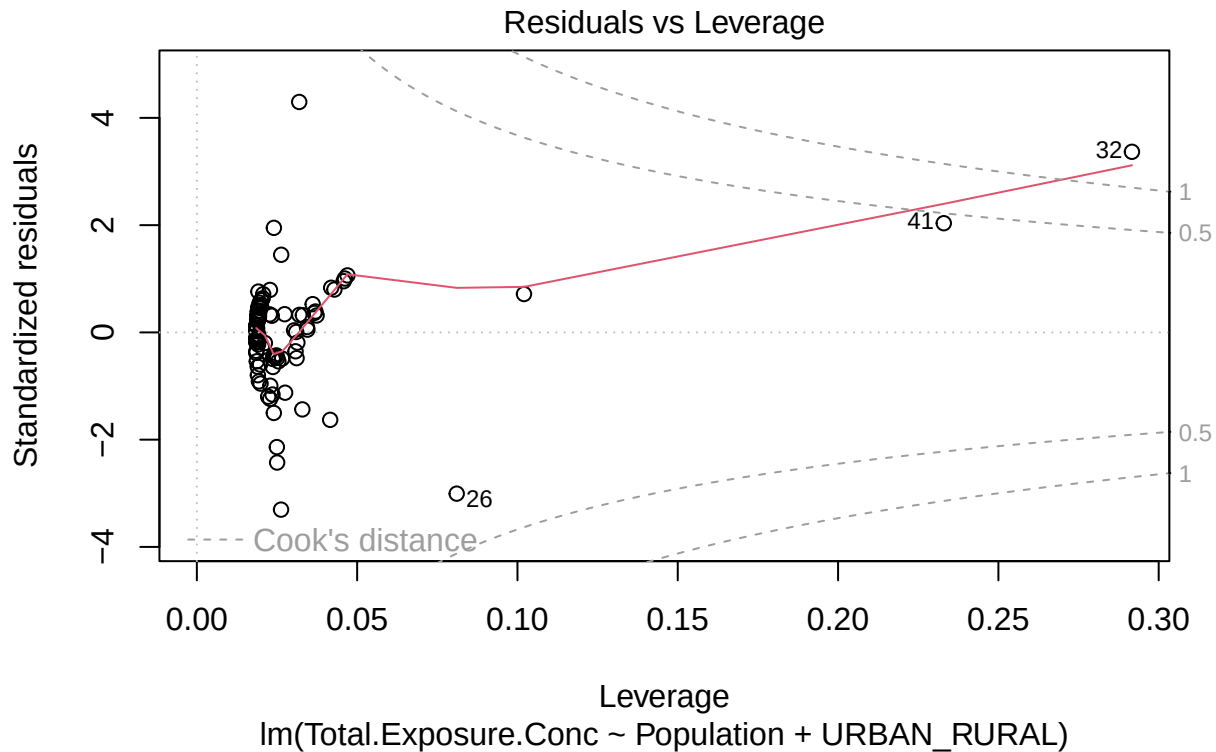
```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.076 on 95 degrees of freedom
## Multiple R-squared:  0.9347, Adjusted R-squared:  0.9333
## F-statistic: 680 on 2 and 95 DF, p-value: < 2.2e-16
```

```
plot(D_model1_nooutlier)
```









```
AIC(D_model1_nooutlier) #297.3357
```

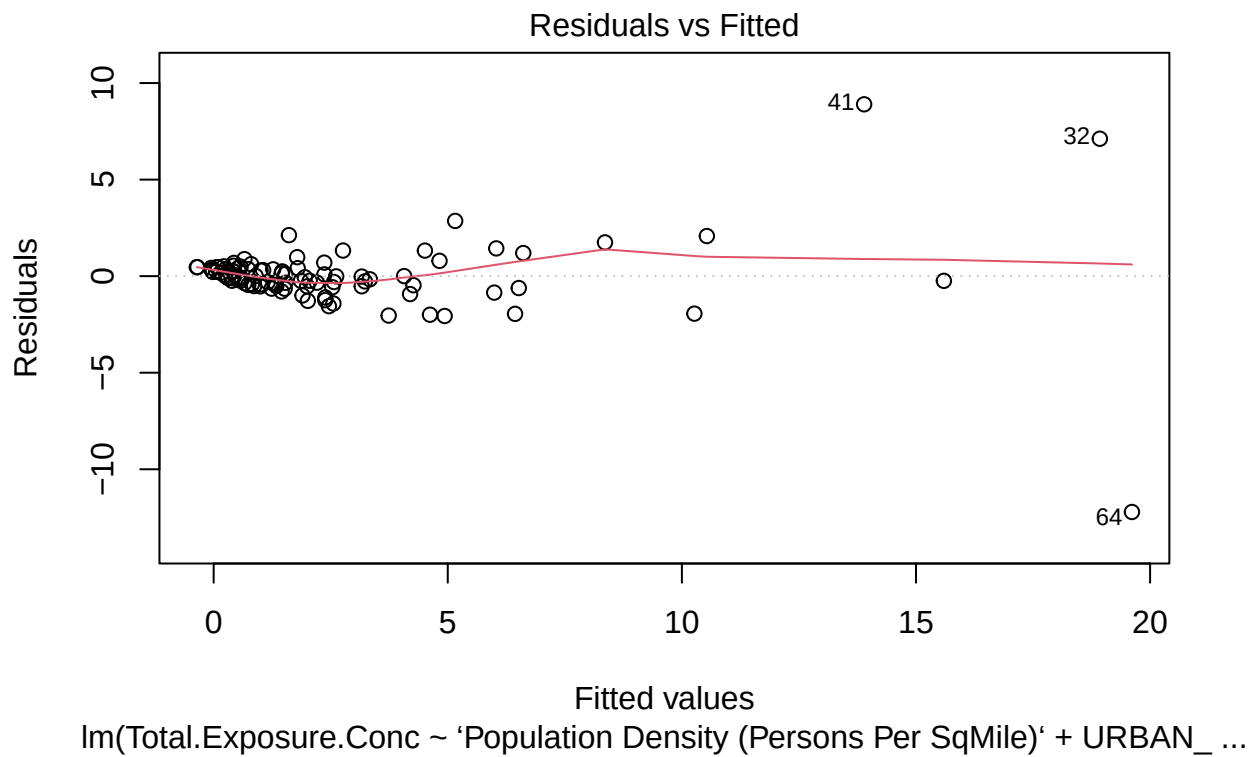
```
## [1] 297.3357
```

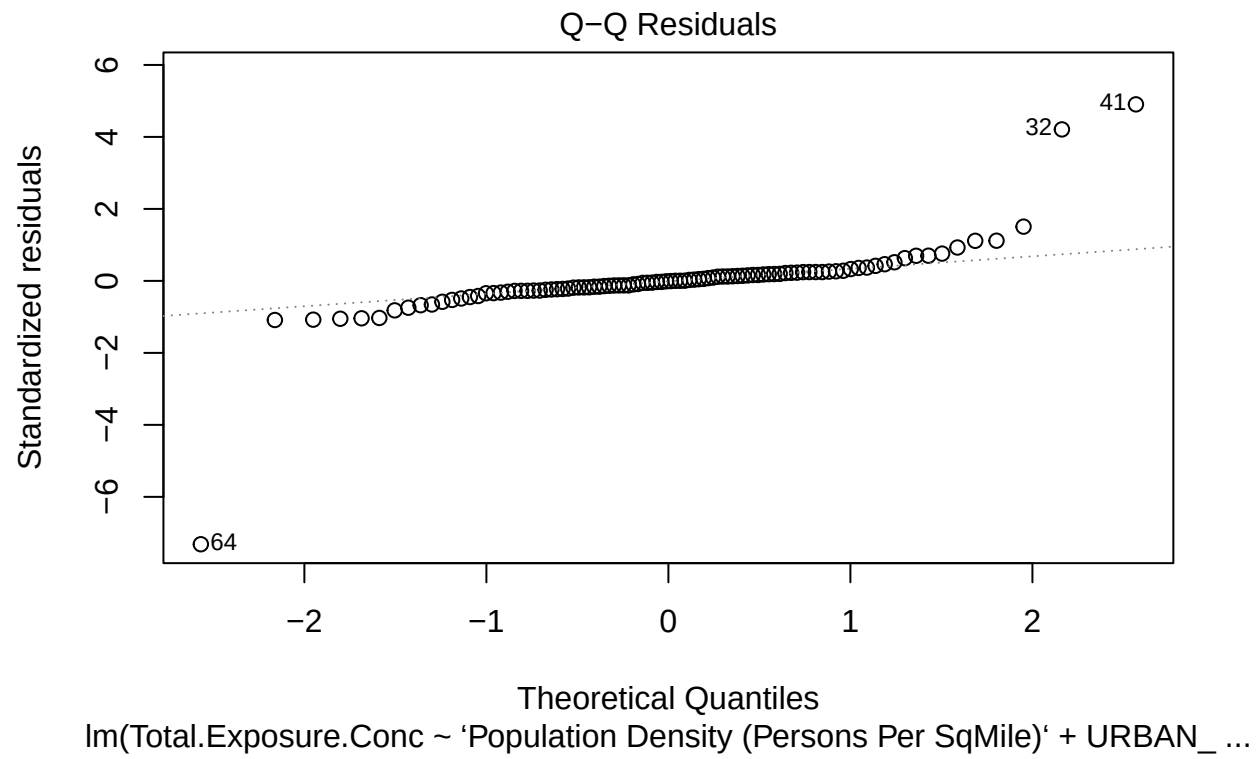
```
D_model2 <- lm(Total.Exposure.Conc ~ `Population Density (Persons Per SqMile)` +
               URBAN_RURAL, diesel_nooutlier)
summary(D_model2)
```

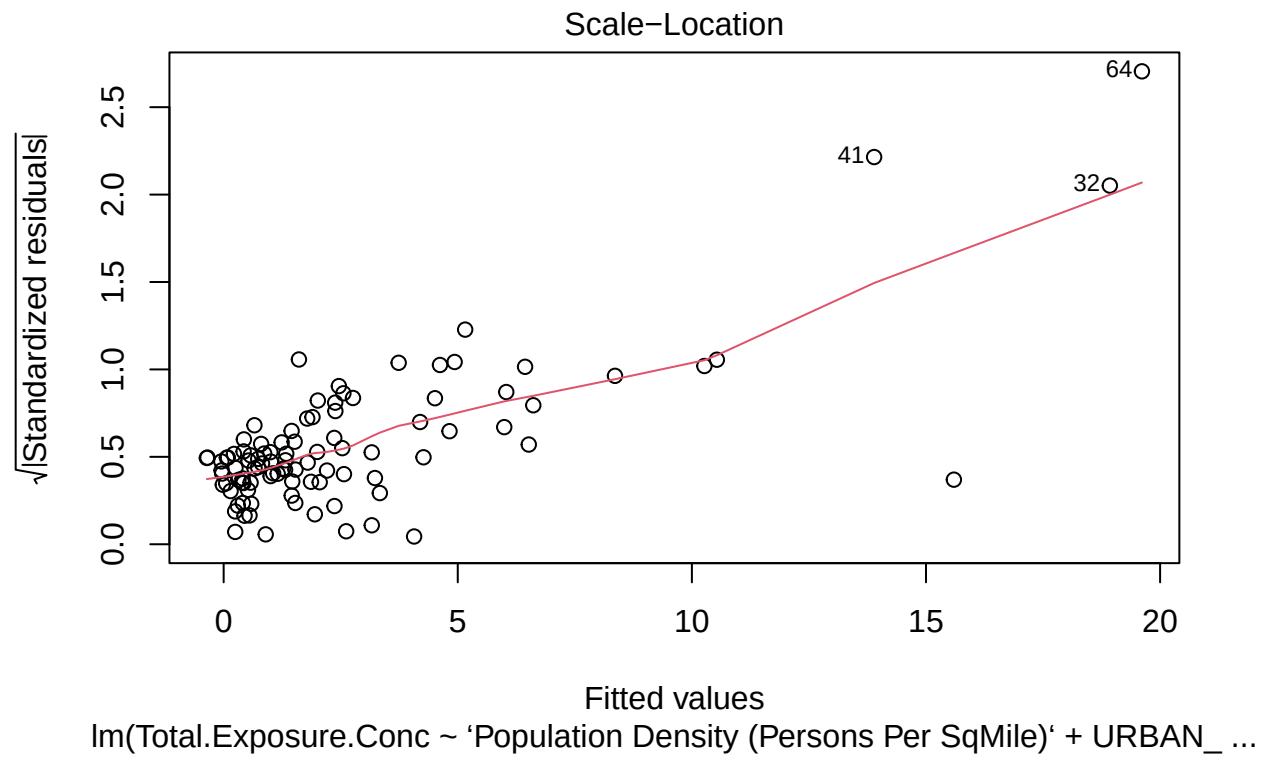
```
##
## Call:
## lm(formula = Total.Exposure.Conc ~ `Population Density (Persons Per SqMile)` +
##    URBAN_RURAL, data = diesel_nooutlier)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.2122  -0.4654  -0.0164   0.4237   8.8986
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -0.48650    0.27654  -1.759   0.0818
## `Population Density (Persons Per SqMile)`  0.01703    0.00103  16.536 <2e-16
## URBAN_RURALUrban    0.11095    0.44512   0.249   0.8037
##
## (Intercept) .
## `Population Density (Persons Per SqMile)` ***
```

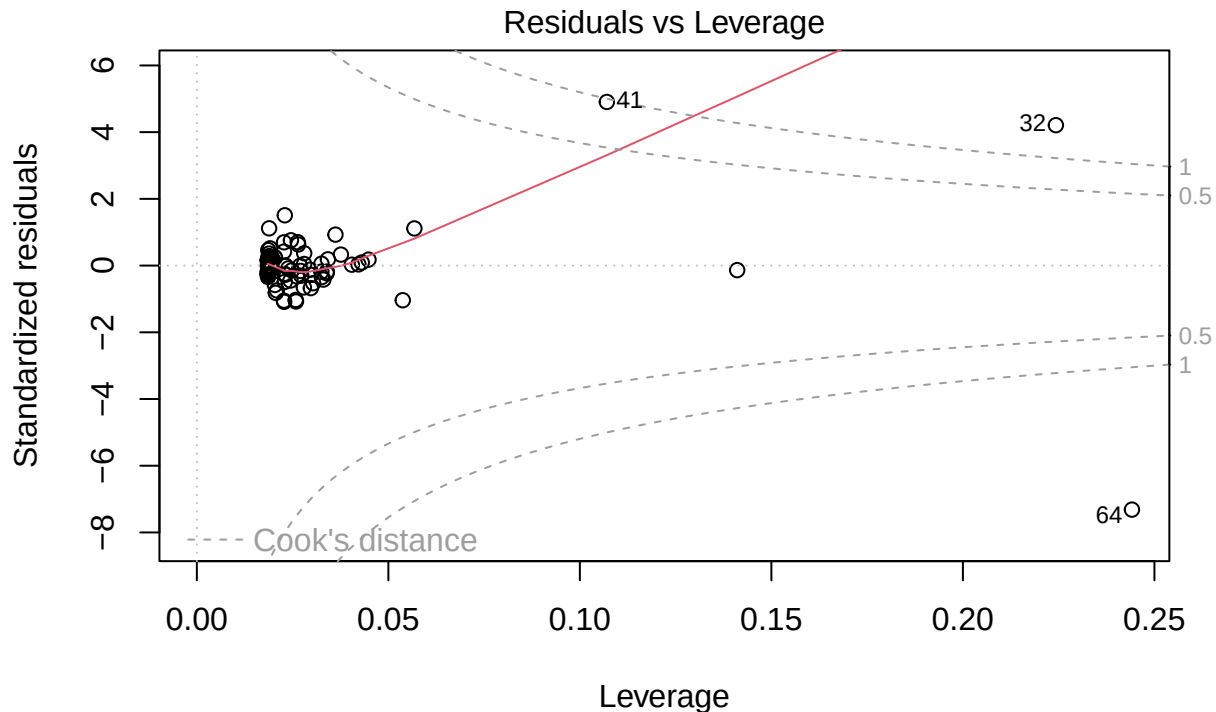
```
## URBAN_RURALUrban
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.92 on 95 degrees of freedom
## Multiple R-squared:  0.792, Adjusted R-squared:  0.7876
## F-statistic: 180.8 on 2 and 95 DF,  p-value: < 2.2e-16
```

```
plot(D_model2)
```









lm(Total.Exposure.Conc ~ 'Population Density (Persons Per SqMile)' + URBAN_ ...

```
AIC(D_model2) #410.9054
```

```
## [1] 410.9054
```

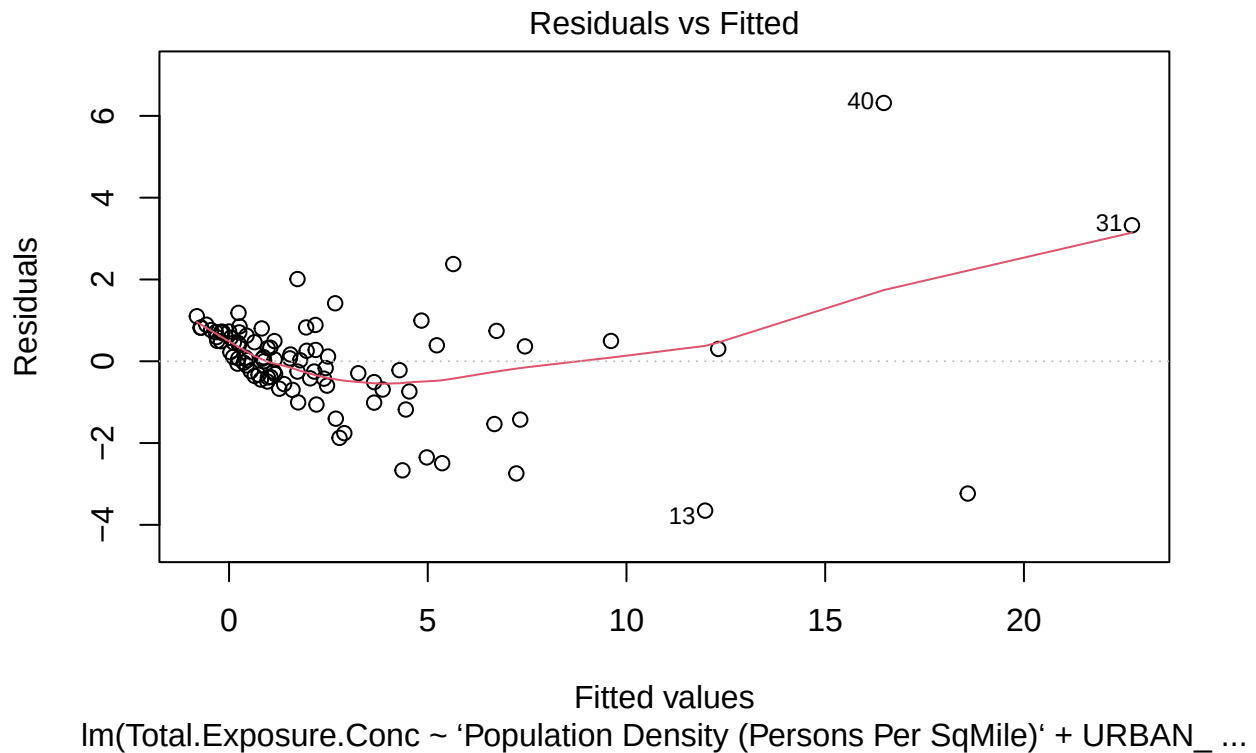
```
D_model2_nooutlier <- diesel_nooutlier[-c(31,42,64),]
```

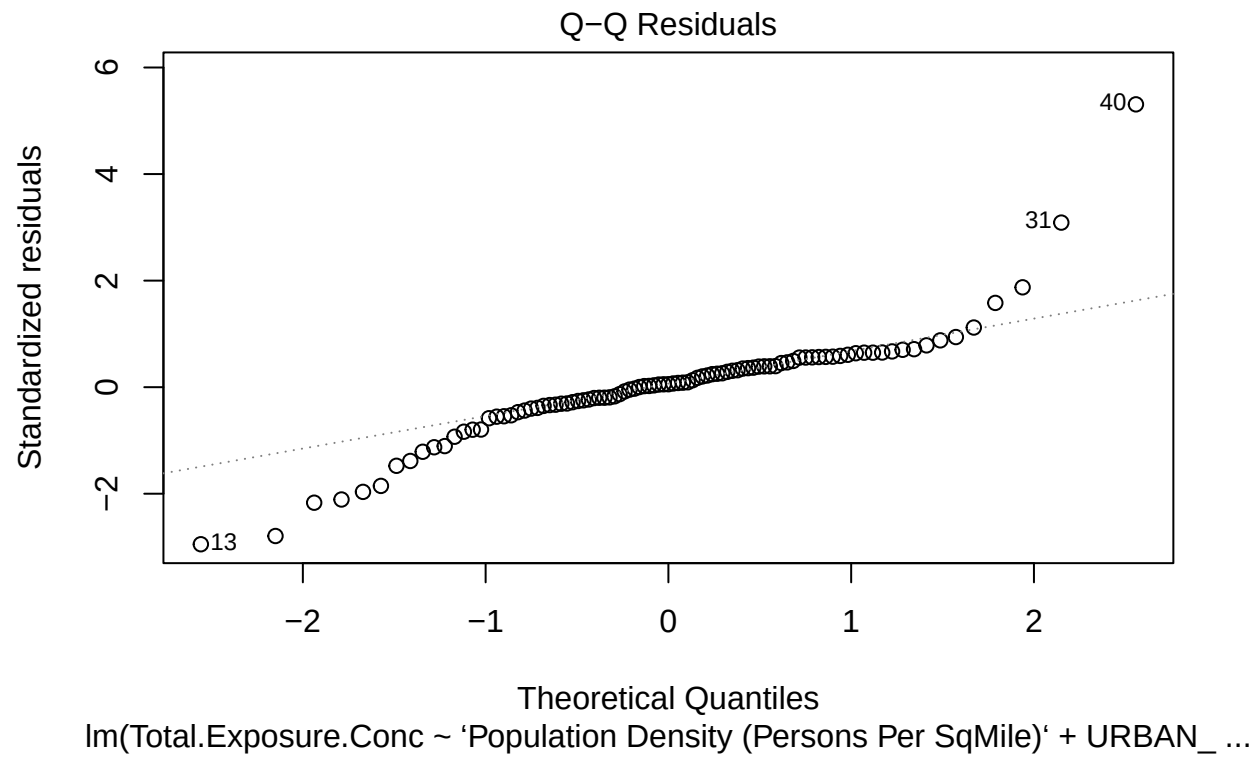
```
D_model3 <- lm(Total.Exposure.Conc ~ 'Population Density (Persons Per SqMile)' +  
               URBAN_RURAL, D_model2_nooutlier)  
summary(D_model3)
```

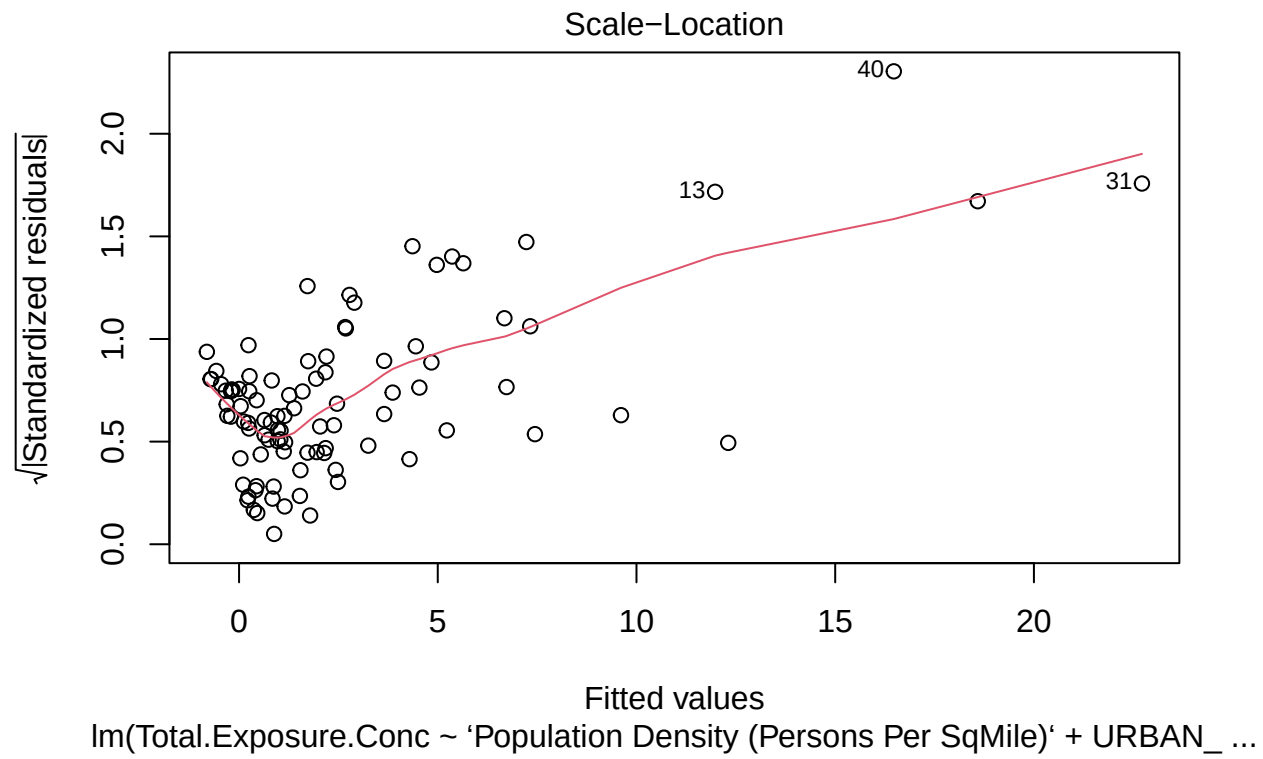
```
##  
## Call:  
## lm(formula = Total.Exposure.Conc ~ 'Population Density (Persons Per SqMile)' +  
##     URBAN_RURAL, data = D_model2_nooutlier)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -3.6520 -0.4362  0.0704  0.6082  6.3162   
##  
## Coefficients:  
##              Estimate Std. Error t value  
## (Intercept)    -0.8758865   0.1911439   -4.582  
## 'Population Density (Persons Per SqMile)'  0.0211168   0.0007829  26.972  
## URBAN_RURALUrban    -0.3455776   0.3024506   -1.143
```

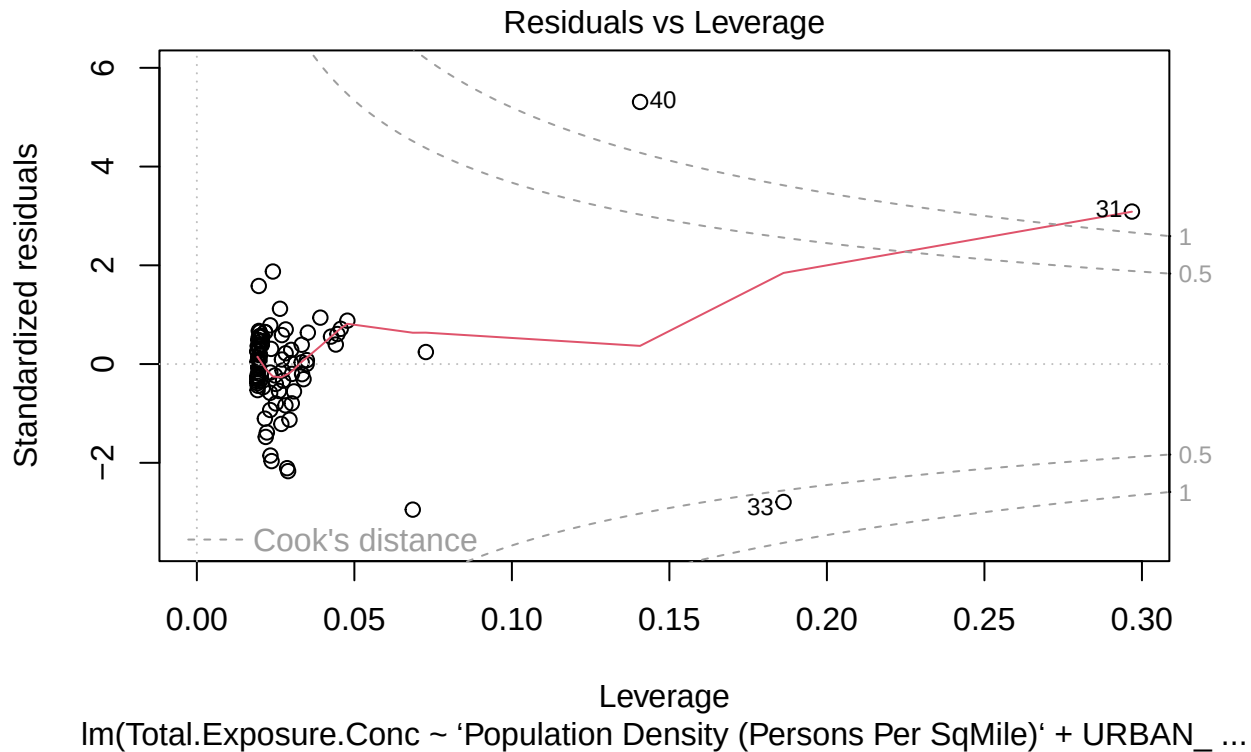
```
##                                Pr(>|t|)
## (Intercept)                   1.44e-05 ***
## 'Population Density (Persons Per SqMile)' < 2e-16 ***
## URBAN_RURALUrban              0.256
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.284 on 92 degrees of freedom
## Multiple R-squared:  0.9085, Adjusted R-squared:  0.9065
## F-statistic: 456.7 on 2 and 92 DF,  p-value: < 2.2e-16
```

```
plot(D_model3)
```









```
AIC(D_model13) #321.977
```

```
## [1] 321.977
```

```
diesel_complete <- diesel_complete %>%
  mutate(log_conc = log10(Total.Exposure.Conc))

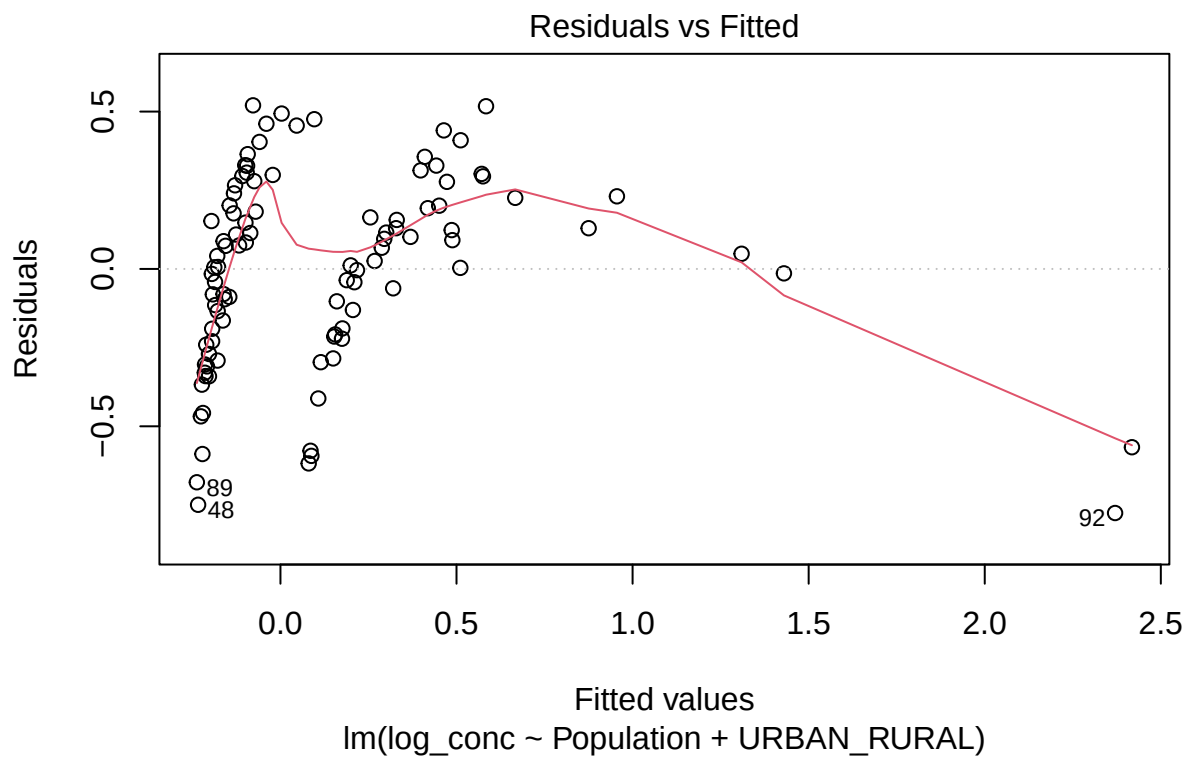
#log concentrations in diesel complete (new column)
#instead of total concentratuion, have log concentration in the model

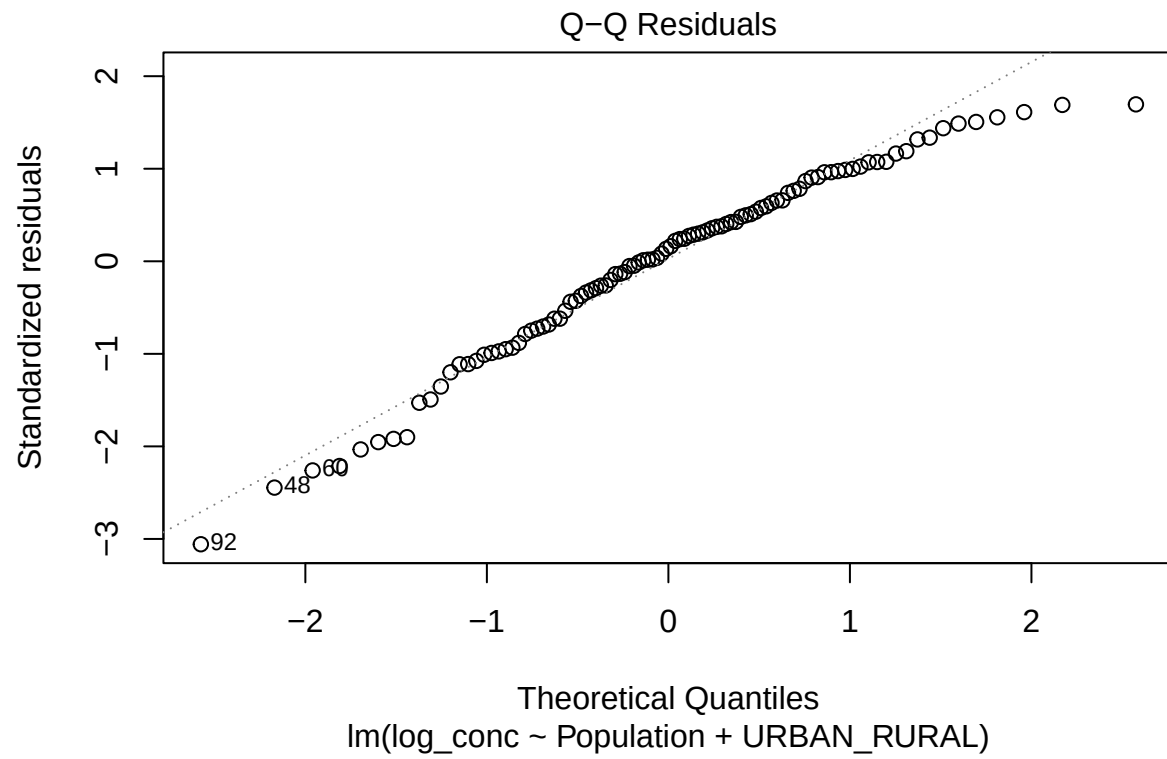
D_model_log <- lm(log_conc ~ Population + URBAN_RURAL,
  diesel_complete)
summary(D_model_log)

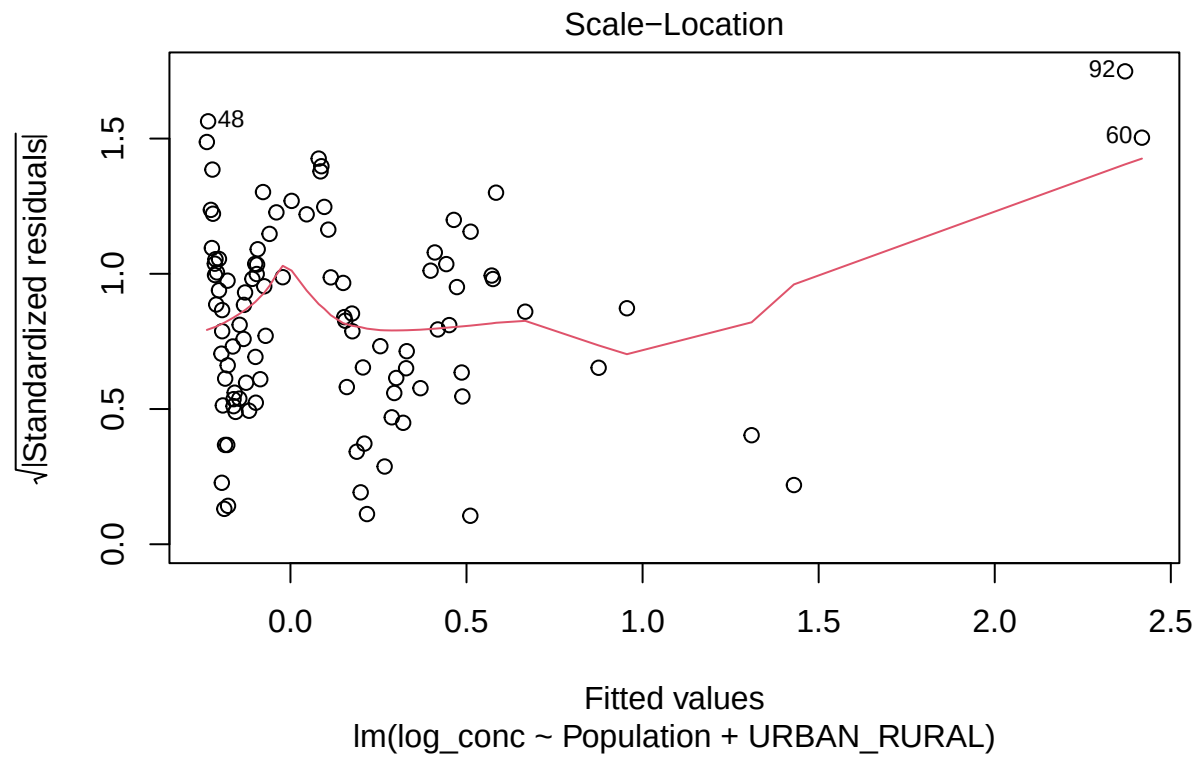
##
## Call:
## lm(formula = log_conc ~ Population + URBAN_RURAL, data = diesel_complete)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.77552 -0.20979  0.04476  0.22712  0.51979
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -2.487e-01  4.317e-02  -5.761 9.84e-08 ***
```

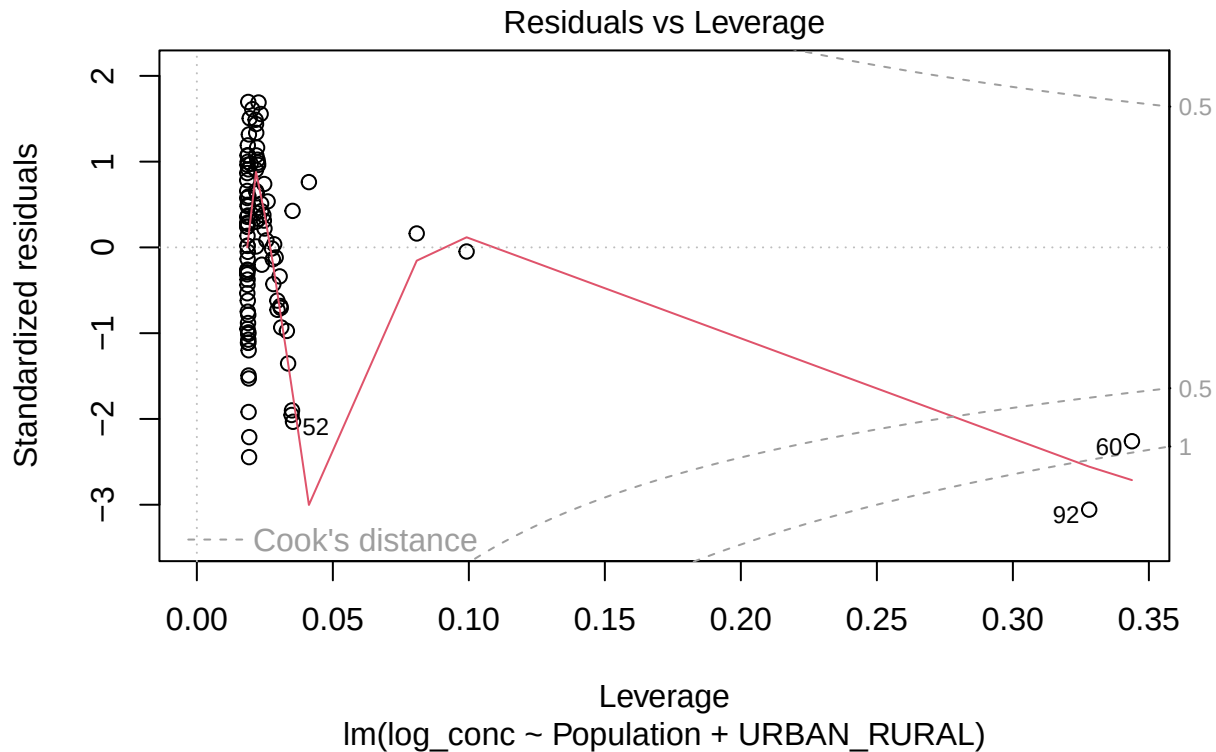
```
## Population      1.285e-06  1.164e-07  11.042  < 2e-16 ***
## URBAN_RURALUrban 3.028e-01  6.852e-02   4.419  2.58e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3095 on 97 degrees of freedom
## Multiple R-squared:  0.6964, Adjusted R-squared:  0.6902
## F-statistic: 111.3 on 2 and 97 DF,  p-value: < 2.2e-16
```

```
plot(D_model_log)
```









```
AIC(D_model_log)
```

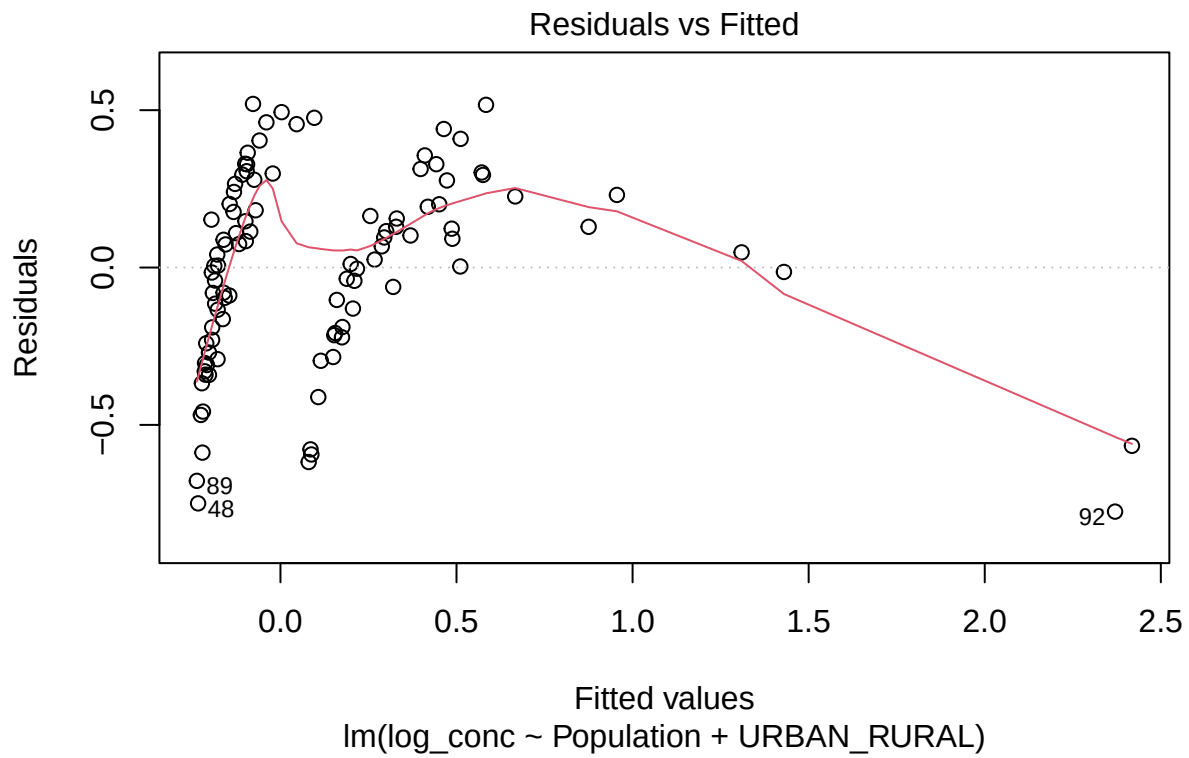
```
## [1] 54.15746
```

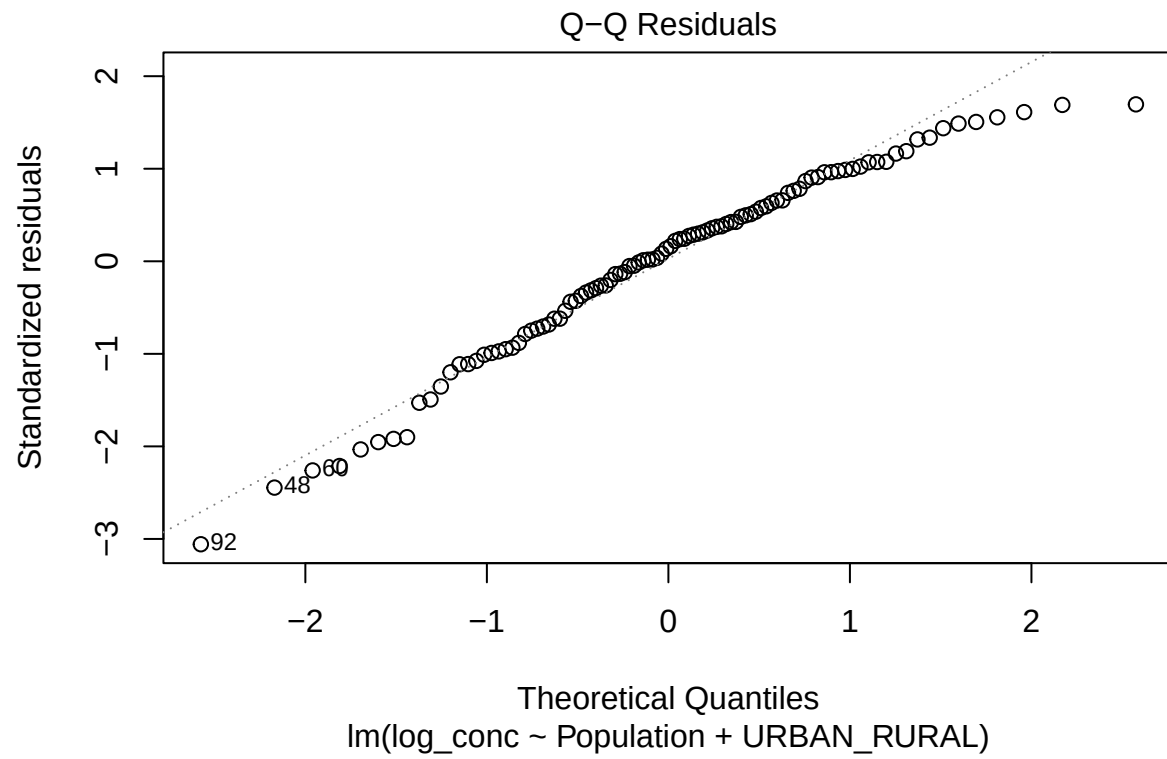
```
log_model2 <- lm(log_conc ~ Population + URBAN_RURAL,
  diesel_complete)
summary(log_model2)
```

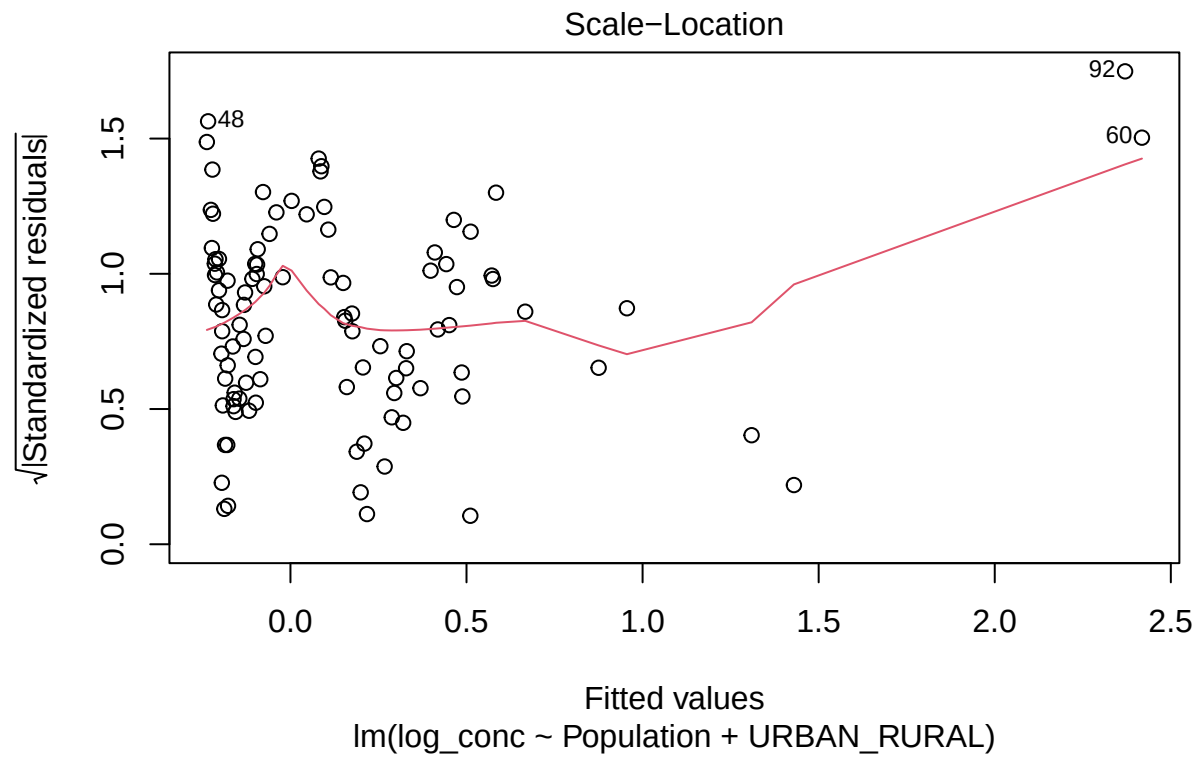
```
##
## Call:
## lm(formula = log_conc ~ Population + URBAN_RURAL, data = diesel_complete)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.77552 -0.20979  0.04476  0.22712  0.51979
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -2.487e-01  4.317e-02  -5.761 9.84e-08 ***
## Population      1.285e-06  1.164e-07  11.042 < 2e-16 ***
## URBAN_RURALUrban  3.028e-01  6.852e-02   4.419 2.58e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

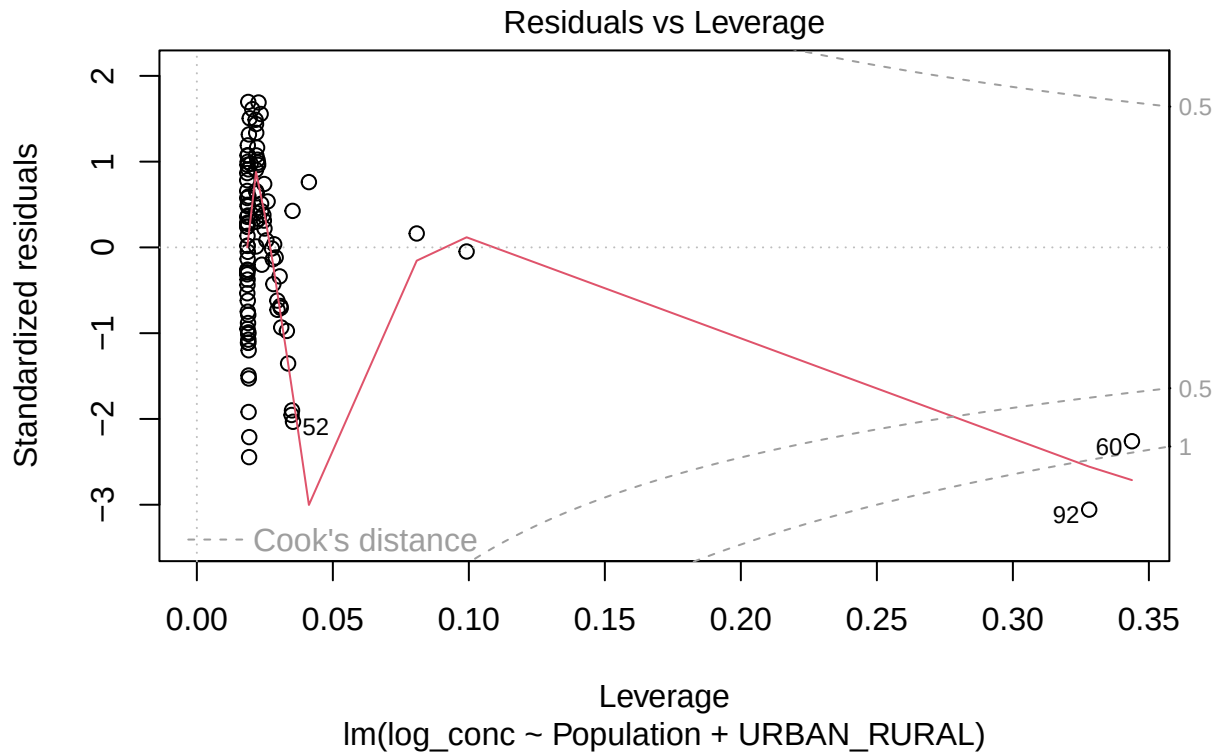
```
## Residual standard error: 0.3095 on 97 degrees of freedom
## Multiple R-squared:  0.6964, Adjusted R-squared:  0.6902
## F-statistic: 111.3 on 2 and 97 DF,  p-value: < 2.2e-16
```

```
plot(log_model2)
```







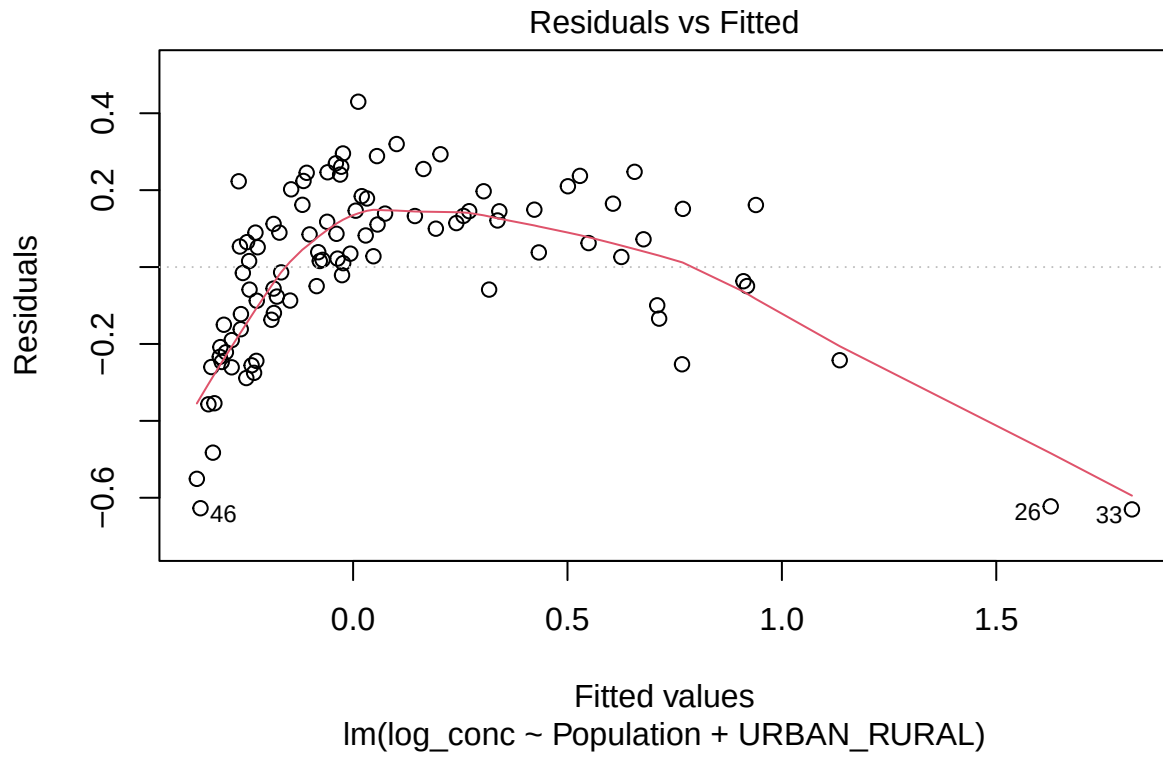


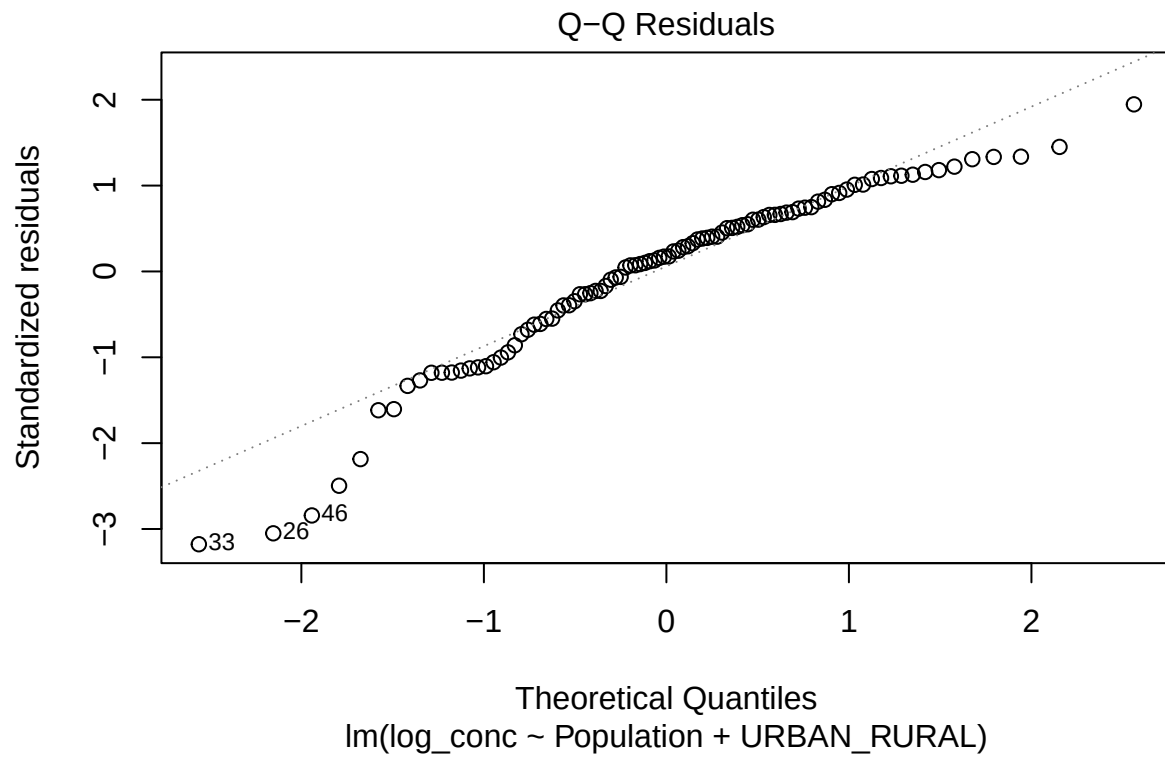
```
diesel_nooutlier2 <- diesel_complete[-c(60,92),]
log_nooutliers <- diesel_nooutlier2[-c(32,41),]

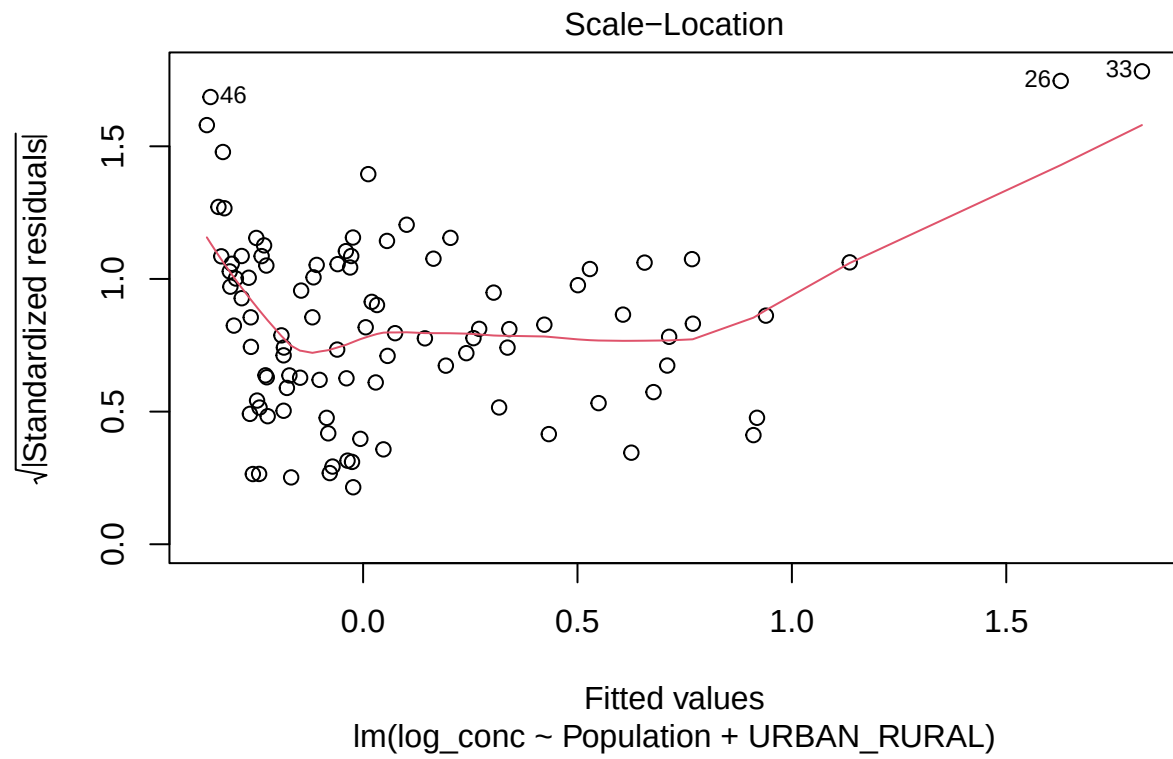
log_model3 <- lm(log_conc ~ Population + URBAN_RURAL,
                 log_nooutliers)
summary(log_model3)
```

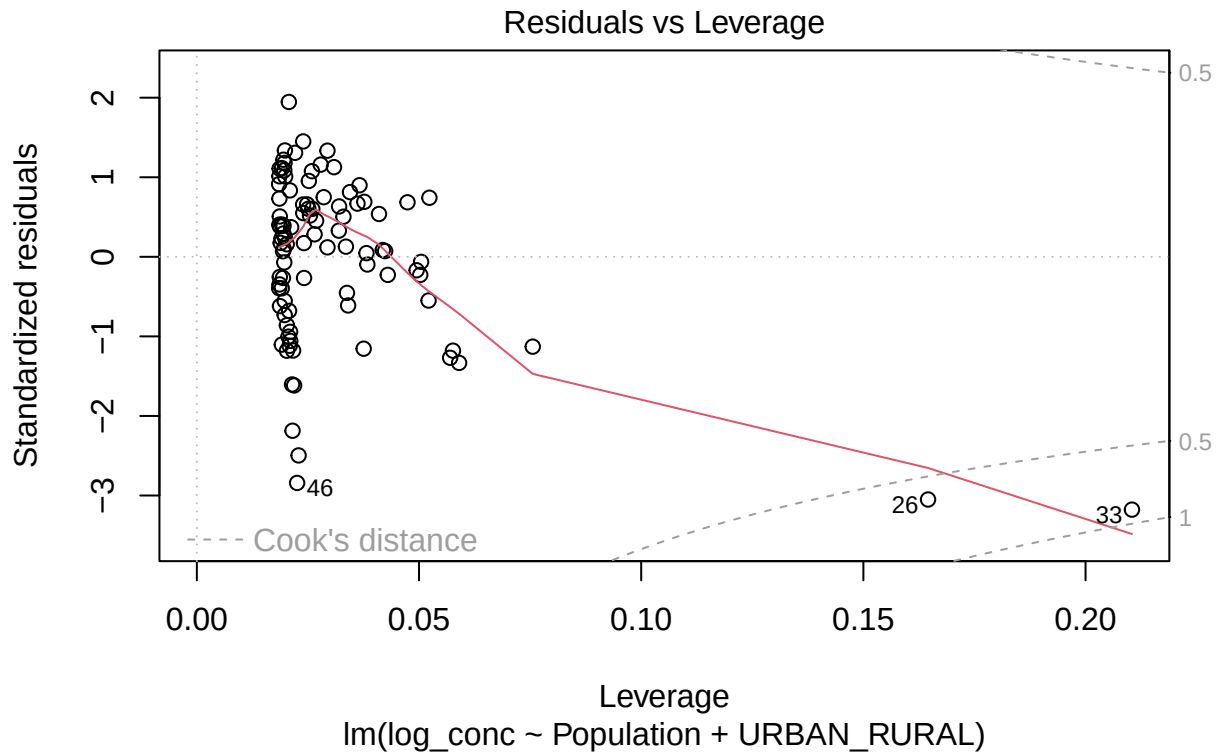
```
##
## Call:
## lm(formula = log_conc ~ Population + URBAN_RURAL, data = log_nooutliers)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.63026 -0.12527  0.03832  0.14978  0.42995
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -3.912e-01  3.460e-02 -11.307  <2e-16 ***
## Population      3.033e-06  2.031e-07  14.933  <2e-16 ***
## URBAN_RURALUrban  8.052e-02  5.456e-02   1.476    0.143
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2232 on 93 degrees of freedom
## Multiple R-squared:  0.7907, Adjusted R-squared:  0.7862
## F-statistic: 175.7 on 2 and 93 DF, p-value: < 2.2e-16
```

```
plot(log_model3)
```









```
log_nooutliers2 <- log_nooutliers[-c(26,33),]
```

```
log_model4 <- lm(log_conc ~ Population + URBAN_RURAL,
                 log_nooutliers2)
summary(log_model4)
```

```
##
## Call:
## lm(formula = log_conc ~ Population + URBAN_RURAL, data = log_nooutliers2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.57574 -0.12229  0.03976  0.12985  0.39210
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -4.512e-01  3.161e-02 -14.273  <2e-16 ***
## Population      3.769e-06  2.165e-07  17.412  <2e-16 ***
## URBAN_RURALUrban 2.141e-02  4.820e-02   0.444    0.658
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1927 on 91 degrees of freedom
## Multiple R-squared:  0.8313, Adjusted R-squared:  0.8275
## F-statistic: 224.1 on 2 and 91 DF,  p-value: < 2.2e-16
```

```
plot(log_model4)
```

